

SSC201 / SSC202

STATISTICS

CHAPTER ONE

STATISTICS, DATA COLLECTION AND SUMMARY

CHAPTER CONTENTS

- (a) Types of Data;
- (b) Types of Data by Generation Source;
- (c) Methods of Data Collection;
- (d) Presentation of Data;
- (e) Sampling Techniques; and
- (f) Errors and Approximation

OBJECTIVES

At the end of the chapter, students and readers should be able to

- a) know the meaning of statistics
- b) know types of data, how to collect and classify them
- c) understand the concept of data presentation in form of charts and graphs
- d) understand various methods of sampling
- e) distinguish between error and approximation.

1.1 Introduction

Statistics is a scientific method that concerns data collection, presentation, analysis, interpretation or inference about data when issues of uncertainties are involved. Definitely, statistics is useful and needed in any human area of endeavour where decision making is of vital importance. Hence, it is useful in Accountancy, engineering, Education, Business, social Sciences, Law, Agriculture, to mention a few.

Statistics can be broadly classified into two:

- (i) Descriptive and (ii) Inferential.

Descriptive statistics deals with data collection, summarizing and comparing numerical data, i.e. descriptive statistics is concerned with summarizing a data set rather than use the data to learn about the population, while inferential statistics deals with techniques and tools for collection of data from a population. These data are then studied by taking a sample from the population. By doing this, knowledge of the population characteristics is gained and vital decisions are made about the sampled population.

1.2 Types of data

Data are the basic raw facts needed for statistical investigations. These investigations may be needed for planning, policy implementation, and other purposes.

Data can be quantitative or qualitative.

It is quantitative when the data set have numerical values. Examples of quantitative data include:

- (a) heights of objects,
- (b) Weights of objects
- (c) prices of goods and services,
- (d) The family expenditure on monthly basis.

The quantitative data, which are also termed numeric data, can be further classified into two:

- i) **Discrete:** The data values are integers (whole numbers). Examples include: (i) number of books sold in bookshop, (ii) number of registered ABWA students.
- ii) **Continuous:** the data values are real numbers which can be integer, fraction or decimal. Examples include: (i) heights of objects; (ii) weights of objects; (iii) price of goods.

For the qualitative data, it is the set of data which have no numerical values and can not be adequately represented by numbers. They are also called non-numerical data. They are further classified into two:

- i) **Ordinal Non-numeric:** The data set is on ordinal scale when ranking is allowed. Example can be seen in the beauty contest judgment.
- ii) **Nominal Non-numeric:** The data set is in categorical form where the facts collected are based on classification by group or category. Examples include: (i) level of education, (ii) sex, (iii) occupation.

1.3 Types of Data by Generation Source

The data source can be broadly classified into: primary and secondary

- (a) **Primary data:** These data originate or are obtained directly from respondents for the purpose of an initial investigation.

(b) Secondary data: These are data obtained, collected, or extracted from already existing records or sources. They are derived from existing published or unpublished records of government agencies, trade associations, research bureaus, magazines and individual research work.

It will be of great benefit to know that data source can be either internal or external. Internal source are the data collected within the organization. Such data are used within the organization. Examples are (a) the sales receipts, (b) invoice, (c) work schedule.

For the external source, the data are collected outside the organization. The data are generated or collected from any other places which are external to the user.

1.4 Methods of Data Collection

The major practising methods of data collection are: The Interview, Questionnaire, Observation, Documents/Reports and Experimental methods.

1.4.1 Interview Method

This is a method in which the medium of data collection is through conversation by face-to-face contact, through a medium like telephone.

In general, the interview method can be accomplished by:

(a) Schedule; (b) Telephone; and (c) Group discussion.

(a) Interview schedule: This is the use of schedule by the interviewer to obtain necessary facts/data. A schedule is a form or document which consists of a set of questions to be completed by interviewer as he/she asks the respondent questions.

(b) Telephone interview: Here the questions are asked through the use of telephone in order to get the needed data. The respondents and interviewers must have access to telephone.

(c) Interview by Group discussion: The interview is conducted with more than one person with a focus on a particular event.

1.4.2 Questionnaire Method

A questionnaire is a document that consists of a set of questions which are logically arranged and are to be filled by the respondent himself. The interviewers

send the copies of their questionnaires to the respondents to be filled without necessarily being present there.

The types of questions in a questionnaire can be classified into two; namely:

- a. Close – ended or coded question is the type of questions in which alternative answers are given for the respondent to pick one.
- b. Open-ended or uncoded questions is the type of questions in which the respondent is free to give his own answer and is not restricted to some particular answers.

In drafting a questionnaire, the following qualities are essentials to note:

- i. A questionnaire must be well structured so that major sections are available. The first section usually deals with personal data while other sections consider necessary and relevant questions on the subject matter of investigation.
- ii. A questionnaire must be clear in language, not ambiguous and not lengthy.
- iii. A question should not be a leading question.

1.4.3 Observation Method

This method enables one to collect data on behaviour, skills etc. of persons, objects that can be observed in their natural ways. Observation method can either be of a controlled or uncontrolled type. It is controlled when issues to be observed are predetermined by rules and procedures; while otherwise, it is termed uncontrolled type.

1.4.4 Experimental Method

Here, experiments are carried out in order to get the necessary data for the desired research. There are occasions when some factors are to be controlled and some not controlled in this method. The method is mostly used in the sciences and engineering.

1.4.5 Documents/Reports

This method allows one to check the existing documents/reports from an organization.

1.5 Presentation of Data

Statistical data are organized and classified into groups before they are presented for analysis. Four important bases of classification are:

- (a) Qualitative – By type or quality of items under consideration.
- (b) Quantitative – By range specified in quantities.
- (c) Chronological – Time series – Monthly or Yearly: An analysis of time series involving a consideration of trend, cyclical, periodic and irregular movements.
- (d) Geographical – By location.

The classified data are then presented in one of the following three methods: (a) Text presentation, (b) Tabular presentation and (c) Diagrammatic presentation:

1.5.1 Text Presentation:

This is a procedure by which text and figures are combined. It is usually a report in which much emphasis is placed on the figures being discussed.

For instance, a text presentation can be presented as follows:

The populations of science and management students are 3000 and 5000 respectively for year 2006 in a Polytechnic in Ghana.

1.5.2 Tabular presentation:

A table is more detailed than the information in the text presentation. It is brief and self explanatory. A number of tables dealt with in statistical analysis are – general reference table, summary table, Time series table, frequency table.

Tables may be simple or complex. A simple table relates a single set of items such as the dependent variable against another single set of items – the independent variable. A complex table on the other hand has a number of items presented and often shows sub-divisions.

Essential features of a table are:

- A title to give adequate information about it.
- Heading for identification of the rows and columns.
- Source i.e. the origin of the figures; and
- Footnote to give some detailed information on some figures in the table.

Illustration 1.1: A typical example of simple table.

Classification of two hundred Polytechnic students on departmental basis.

Department	No of Students
Accountancy	60
Business Administration	50
Marketing	40
Banking / Finance	50
Total	200

Illustration 1.2: A Typical Example of Complex Table

Departmental classification of 200 University students on the basis of gender.

Department	No of Students		Total
	Male	Female	
Accountancy	40	20	60
Business Administration	36	14	50
Marketing	30	10	40
Banking / Finance	24	26	50
Total	130	70	200

Illustration 1.3: Formulation of Frequency Table

The following marks are obtained by 30 students in accountancy department out of a total mark of 100:

6, 10, 16, 60, 50, 30, 70, 14, 32, 80, 98, 80, 28,
 32, 60, 46, 42, 44, 58, 54, 50, 64, 30, 22, 18, 16,
 86, 56, 64, 28.

Prepare a frequency table with class intervals of 1 – 10, 11 – 20, 21 – 30, etc.

Solution

Usually a Tally method is used. A tally is a stroke (|) for each appearance of a number. The fifth stroke is drawn across the first four strokes (||||). This allows for easy counting.

The great advantage of the tally method is that you run across the data only once.

Class Intervals	Tally Bars	Frequency
1 – 10		2
11 – 20		4
21 – 30		5
31 – 40		2
41 – 50		5
51 – 60		5
61 – 70		3
71 – 80		2
81 – 90		1
91 – 100		1
		N = 30

1.5.3 Diagrammatic Presentation:

Diagrams are used to reflect the relationship, trends and comparisons among variables presented on a table. The diagrams are in form of charts and graphs.

Examples are:

(a) Bar charts

- (i) Simple bar chart; component bar chart
- (ii) Percentage component bar charts;
- (iii) Multiple bar charts

(b) Pie Charts.

(c) Pictograms

(d) Statistical maps and area diagrams

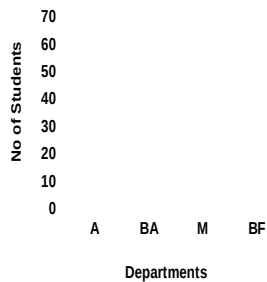
(e) Graphs – a graph shows relationship between variables concerned by means of a curve or a straight line. A graph will, for example, show the relationship between output and cost, or the amount of sales to the time the sales were made.

Typical graphs used in business are the cumulative frequency curve, histogram based on a time series data; the z – charts and the net balance charts; break-even chart and progress chart.

Illustration 1.4: Simple Bar Chart

Draw simple bar chart for the table in Example 1.4.2.1. i.e:

Department	No of Students
Accountancy	60
Business Administration	50
Marketing	40
Banking / Finance	50
Total	200



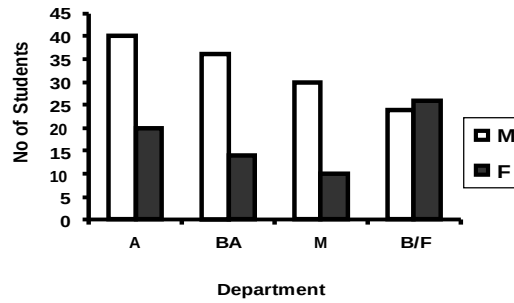
Legend

Departments	Key
Accountancy	A
Business Admin	BA
Marketing	M
Banking/Finance	B/F

Illustration 1.5:(a) Multiple Bar Chart

Draw Multiple bar chart for the table in Example 1.4.2.2. i.e:

Department	No of Students		Total
	Male	Female	
Accountancy	40	20	60
Business Administration	36	14	50
Marketing	30	10	40
Banking / Finance	24	26	50
Total	130	70	200



(b) Draw Component bar chart for the table in (a)

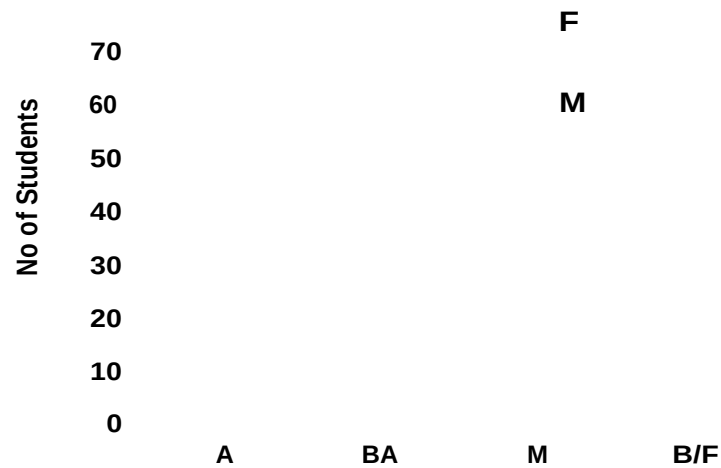


Illustration 1.6: Pie Chart

Draw Pie chart for the table in Example 1.4.2.1.

Solution

Calculation of angles

Total no of students is 200

$$\therefore \text{For Accountancy, the corresponding angle is} = \frac{60}{200} \times \frac{360}{1} = 108^{\circ}$$

$$\text{For Business Administration, the corresponding angle is} = \frac{50}{200} \times \frac{360}{1} = 90^{\circ}$$

$$\text{For Marketing, the corresponding angle is} = \frac{40}{200} \times \frac{360}{1} = 72^{\circ}$$

$$\text{For Banking / Finance, the corresponding angle is} = \frac{50}{200} \times \frac{360}{1} = 90^{\circ}$$

Check: $108^{\circ} + 90^{\circ} + 72^{\circ} + 90^{\circ} = 360^{\circ}$

Pie chart

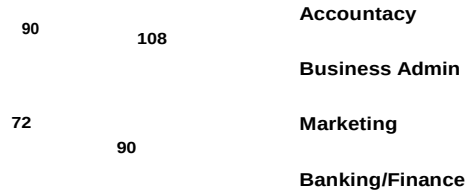


Illustration 1.7: Component Bar Chart

Draw the percentage component bar chart for the table below.

	Accountancy	Banking/Finance	Total
Male	40	20	60
Female	30	10	40
	70	30	100

Solution

	Accountancy	%	Banking/Finance	%	Total
Male	40	57%	20	67%	60
Female	30	43%	10	33%	40
	70	100	30	100	100



Illustration 1.8:

From Example 1.4.2.3 draw the Histogram for the frequency distribution.

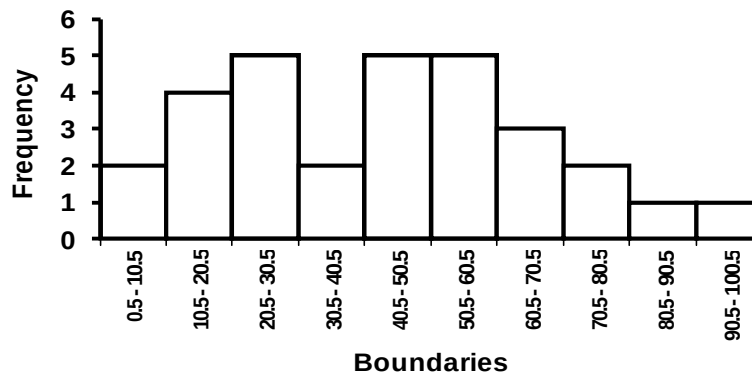
Class Interval	Frequency
1 – 10	2
11 – 20	4
21 – 30	5
31 – 40	2
41 – 50	5
51 – 60	5
61 – 70	3
71 – 80	2
81 – 90	1
91 – 100	1
	30

Solution

Class Interval	Class Boundaries	Frequency
1 – 10	0.5 – 10.5	2
11 – 20	10.5 – 20.5	4
21 – 30	20.5 – 30.5	5
31 – 40	30.5 – 40.5	2
41 – 50	40.5 – 50.5	5
51 – 60	50.5 – 60.5	5
61 – 70	60.5 – 70.5	3
71 – 80	70.5 – 80.5	2
81 – 90	80.5 – 90.5	1
91 – 100	90.5 – 100.5	1

e.g. the lower class boundary of the first class interval is $\frac{0+1}{2} = 0.5$, while the upper class boundary is $\frac{10+11}{2} = 10.5$

The Histogram

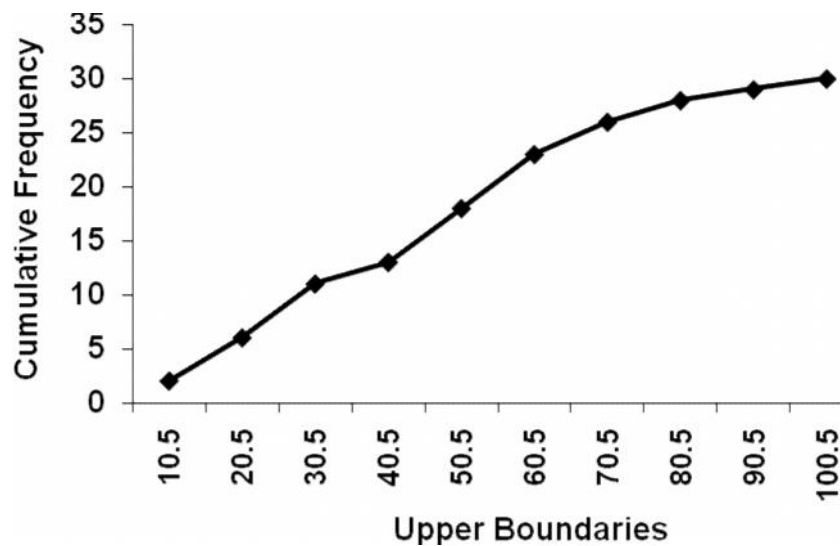


Note: The bars in the histogram must touch each other unlike in the bar chart where the bars may or may not touch each other.

Illustration 1.9:

From Example 1.4.2.3 construct the cumulative frequency (Ogive).

Boundaries	Frequency	C. F
0.5 – 10.5	2	2
10.5 – 20.5	4	6
20.5 – 30.5	5	11
30.5 – 40.5	2	13
40.5 – 50.5	5	18
50.5 – 60.5	5	23
60.5 – 70.5	3	26
70.5 – 80.5	2	28
80.5 – 90.5	1	29
90.5 – 100.5	1	30
	30	



1.6 Sampling Techniques

Sampling technique is an important statistical method that entails the use of fractional part of a population to study and make decisions about the characteristics of the given population. As an introduction to the sampling techniques, the following concept shall be discussed:

1.6.1 Definition of Some Important Concepts on Sampling Techniques

- (a) **Population:** a population is a collection or group of items satisfying a definite characteristic. Items in the definition can be referred to
- (i) human beings (as population census);
 - (ii) animate objects as cattle, sheep, goats etc
 - (iii) inanimate objects such as chairs, tables, etc.
 - (iv) a given population like students of ATS.

There are two types of population, namely: finite and infinite populations:

By finite population, we mean the population which has countable number of items; while the items in infinite population are uncountable. Population of students in a college is an example of finite population while sand particles on the beach are that of infinite population.

- (b) **Sample:** A sample is the fractional part of a population selected in order to observe or study the population for the purpose of making scientific statement or decision about the said population.
- (c) **Census:** This is a complete enumeration of all units in the entire concerned population with the aim of collecting vital information on each unit. Year 2006 head count in Nigeria is a good example of census.
- (d) **Sample Survey:** A sample survey is a statistical method of collecting information by the use of fractional part of population as a representative sample.
- (e) **Sampling Frame:** This is a list containing all the items or units in the target population and it is normally used as basis for the selection of sample. Examples are: list of church members, telephone directory, etc.
- (f) **Some sampling notations and terms:**
 - (i) Let “N” represent the whole population, then the sample size is denoted by “n”.
 - (ii) $f = n/N$, where f is called the sampling fraction, N and n are as defined in vi(a).
 - (iii) The expansion or raising factor is given as $g = N/n$, where g is the expansion factor, N and n are as defined in vi(a).

1.6.2 Types of Sampling Techniques

Sampling techniques or methods can be broadly classified into two types; namely:

- (a) probability sampling methods; and
- (b) purposive sampling method.

(i) The Probability Sampling Method

The probability sampling technique is a method involving random selection. It is done in such a way that each unit of the population is given a definite chance or probability of being included in the sample. Among these sampling techniques are simple random sampling, systematic sampling, stratified sampling, multi-stage sampling and cluster sampling.

- (ii) **Simple Random Sampling (SRS):** This is a method of selecting a sample from the population with every member of the population having equal chance of being selected. The sampling frame is required in this technique.

The method can be achieved by the use of

- table of random numbers;
- lottery or raffle draw approach.

We are to note that SRS can be with replacement or without replacement. When it is with replacement, a sample can be repeatedly taken or selected; while in the SRS without replacement, a sample can not be repeatedly taken or selected.

Advantages of Simple Random Sampling

- SRS gives equal chance to each unit in the population which can be termed a fair deal.
- SRS method is simple and straight forward.
- SRS estimates are unbiased.

Disadvantages of Simple Random Sampling

- The method is not good for a survey if sampling frame is not available.
- There are occasions when heavy drawings are made from one part of the population, the idea of fairness is defeated.

- iii) **Systematic Sampling (SS):** This is a method that demands for the availability of the sampling frame with each unit numbered serially. Then from the frame, selection of the samples is done on regular interval K after the first sample is selected randomly within the first given integer number K; where

$K = \frac{N}{n}$, and must be a whole number, and any decimal part of it should be cut off or cancelled. N and n are as earlier defined. That is, after the first selection within the K interval, other units or samples of selection will be successfully equi-distant from the first sample (with an interval of K).

For instance, in a population $N = 120$ and 15 samples are needed, the SS method will give us

$$K = 120/15 = 8$$

It means that the first sample number will be between 1 and 8. Suppose serial number 5 is picked, then the subsequent numbers will be 13, 21, 29, 37, ---. (Note that the numbers are obtained as follows: $5 + 8 = 13$, $13 + 8 = 21$, $21 + 8 = 29$, $29 + 8 = 37$ etc).

Advantages of Systematic Sampling

- o It is an easy method to handle.
- o The method gives a good representative if the sampling frame is available.

Disadvantages

- If there is no sampling frame, the method will not be appropriate.

- (iv) **Stratified Sampling (STS):** This is a method that is commonly used in a situation where the population is heterogeneous. The principle of breaking the heterogeneous population into a number of homogeneous groups is called stratification. Each group of unique characteristics is known as stratum and there must not be any overlapping in the groups or strata. Some of the common factors that are used to determine the division or breaking into strata are income levels, employment status, etc.

The next step in stratification is the selection of samples from each group (stratum) by simple random sample. An example of where to apply STS can be seen in the survey concerning standard of living or housing pattern of a town.

Advantages of STS

- By pooling the samples from the strata, a more and better representative of the population is considered for the survey or investigation.
- Breaking the population into homogeneous goes a long way to have more precise and accurate results.

Disadvantages of STS

- There are difficulties in deciding the basis for stratification into homogeneous groups.
- STS suffers from the problems of assigning weights to different groups (strata) when the condition of the population demands it.

- (v) **Multi-Stage Sampling:** This is a sampling method involving two or more stages. The first stage consists of breaking down the population into first set of distinct groups and then select some groups randomly.

The list of selected groups is termed the primary sampling units. Next, each group selected is further broken down into smaller units from which samples are taken to form a frame of the second-stage sampling units. If we stop at this stage, we have a two-stage sampling.

Further stages may be added and the number of stages involved is used to indicate the name of the sampling. For instance, five-stage sampling indicates that five stages are involved.

An example where the sampling method [MSS] can be applied is the survey of students' activities in higher institutions of a country. Here, first get the list of all higher institutions in the country and then select some randomly. The next stage is to break the selected institutions into faculties and select some. One can go further to break the selected faculties into departments and select some departments to have the third-stage sampling.

Advantages Multi-stage sampling(MSS)

- The method is simple if the sampling frame is available at all stages.
- It involves little cost of implementation because of the ready-made availability of sampling frame.

Disadvantages of MSS

- If it is difficult to obtain the sampling frame, the method may be tedious.
- Estimation of variance and other statistical parameters may be very complicated.

- (vi) **Cluster Sampling [CS]:** Some populations are characterized by having their units existing in natural clusters. Examples of these can be seen in farmers' settlement (farm settlements are clusters); students in schools (schools are clusters). Also, there could be artificial clusters when higher institutions are divided into faculties.

The cluster method involves random selection of A clusters from M clusters in the population which represents the samples.

(vii) **Non-Probability Sampling**

The sampling techniques which are not probabilistic are **Purposive Sampling (PS)**.

In this sampling, no particular probability for each element is included in sample. It is at times called the judgment sampling.

In purposive sampling, selection of the sample depends on the discretion or judgment of the investigator. For instance, if an investigation is to be carried out on students' expenditures in the university hostels, the investigator may select those students who are neither miser nor extravagant in order to have a good representation of the targeted population.

Advantage of PS

- It is very cheap to handle.
- No need of sampling frame.

Disadvantage of PS

- The method is affected by the prejudices of the investigator.
- Estimates of sampling errors cannot be possible by this method.

(viii) **Quota Sampling (QS)**

This is a sampling technique in which the investigator aimed at obtaining some balance among the different categories of units in the population as selected samples. A good example of this can be seen in the quota selection of students to Federal Colleges in Nigeria.

Advantages of QS

- It has a fair representation of various categories without probability basis.
- No need for sampling frame.

Disadvantages of QS

- Sampling error cannot be estimated.

1.7 Errors and Approximation

Approximation and errors are inevitable phenomena in data. Hence, the two concepts shall be examined in the following subsequent sections.

1.7.1 Approximation:

The results of the measurements, counting, recordings and observation in data collection, and even in computations are subjected to approximations due to a certain degree of accuracy desired. In the light of this, the concept of approximation is relevant and necessary to know or determine the degree of accuracy to be maintained in data collection and various computations.

Formally, approximation can be defined as a technique of rounding off a number so that the last digit(s) is(are) affected in order to make the number clearer and more understandable.

For instance, let's compare these two statements:

- a. the total number of people in a town is 515,176;
- b. the total number of people in a town is 515,000

There is no doubt that statement (b) is more comprehensive than statement (a) even though statement (a) is more accurate.

Basically, the approximation concept is achieved by rounding off statistical figures or numbers to:

- specific nearest units e.g to the nearest 100,
- specific significant figures e.g to three significant figures.

Furthermore, the principle of rounding off can be done by either

- rounding up; where any figure which is up to half of the required specification unit is rounded up to a whole unit; or
- rounding down, where any figure or number is not up to half of the required specification unit is rounded down to zero unit.

The following examples shall be used to demonstrate the principle of approximation.

Illustration 1.10

- Approximate 514,371 to:
- (a) to the nearest 1,000
 - (b) to the nearest 100
 - (c) to 3 significant figures
 - (d) to 4 significant figures.

Solution:

- a. 514,000 (because 371 is not up to half of 1000)
- b. 514,400 (because 71 is up to half of 100)
- c. 514,000 (because 3 of hundred unit is the fourth digit and is not up to 5)
- d. 514,400 (because 7 of ten unit is the fifth digit and is up to 5)

Illustration 1.11

Approximate the following:

- a. 154.235 to 2 decimal places and 2 significant figures.
- b. 0.02567 to 3 decimal places and 3 significant figures.

Solution

- a.
 - i. To 2 decimal places = 154.24
 - ii. To 2 significant figures = 150.000
- b.
 - i. To 3 decimal places = 0.026
 - ii. To 3 significant figures = 0.0257

Note: Leading and trailing Zeros are not significant.

1.7.2 Errors

- (a) In statistical thinking, the word “error” is used in a very restricted sense. It denotes the difference between the estimation (through measurements or computations as obtained by the investigator) and the actual size of the object or item under investigation.

Sources of some errors can be traced to:

- (i) Origin: where for example, precision in measuring is not possible due to limitations of the measuring instruments;
- (ii) Inadequacy: which can be due to small samples or insufficient coverage;
- (iii) Manipulation: which can be seen as errors unconsciously committed by enumerators in measuring the object;
- (iv) Calculation: with all arithmetic mistakes and other calculation errors.

Errors can be broadly classified into two; namely:

- (a) Sampling error; and (b) Non-sampling error.
 - **Sampling Error:** these are series of errors that can be committed due to sampling principles. When these accumulated errors are combined, they form the sampling errors. Suppose in a class of 30, a sample of five is taken in order to obtain necessary information about the class. The difference between the estimate based on the sample (of five students) and the actual class value is termed the sampling error. At times, sampling error is called “sampling variability”.
 - **Non-Sampling Errors:** these are errors committed which are not due to sampling principles. Therefore, the term “non-sampling errors” refers to errors in measurements, counting, assessments, judgments, calculations (including arithmetic mistakes), interpretations, instructions and even in machines. For instance, there exists only non-sampling error in census while both sampling and non-sampling errors are present in sample survey.

(b) Measurement of errors:

Errors can be measured as absolute, relative and percentage.

Let $\underline{x}$ be the observed (estimated) value and

$\underline{x} + \underline{e}$ be the true value, then

$$\text{error} = (\underline{x} + \underline{e}) - \underline{x} = \underline{e} \quad \text{---1.6.2.2.1}$$

This error($\underline{e}$) in equation (1.6.2.2.1), which is the difference between observed and true value is known as the absolute Error; and it is dependent upon the $\underline{x}$ unit of measurement.

The Relative Error is the absolute error divided by the true value.

$$\text{i.e. relative Error} = \frac{\underline{e}}{(\underline{x} + \underline{e})} \quad \text{-----1.6.2.2.2}$$

While the Percentage Error = Relative Error x 100

The relative and percentage errors are both independent of the unit of measurement. Therefore, they are good and true indices of measurement or calculation accuracy.

Illustration 1.12

A population with 518,413 as actual number has an estimated figure of 518,000.

Determine the:

- a. Absolute Error;
- b. Relative Error
- c. Percentage Error.

Solution

- a. Using equation (1.6.2.2.1), the
Absolute error = $518,413 - 518,000 = 413$.
- b. Using equation (1.6.2.2.2), the
Relative error = $\frac{413}{518000} = 0.0007973$
- c. Percentage error = $0.0007973 \times 100\%$
= 0.07973%

1.8 Summary

Data are defined as raw facts in numerical form. Its classification by types, method of collection and the forms of presentation are discussed. Some sampling techniques with their advantages and disadvantages are presented. The issues of approximation and errors are also presented.

Multiple Choice And Short – Answer Questions

1. Which of the following is a non-numeric ordinal data?
 - A. Income
 - B. Price of commodity
 - C. Occupation
 - D. Rating in beauty contest
 - E. Students number in a class

2. A schedule in statistics refers to_____
 - A. Examination time table
 - B. set of questions used to gather pieces of information which is filled by an informant or respondent.
 - C. A set of questions used to gather information and filled by the investigator him/herself.
 - D. A set of past examination questions.
 - E. Paper used by bankers to carry out investigation.

3. Which of the following sampling methods does not need sampling frame?
 - A. Simple random sampling.
 - B. Purposive sampling
 - C. Systematic sampling
 - D. Cluster sampling
 - E. Stratified sampling.

4. The following are qualities of a good questionnaire except
 - A. That each question in the questionnaire must be precise and unambiguous.
 - B. Avoidance of leading question in the questionnaire.
 - C. That a questionnaire must be lengthy in order to accommodate many questions.
 - D. That a questionnaire must be well structured into sections such that questions in each section are related.
 - E. Avoidance of double barrelled questions in the questionnaire.

SHORT ANSWER QUESTIONS

5. In a sampling technique, a list consisting of all units in a target population is known as__
6. A small or fractional part of a population selected to meet some set objective is known as_____
7. Data presentation in form of charts and graphs is known as_____
8. A procedure in the form of report involving a combination of text and figures is known as_____
9. The act of rounding off data is known as_____
10. The difference between the actual and estimated value is known as_____

Answers

1. D,
2. C,
3. B,
4. C,
5. Sampling frame.
6. Sample,
7. Diagrammatic presentation
8. Text presentation,
9. Approximation
10. Absolute error

CHAPTER TWO

MEASURES OF LOCATIONS

Chapter Contents

- (a) Introduction;
- (b) Mean;
- (c) Mode and Median; and
- (d) Measures of Partition.

Objectives

At the end of the chapter, readers or readers should be able to:

- a) know the meaning of “Measure of Central Tendency”;
- b) understand and solve problems on various means such as Arithmetic mean, Geometric mean and Harmonic mean;
- c) know and handle problems on the mode and median;
- d) understand the concept of measures of partition; and
- e) handle problems on quintiles, deciles and percentiles.

2.1 Introduction

Measure of location is a summary statistic which is concerned with a figure which represents a series of values. Measures of location can be further classified into

- a) Measures of Average or Measures of Central Tendency.
- b) Measures of Partition.

In average or measures of central tendency, there is an average value which is a representative of all the values in a group of data. It is usual to give these typical values of averages which tend to lie centrally within the set of data arranged in an array, hence they are called measures of central tendency.

The common and usable types of averages or measures of central tendency are

- i. Mean; consisting of the following types: Arithmetic Mean (AM), Geometric Mean (GM) and Harmonic Mean (HM);
- ii. Mode; and
- iii. Median;

Each of the above averages will be discussed in the subsequent sections.

2.2 Mean

2.2.1 Arithmetical Mean (A.M):-

This is the sum of all the values ($x_1, x_2, \dots, x_n$) in the data group divided by the total number of the values. In symbolic form, for individual series (as case 1) the mean can be expressed as

$$\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n} \quad \dots 2.2.1$$

where $\bar{x}$ represents the arithmetical mean,

Σ is the summation usually called “sigma” and

n is the number of the values

In the case of discrete or ungrouped values with frequency distribution, the above formula can be modified as:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad \text{or simply} \quad \bar{x} = \frac{\sum fx}{\sum f} \quad \dots 2.2.2$$

where, $\sum f = n$ (as defined as 2.2.1) and f represents frequency.

Also for grouped frequency distribution table, formula 2.2.2 will still be equally applied. However, the x_i values in the formula are obtained by the use of Mid-Value. This is also called class mark. The class mark is obtained by adding the lower and upper boundary limits of the class interval and then divide by two.

$$\text{i.e. Class Mark} = \frac{L_B + U_B}{2} \quad \dots (2.2.3)$$

where L_B = Lower boundary / limit.

U_B = Upper boundary / limit.

Illustration 2.1(a)

From the underlisted data generated on the number of sales of cement (in bags):

12, 7, 2, 6, 13, 17, 14, 5, 9, 4, determine the mean

$$\begin{aligned} \bar{x} &= \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_{10}}{n} \\ \bar{x} &= \frac{12 + 7 + 2 + 6 + 13 + 17 + 14 + 5 + 9 + 4}{10} \end{aligned}$$

$$\bar{x} = \frac{89}{10} = 8.9 \text{ bags} \simeq 9 \text{ bags}$$

Illustration 2.1(b)

From the underlisted data generated on the volume (litres) of water: 2.7, 6.3, 4.1, 6.4, 3.5, 4.7, 13.8, 7.9, 12.1, 10.4, determine the mean

$$\begin{aligned}\bar{x} &= \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_{10}}{n} \\ \bar{x} &= \frac{2.7 + 6.3 + 4.1 + 6.4 + 3.5 + 4.7 + 13.8 + 7.9 + 12.1 + 10.4}{10} \\ &= \frac{71.9}{10} = 7.19 \text{ litres}\end{aligned}$$

Note: The units of the mean is the same as that of the data

Illustration 2.2(a)

These are the data collected on the number of students that register for QA at some examination centres, obtain the mean.

x	1	2	3	4	5
f	8	7	10	6	2

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

X	f	fx
1	8	8
2	7	14
3	10	30
4	6	24
5	2	10
	33	86

$$\text{Mean } (\bar{x}) = \frac{86}{33} = 2.6061 \text{ students}$$

Note: This is an unrealistic figure since 2.6061 students do not exist. This is one of the shortcomings of the mean

Illustration 2.2(b)

These are the data generated on the weight (kilogram) of soap produced in the production section of a company. Obtain the mean

Weight (x)	1.0	2.0	3.0	4.0	5.0
Frequency (f)	4	3	7	5	6

Solution

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

<i>x</i>	<i>f</i>	<i>fx</i>
1	4	4
2	3	6
3	7	21
4	5	20
5	6	30
	25	81

$$\text{Mean } (\bar{x}) = \frac{81}{25} = 3.24\text{kg}$$

Illustration 2.3(a)

The following are data collected on the ages (in years) of the people living in Opomulero village.

<i>Class</i>	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	12	8	7	7	6

Obtain the mean age of the people.

Solution

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

Class	x	f	fx
0 – 10	5.0	12	60
10 – 20	15.0	8	120
20 – 30	25.0	7	175
30 – 40	35.0	7	245
40 – 50	45.0	6	270
		40	870

$$\text{Mean } (\bar{x}) = \frac{870}{40} = 21.75 \text{ years}$$

Illustration 2.3(b)

From the following data generated on the ages (in years)of people living in an Estate.

Class	1 – 10	11 – 20	21 – 30	31 – 40	41 – 50
Frequency	6	17	7	12	8

Obtain the mean age of the data.

Solution

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i}$$

sss	Class	x	f	fx
	1 – 10	5.5	6	33.0
	11 – 20	15.5	17	263.5
	21 – 30	25.5	7	178.5
	31 – 40	35.5	12	426.0
	41 – 50	45.5	6	273
			50	1174

$$\text{Mean } (\bar{x}) = \frac{1174}{50} = 23.48 \text{ years}$$

2.2.2. Assumed Mean: It is possible to reduce the volume of figures involved in the computation of the mean by the use of “assumed mean” method. In this method, it requires that one of the observed values of x_i is taken, preferably the middle one. If A denotes the assumed mean, a new variable d is introduced by the expression $d_i = x_i - A$, then the actual mean is obtained by

$$\bar{x} = A + \frac{\sum d_i}{N} \quad 2.2.4$$

$$\text{or} \quad A + \frac{\sum f_i d_i}{\sum f_i} \quad 2.2.5$$

Illustration 2.4(a): Using the data in Illustration 2.3(a), obtain the arithmetic mean by the assumed mean method.

Solution

Let the assumed mean be $A=9$

x_i	$d_i = x_i - 9$	
12	3	
7	-2	
2	-7	
6	-3	
13	4	
17	8	
14	5	
5	-4	
9	0	
4	-5	
	$\sum d_i = -1$	

$$\begin{aligned} \bar{x} &= A + \frac{\sum d_i}{N} \\ &= 9 + \frac{(-1)}{10} \\ &= 9 - 0.1 \\ &= 8.9 \\ &\approx 9 \end{aligned}$$

Note that it gives the same answer as Illustration 2.1(a)

Illustration 2.4(b): Using the data in Illustration 2.3(a), obtain the mean age by the assumed mean method.

Solution

Class	x	F	If $A = 25$ $d_i = x - 25$	fd_i
0 – 10	5	12	-20	-240
10 – 20	15	8	-10	-80
20 – 30	25	7	0	0
30 – 40	35	7	10	70
40 – 50	45	6	20	120
		40		$\Sigma fd_i = -130$

$$\begin{aligned}x &= A + \frac{\Sigma fd_i}{\Sigma f_i} \\&= 25 + \frac{-130}{40} \\&= 25 - 3.25 \\&= 21.75\end{aligned}$$

which is the same as the answer obtained in Illustration 2.3(a)

Assumed mean and common factor approach can equally be used to obtain the mean. Here, it is only applicable to group data with equal class intervals.

By this approach, equation 2.2.5 becomes:

$$\bar{x} = A + C \left(\frac{\Sigma f_i d_i}{\Sigma f_i} \right) \quad 2.2.6$$

where C is common factor or the class interval.

Example: Using the data in Illustration 2.3(a), obtain the mean, using the assumed mean and common factor method.

Solution

Class	x	f	If $A = 25$ $d_i = x - 25$	If $C = 10$ $d_i' = d_i/10$	fd_i'
0 – 10	5	12	-20	-2	-24
10 – 20	15	8	-10	-1	-8
20 – 30	25	7	0	0	0
30 – 40	35	7	10	1	7
40 – 50	45	6	20	2	12
		40			$\Sigma fd_i' = -13$

$$\begin{aligned}\bar{x} &= A + \frac{C \Sigma f_i d_i'}{\Sigma f_i} \\ &= 25 + \frac{10(-13)}{40} \\ &= 25 - 3.25 \\ &= 21.75\end{aligned}$$

2.2.2 Geometric Mean (G.M)

This is the n^{th} root of the product of the values of variable x_i , where $i = 1, 2, \dots, n$

$$\begin{aligned}\text{i.e. } GM(x) &= \sqrt[n]{x_1 \times x_2 \times \dots \times x_n} \\ &= (\Pi x_i)^{1/n} \quad \text{---2.2.2.1}\end{aligned}$$

where Π stands for the product of the data values

Illustration 2.5

The following are the data generated on the number of sales of cement (in bags):

12, 7, 2, 6, 13, 17, 14, 5, 9, 4. Compute the Geometric Mean.

Solution

$$\begin{aligned}GM(x) &= \sqrt[n]{x_i}, \quad i = 1, 2, \dots, n \\ GM(x) &= \sqrt[10]{12 \times 7 \times 2 \times 6 \times 13 \times 17 \times 14 \times 5 \times 9 \times 4} \\ &= \sqrt[10]{561375360} \\ &= 7.4977\end{aligned}$$

Remark:

The GM is more appropriate for data in the form of rates, ratios and percentages.

In the computation of GM, logarithm approach can be applied. Here, one takes the logarithm of both sides of the equation 2.2.2.1 to give

$$\text{Log GM}(x) = \frac{\text{Log } x}{n} = y$$

The antilogarithm of y gives the $\text{GM}(x)$

$$\text{i.e. } \text{GM}(x) = 10^y$$

Logarithm to base 10 or e i.e Napierian logarithm can be used, but one must be consistent with the base used.

2.2.3 Harmonic Mean (HM)

This is an average often used in engineering and related disciplines. It is good for calculation of average speed of machines.

For the values of variable x_i , $i = 1, 2, \dots, n$, the HM of these values is given by

$$\text{HM}(x) = \frac{n}{\sum \frac{1}{x_i}} \quad \text{or} \quad n \left(\sum \frac{1}{x_i} \right)^{-1}$$

Illustration 2.6

Using the data in Illustration 2.5, obtain the Harmonic mean.

$$\begin{aligned} \text{HM}(x) &= \frac{n}{\sum \frac{1}{x_i}} \\ &= \frac{10}{\frac{1}{12} + \frac{1}{7} + \frac{1}{2} + \frac{1}{6} + \frac{1}{13} + \frac{1}{17} + \frac{1}{14} + \frac{1}{5} + \frac{1}{9} + \frac{1}{4}} \\ &= \frac{10}{0.0833 + 0.1429 + 0.5 + 0.1667 + 0.0769 + 0.0588 + 0.0714 + 0.2 + 0.1111 + 0.25} \\ &= \frac{10}{1.6611} \end{aligned}$$

$$\therefore \text{HM}(x) = 6.0201$$

2.3 Mode and Median

2.3.1 The MODE is the value which occurs most frequently in a set of data. For a data set in which no measured values are repeated, there is no mode. Otherwise, a distribution can have one mode(Unimodal)two modes(Bimodal) or many modes(Multimodal) depending on the data set.

For a grouped data with frequency distribution, the mode is determined by either graphical method through the use of Histogram or the use of formula.

(a) In the graphical method, the following steps are to be taken:

- i) Draw the histogram of the given distribution.
- ii) Identify the highest bar (the modal class).
- iii) Identify the two linking bars to the modal class, i.e the bar before and after the modal class.
- iv) Use the two flanking bars to draw diagonals on the modal class.
- v) Locate the point of intersection of the diagonals drawn in step (iv) above, and vertically draw a line from the point of intersection to the boundary axis which gives the desired mode.

(b) By formula, the mode is defined by

$$\text{Mode} = L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) C$$

where

L_1 = Lower class boundary of the modal class

Δ_1 = Modal class frequency – Frequency of the class before the modal class.

Δ_2 = Modal class frequency – Frequency of the class after the modal class.

C = Modal class size

2.3.2 The MEDIAN of a data set is the value of the middle item of the data when all the items in the data set are arranged in an array form (either ascending or descending order).

For an ungrouped data, the position (Median) is located by $(N + 1)/2$, where N is the total number of items in the data set.

In a grouped data, the position of the Median is located by $N/2$. Here, the specific value of the Median is determined by either graphical method through the use of Ogive (Cumulative frequency curve) or the use of formula

a) In the graphical method, the following procedure is used:

- i) Draw the Ogive of the given distribution.
- ii) Locate the point $N/2$ on the cumulative frequency axis of the Ogive.
- iii) Draw a parallel line through the value of $N/2$ to the curve and then draw a perpendicular line from curve intercept to the boundary axis in order to get the Median value.

b) The Median is obtained by the use of the following formula:

$$\text{Median} = L_i + \left(\frac{N/2 - (F_i)}{f_i} \right) C$$

where L_i = Lower class boundary of median class.
 N = Total number of items in the data set.
 F_i = Summation of all frequencies before the median class.
 f_i = Frequency of the median class.
 C = Median class size or interval.

Note: The units of the Mode and Median are the same as that of the data

Worked Examples:

Illustration 2.7 (a): Case I (n is odd and discrete variable is involved)

The following are the data generated from the sales of oranges within eleven days: 4, 7, 12, 3, 5, 3, 6, 2, 3, 1, 3, calculate both the mode and median.

Solution

Mode = Most occurring number

Mode = 3

Median = $\frac{\{n + 1\}}{2}$ position

Put the data in an array i.e 1, 2, 3, 3, 3, 3, 4, 5, 6, 7, 12

Median ($n = 11$) = $\frac{11 + 1}{2} = \frac{12}{2} = 6^{\text{th}}$ position

∴ The median = 3

Illustration 2.7(b): (n is odd and continuous variable is involved)

The following are the data generated from the measurement of iron-rods; 2.5, 2.0, 2.1, 3.5, 2.5, 2.5, 1.0, 2.5, 2.2, calculate both the mode and median.

Solution

1.0, 2.0, 2.1, 2.2, 2.5, 2.5, 2.5, 2.5, 3.5

Mode = 2.5

Median ($n = 9$) = $\frac{n + 1}{2} = \frac{9 + 1}{2} = \frac{10}{2} = 5^{\text{th}}$ position

∴ The median = 2.5

Illustration 2.7(c): (n is even and discrete variable is involved)

The following are the data generated from the sales of orange within 12 days

4, 7, 12, 3, 5, 3, 6, 2, 3, 1, 3, 4, calculate the mode and median.

Solution

Mode = Most occurring number

Mode = 3

Median = $\frac{(n + 1)^{\text{th}}}{2}$ position

If $n = 12$, $\frac{12 + 1}{2} = \frac{13}{2} = 6.5^{\text{th}}$ position

Rearrange the data first and take the mean of the middle values.

1, 2, 3, 3, 3, 3, 4, 4, 5, 6, 7, 12

The 6.5^{th} position gives $\frac{3 + 4}{2} = 3.5$

Illustration 2.7(d) (n is even and continuous variable is involved):

The following are the data generated from the measurement of an iron-rod; 2.5, 2.0, 2.1, 3.5, 2.5,

2.5, 1.0, 2.5, 2.2, 2.6, calculate the mode and median

Solution

Mode = Most occurring number

Mode = 2.5

Median = $\frac{(n + 1)^{\text{th}}}{2}$ position

If $n = 12$, $\frac{12 + 1}{2} = \frac{13}{2} = 6.5^{\text{th}}$ position

Rearrange the data first and take the mean of the middle values.

1.0, 2.0, 2.1, 2.2, 2.5, 2.5, 2.5, 2.5, 2.6, 3.5

The 6.5^{th} position gives $\frac{2.5 + 2.5}{2} = 2.5$

$\therefore$ Median = 2.5

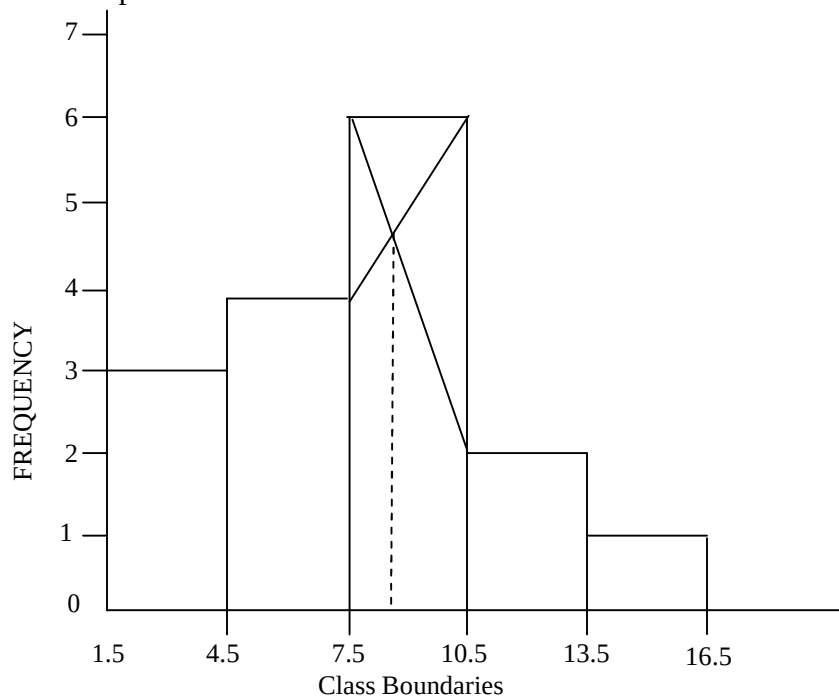
Illustration 2.8: Case III: Grouped data by graphical method

Use the graphical method to determine the mode of the following data

Tyre range	Frequency	C. F	Class Boundaries
2 – 4	3	3	1.5 – 4.5
5 – 7	4	7	4.5 – 7.5
8 – 10	6	13	7.5 – 10.5
11 – 13	2	15	10.5 – 13.5
14 – 16	1	20	13.5 – 16.5
	20		

Solution

- Draw the histogram of the given table
- Interpolate on the modal class as discussed earlier in order to obtain the mode.



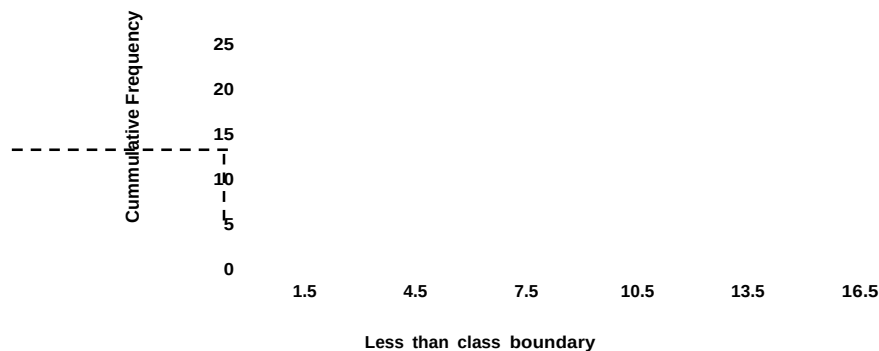
Mode = 8.5

Illustration 2.9: Case IV: Graphical method

Use graphical method to find the median using case III data

Solution:

Tyre range	Frequency	C. F	Less than Class Boundary
2 – 4	3	3	4.5
5 – 7	4	7	7.5
8 – 10	6	13	10.5
11 – 13	2	15	13.5
14 – 16	1	20	16.5
	20		

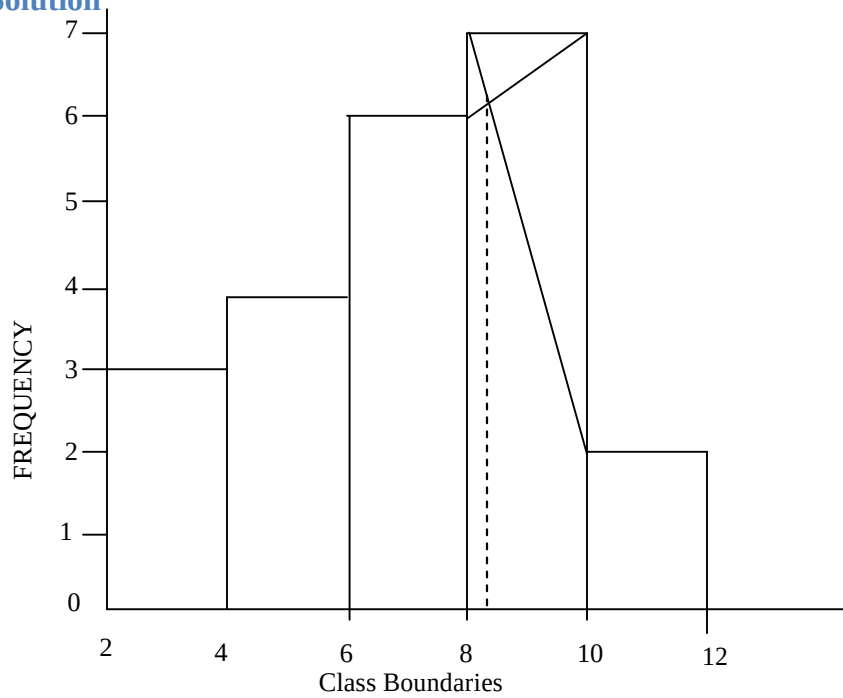


$$\begin{aligned}\text{Median frequency} &= \frac{N}{2} = \frac{20}{2} = 10 \\ \text{Median} &= 9.0\end{aligned}$$

Illustration 2.10: Graphical method to find mode and median.

Determine the mode and median by graphical method from the following data.

Class interval	Frequency(f)	C. F
2 – 4	3	3
4 – 6	4	7
6 – 8	6	13
8 – 10	7	20
10 – 12	2	22
	22	

Solution.

Mode = 8.50

To find the Median, we proceed as follows:

Table	Frequency (f)	C. F	Less than Boundary
2 – 4	3	3	4
4 – 6	4	7	6
6 – 8	6	13	8
8 – 10	7	20	10
10 – 12	2	22	12
	22		

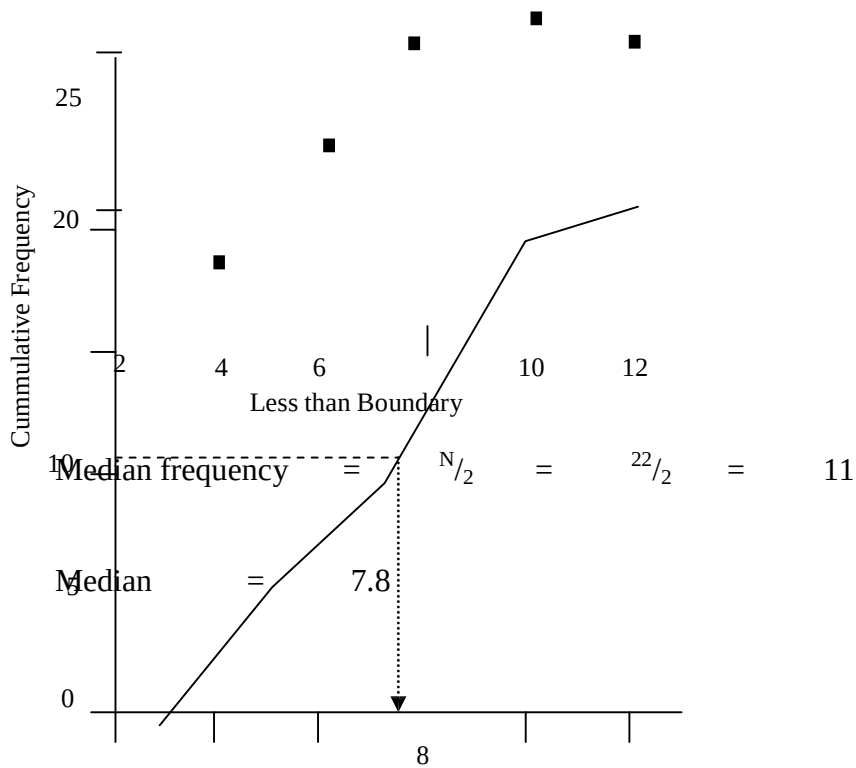


Illustration 2.11: Case III: Grouped data using formula

The following are the data generated on the sales of electric components

Class interval	Frequency
2 – 4	3
4 – 6	4
6 – 8	6
8 – 10	7
10 – 12	2

Determine the mode and median using the formula.

Solution

Class	Frequency	Cumulative Frequency
2 – 4	3	3
4 – 6	4	7
6 – 8	6	13 → Median class
8 – 10	7 → Modal class	20
10 – 12	2	22
	22	

$$N = 22; \quad \frac{N}{2} = \frac{22}{2} = 11$$

$$\text{Mode} = L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) C$$

where

L_1 = Lower class boundary of the modal class

Δ_1 = Modal class frequency – Frequency of the class before the modal class.

Δ_2 = Modal class frequency – Frequency of the class after the modal class.

C = Modal class size

$$L_1 = 8, \quad \Delta_1 = 7 - 6 = 1, \quad \Delta_2 = 7 - 2 = 5, \quad C = 2$$

$$\begin{aligned} \text{Mode} &= 8 + \left(\frac{1}{1 + 5} \right) (2) \\ &= 8 + \left(\frac{1}{6} \right) (2) \\ &= 8 + \frac{1}{3} \\ &= 8 + 0.33 = 8.33 \\ &= 8.33 \end{aligned}$$

$$\text{Median} = L_1 + \left(\frac{\frac{N}{2} - (F_i)}{f_i} \right) C$$

where L_1 = Lower class boundary of the median class

f_i = Frequency of the median class

C = Median class size

F_i = Summation of all frequencies before the median class

$$L_1 = 6, \quad F_{\text{med}} = 6, \quad C = 2, \quad \Sigma f_i = 3 + 4 = 7, \quad N = 22$$

$$\begin{aligned} \text{Median} &= 6 + \left(\frac{\frac{22}{2} - 7}{6} \right) (2) \\ &= 6 + \left(\frac{11 - 7}{6} \right) (2) \\ &= 6 + \left(\frac{4}{6} \right) (2) \\ &= 6 + \frac{4}{3} \\ &= 6 + 1.3 \\ &= 7.3 \end{aligned}$$

Illustration 2.11

The following are the data generated on the sales of tyres:

Tyre range	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12
Frequency	3	4	6	2	5

Calculate the mode and median.

Solution

Tyre range	Frequency	Cumulative Frequency
2 – 4	3	3
4 – 6	4	7
6 – 8	6 Modal class	13 → Median class
8 – 10	2	15
10 – 12	5	20
	20	

$$\underline{N} = 20; \quad \frac{N}{2} = \frac{20}{2} = 10$$

$$\text{Mode} = L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) C$$

$$L_1 = 6.0, \quad \Delta_1 = 6 - 4 = 2, \quad \Delta_2 = 6 - 2 = 4, \quad C = 3 \text{ (i.e } 4.5 - 1.5)$$

$$\begin{aligned} \text{Mode} &= 6.0 + \left(\frac{2}{2 + 4} \right) (3) \\ &= 6.0 + 1 \\ &= 7.0 \end{aligned}$$

$$\text{Median} = L_1 + \left(\frac{\frac{N}{2} - (F_i)}{f_i} \right) C$$

$$L_1 = 6.0, \quad F_{\text{med}} = 6, \quad C = 3, \quad F_i = \Sigma f_i = 3 + 4 = 7, \quad N = 20$$

$$\begin{aligned} \text{Median} &= 6.0 + \left(\frac{\frac{20}{2} - 7}{6} \right) (3) \\ &= 6.0 + \left(\frac{10 - 7}{6} \right) (3) \end{aligned}$$

$$\begin{aligned}
&= 6.0 + \left(\frac{3}{6}\right)(3) \\
&= 6.0 + \frac{3}{2} \\
&= 6.0 + 1.5 \\
&= 7.5
\end{aligned}$$

2.4 Measures of Partition

These are other means of measuring location. We will recall that the median characterises a series of values by its midway position. By extension of this idea, there are other measures which divide a series into a number of equal parts but which are not measures of central tendency. These are the measures of partition. The common ones are the quartiles, deciles and the percentiles. They are collectively called the QUANTILES.

The methods of computation for the quantiles follow the same procedure for the Median according to what is being measured. For instance, the position of the first quartile from a grouped data is located by $N/4$ and the value is determined graphically from Ogive or by use of formula as done in the Median.

Also the position for locating third quartile is $3N/4$, for the seventh decile is $7N/10$ and tenth percentile is $10N/100$. We should note that the second quartile, fifth decile and fiftieth percentile coincide with the Median.

The use of formulae to compute the quantiles are summarized below:

$$\text{Quartile (Q}_i\text{)} = L_i + \left(\frac{N/4 - F_i}{f_i}\right)C, \quad i = 1, 2, 3$$

$$\text{Decile (D}_i\text{)} = L_i + \left(\frac{N/10 - F_i}{f_i}\right)C, \quad i = 1, 2, 3, 4, 5, 6, 7, 8, 9$$

$$\text{Percentile (P}_i\text{)} = L_i + \left(\frac{N/100 - F_i}{f_i}\right)C, \quad i = 1, 2, 3, 4, 5, \dots, 98, 99.$$

where

L_i = Lower Class boundary of the i^{th} quantile class.

f_i = frequency of i^{th} quantile class.

$F_i = \Sigma f_i$ = Sum of the frequencies of all classes lower than the i^{th} quantile class.

C = Class interval of the i^{th} quantile class.

N = Total number of items in the distribution.

Example 2.4.1:

Find Q_1 , Q_3 , D_7 , P_{20} for the following data:

Interval	x	f	c. f	Less than Class Boundaries
0 – 2	1	2	2	2.5
3 – 5	4	4	6	5.5
6 – 8	7	8	14	8.5
9 – 11	10	4	18	11.5
12 – 14	13	2	20	14.5
		20		

Position of Q_1 = $(N/4)^{\text{th}}$

Frequency of Q_1 = $20/4 = 5$

Q_1 Class = 3 – 5 class

$$Q_1 \text{ Class} = L_1 + \left(\frac{\frac{N}{4} - F_i}{f} \right) C$$

$L_1 = 2.5$, $N/4 = 5$, $F_i = 2$, $f_i = 4$, $C = 3$

$$Q_1 = 2.5 + \left(\frac{5 - 2}{4} \right) (3)$$

$$Q_1 = 2.5 + (3/4)3$$

$$Q_1 = 2.5 + 2.25$$

$$Q_1 = 4.75$$

Q_3 's position; $\frac{3N}{4} = \frac{3 \times 20}{4} = 15^{\text{th}}$

$$Q_3 = L_1 + \left(\frac{\frac{3N}{4} - F_i}{f_i} \right) C$$

Q_3 's class = 9 – 11

$L_1 = 8.5$, $F_i = 2 + 4 + 8$, $f_i = 4$, $C = 3$

$$\begin{aligned}
Q_3 &= 8.5 + \left(\frac{15-14}{4_i} \right) (3) \\
&= 8.5 + (1/4)3 \\
&= 8.5 + 0.75 \\
&= 9.25
\end{aligned}$$

$$\text{Decile } (D_i) = L_i + \left(\frac{N/10 - F_i}{f_i} \right) C, \quad i = 1, 2, 3, \dots, 9$$

$$\therefore \text{Decile } (D_7) = L_7 + \left(\frac{7N/10 - F_i}{f_i} \right) C, \quad i = 1, 2, 3, \dots, 9$$

$$\therefore \text{its frequency } \frac{N(i)}{10} = \frac{20(7)}{10} = \frac{140}{10} = 14$$

$$L_7 = 5.5; \quad f_7 = 8, \quad F_7 = 6, \quad C = 8.5 - 5.5 = 3$$

$$\begin{aligned}
D_7 &= 5.5 + \frac{3(14-6)}{8} \\
&= 5.5 + 0.375(8) \\
&= 5.5 + 3 \\
&= 8.5
\end{aligned}$$

$$\text{Percentile } (P_i) = L_i + \left(\frac{N/100 - F_i}{f_i} \right) C, \quad i = 1, 2, 3, \dots, 99$$

$$\therefore \frac{N(i)}{100} = \frac{20(20)}{100} = \frac{400}{100} = 4$$

$$P_{20} = L_{20} + \frac{C}{f_{20}} \left(\frac{N(20)}{100} - F_{20} \right), \quad i = 1, 2, 3, \dots, 99$$

$$L_{20} = 2.5, \quad f_{20} = 4, \quad F_{20} = 2, \quad C = 5.5 - 2.5 = 3$$

$$\begin{aligned}
P_{20} &= 2.5 + \frac{3}{4}(4-2) = 2.5 + 0.75(2) = 2.5 + 1.5 \\
&= 4
\end{aligned}$$

2.5 Summary

In this chapter, the measures of central tendency, which consist of mean, mode and median, were considered for both grouped and ungrouped data. The measures of location such as Quartiles, Deciles and Percentiles were also discussed in the chapter.

QUESTIONS (MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS)

1. Which of the following means is often used in engineering and related disciplines?
 - A. Arithmetic mean.
 - B. Geometric mean
 - C. Harmonic mean
 - D. Arithmetic and Geometric mean.
 - E. Hyperbolic mean.
2. Which of the following is not a measure of central tendency?
 - A. Arithmetic mean
 - B. Mode
 - C. Median
 - D. Decile
 - E. Geometric mean
3. Which of the following is not a measure of partition?
 - A. Median
 - B. Mode
 - C. Percentile
 - D. Quantile
 - E. Quintile.
4. Which of the following formulae is used for the computation of quintile?
 - A. $L_i + C \left(\frac{N(i)/2 - F_i}{f_i} \right)$
 - B. $L_i + C \left(\frac{N(i)/4 - F_i}{f_i} \right)$
 - C. $L_i + C \left(\frac{N(i)/5 - F_i}{f_i} \right)$
 - D. $L_i + C \left(\frac{N(i)/10 - F_i}{f_i} \right)$
 - E. $L_i + C \left(\frac{N(i)/100 - F_i}{f_i} \right)$

5. In the graphical method of obtaining the quartiles, which of the following diagrams is used?
- Bar chart
 - Histogram
 - Pie chart
 - Ogive
 - component bar chart.

Use the following data to answer questions 6 to 10:

6, 3, 8, 8, 5

- Calculate the Arithmetic mean.
- Calculate the Geometric mean.
- Calculate the Harmonic mean.
- Determine the Mode.
- Determine the Median.

Answers

- C,
- D;
- B;
- C;
- D

$$6. \quad \bar{x} = \frac{6 + 3 + 8 + 8 + 5}{5} = \frac{30}{5} = 6$$

$$7. \quad G_{xi} = \sqrt[n]{\prod x_i} = \sqrt[5]{6 \times 3 \times 8 \times 8 \times 5}$$

$$= \sqrt[5]{5760} = 5.65$$

$$\begin{aligned}
 8. \quad \text{Hm}(\bar{x}) &= \frac{n}{\sum \frac{1}{x_i}} \\
 &= \frac{5}{\frac{1}{6} + \frac{1}{3} + \frac{1}{8} + \frac{1}{8} + \frac{1}{5}} = \frac{5}{0.1667 + 0.3333 + 0.125 + 0.125 + 0.2} \\
 &= {}^5_{/0.95} = 5.263 \approx 5.3
 \end{aligned}$$

9. Mode is 8

10. 6, 3, 8, 8, 5

Rearrange $\rightarrow$ 3, 5, 6, 8, 8

$\therefore$ Median = 6.

CHAPTER THREE

MEASURES OF VARIATION

CHAPTER CONTENT

- (a) Introduction
- (b) Measures of Variation
- (c) Coefficient of Variation and of Skewness

OBJECTIVES

At the end of the chapter, readers should be able to

- (a) know the meaning of measure of variation;
- (b) determine various measures of variation such as mean deviation, variance, standard deviation and quantile deviation for both grouped and ungrouped data; and
- (c) compute the coefficient of variation and of skewness.

3.1 Introduction

The degree to which numerical data tend to spread about an average value is referred to as measure of variation or dispersion. At times, it is called measure of spread. The purpose and significant uses of these measures are of paramount value in statistics.

The popular ones among these measures of variation are the

- (a) Range;
- (b) Mean Deviation;
- (c) Variance/Standard Deviation;
- (d) Semi – interquantile range

3.2 Measures of Variation

3.2.1 Range

The range for a set of data, is the difference between the highest number and the lowest number of the data. That is,

Range = Highest number – Lowest number (for ungrouped data); OR

Range = Upper bound of the highest class – Least bound of the lowest class
(for grouped data)

Illustration 3.1

Determine the range for the following set of numbers:

- a. 32, 6, 10, 27, 30, 5, 45
- b. 09, 14, 16, 13, 14, 21

Solution 3.1

The data here is ungrouped type

- a. Range = Highest – Lowest
 = 45 – 05 = 40
- b. Range = 21 – 09 = 12

Illustration 3.2

Obtain the range for the following table

Class	Frequency
1 – 10	4
11 – 20	10
21 – 30	12
31 – 40	12
41 – 50	9

Solution 3.2

The data in this question is grouped.

$$\text{Range} = 50 - 1 = 49$$

3.2.2 Mean Deviation (MD)

The Mean Deviation is the arithmetic mean of the absolute deviation values from the mean. For ungrouped data (of $x_1, x_2, \dots, x_n$), the M.D is given as

$$\text{M.D} = \frac{\sum |x_i - \bar{x}|}{N} = \frac{\sum |d_i|}{N} \quad \text{---3.2.2.1}$$

where $d_i = x_i - \bar{x}$, the symbol $| |$ stands for modulus or absolute value. For grouped data, it is written as:

$$\text{M.D} = \frac{\sum f |x_i - \bar{x}|}{\sum f} = \frac{\sum f |d_i|}{\sum f} \quad \text{---3.2.2.2}$$

where $\sum f = N$

Illustration 3.3

Calculate the mean deviation from number of books sold in a small bookshop as given below: 15, 17, 14, 16, 18

Solution 3.3

$$\text{The mean } (\bar{x}) = \frac{\sum x}{N} = \frac{15 + 17 + 14 + 16 + 18}{5} = \frac{80}{5} = 16$$

The following format shall be used for MD:

x	$d_i = x_i - \bar{x}$	$ x_i - \bar{x} $
15	-1	1
17	1	1
14	-2	2
16	0	0
18	2	2
80		6

$$\text{Thus, MD} = \frac{\sum |d_i|}{N} = \frac{6}{5} = 1.20$$

Illustration 3.4

The distribution of book sales (in hundreds) in large bookshop is given in the table below. Determine the M. D from this table

Class	Frequency
1 – 10	10
11 – 20	15
21 – 30	17
31 – 40	13
41 – 50	05

Solution 3.4

Class	Frequency	x	fx	$d_i = x_i - \bar{x} $	$f d_i $
1 – 10	10	5.5	55	18	180
11 – 20	15	15.5	232.5	8	120
21 – 30	17	25.5	433.5	2	34
31 – 40	13	35.5	461.5	12	156
41 – 50	05	45.5	227.5	22	110
	60		1410		600

$$\begin{aligned} \text{Mean } (\bar{x}) &= \frac{\sum fx}{\sum f} \\ &= \frac{1410}{60} = 23.5 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Mean deviation (MD)} &= \frac{\sum f_i |d_i|}{\sum f_i} \\
 &= \frac{600}{60} = 10
 \end{aligned}$$

3.2.3 Variance and Standard Deviation

The variance is similar to the mean deviation (M.D) in the sense that it is based on the difference between each value in a data set and the mean of the group. It differs in that these differences are squared before being summed.

It is important to know further that the standard deviation is the positive square root of the variance.

In an ungrouped data, the population variance is given as

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \text{---3.2.3.1}$$

Where σ^2 is the population variance.

μ is the population mean.

N is the total number of items.

From the above, the population standard deviation is

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}} \quad \text{---3.2.3.2}$$

The variance for a sample is not exactly equal to the population variance. It is given as:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad \text{---3.2.3.3}$$

where $\bar{x}$ is the sample mean and n is the sample size.

Also for grouped data, the population variance is:

$$\sigma^2 = \frac{\sum f(x - \mu)^2}{\sum f - 1} \quad \text{---3.2.3.4}$$

and for sample variance, we have

$$S^2 = \frac{\sum f(x_i - \bar{x})^2}{\sum f - 1} = \frac{\sum f(x_i - \bar{x})^2}{n - 1} \quad \text{---3.2.3.5}$$

where f stands for frequency and $\sum f = n$

Illustration 3.5

The figures 9, 5, 9, 7, 10, 14, 12, 10, 6, 17 represent the volumes of sales (₦'000) of Adebayo Spring Water within the first ten days of operation. Calculate the population variance and hence the standard deviation of sales.

Solution 3.5

Let the sales be represented by x , then,

$$\sigma^2 = V(x) = \frac{\sum(x - \mu)^2}{N} \quad \text{where } \mu = \frac{\sum x_i}{N}$$

$$\mu = \frac{9 + 5 + 9 + 7 + 10 + 14 + 12 + 10 + 6 + 17}{10}$$

$$= \frac{99}{10} = 9.9$$

$$\therefore \sigma^2 = \frac{\{(9 - 9.9)^2 + (5 - 9.9)^2 + (9 - 9.9)^2 + (7 - 9.9)^2 + (10 - 9.9)^2 + (14 - 9.9)^2 + (12 - 9.9)^2 + (10 - 9.9)^2 + (6 - 9.9)^2 + (17 - 9.9)^2\}}{10}$$

$$\therefore \sigma^2 = \frac{(0.81 + 24.01 + 0.81 + 8.41 + 0.01 + 16.81 + 4.41 + 0.01 + 37.21 + 50.41)}{10}$$

$$= \frac{142.9}{10}$$

$$\sigma^2 = 14.29$$

$$\sigma = \text{S.D} = \sqrt{V(x)}$$
$$\text{S.D} = \longleftarrow \sqrt{14.29} = 3.78$$

Illustration 3.6

The table below shows the frequency table of the age distribution of 25 children in a family.

Age (x)	5	8	11	14	17
Frequency (f)	6	7	5	4	3

Determine the sample variance and its standard deviation

Solution 3.6

x	f	fx	$ d_i = x_i - \bar{x}$	d_i^2	fd_i^2
5	6	30	-4.92	24.2064	145.2384
8	7	56	-1.9	3.61	25.27
11	5	55	1.1	1.21	6.05
14	4	56	4.1	16.81	67.24
17	3	51	7.1	50.41	151.23
	25	248			395.0284

$$\begin{aligned}\bar{x} &= \frac{\Sigma fx}{\Sigma f} = \frac{248}{25} = 9.92 & d_i &= x_i - 9.92 \\ S^2 &= \frac{\Sigma f_i(x_i - 9.92)^2}{25 - 1} = \frac{395.0284}{24} = 16.4595\end{aligned}$$

Hence,

$$S = \sqrt{16.4595} = 4.0570$$

Remark:

From the above computations, it is clearly seen that the set of formulae used above is tedious. Hence, the call or demand for a short cut and easier method is necessary. The following short cut formulae will be used:

For ungrouped data:

$$S^2 = \frac{\Sigma x_i^2 - n(\bar{x})^2}{n - 1} \quad \text{or} \quad \frac{\Sigma x_i^2 - (\Sigma x)^2/n}{n - 1} \quad \text{---3.2.3.6}$$

Also for grouped data:

$$S^2 = \frac{\Sigma fx_i^2 - n(\bar{x})^2}{\Sigma f - 1} \quad \text{or} \quad \frac{\Sigma fx_i^2 - (\Sigma fx)^2/n}{n - 1} \quad \text{---3.2.3.7}$$

where $n = \Sigma f$

Illustration 3.7: Another method for ungrouped data

The values 9, 5, 9, 7, 10, 14, 12, 10, 6, 17 represent the volume of sales (₹'000) of Stephen Pure Water within the first ten days of operation. Calculate the population variance and hence the standard deviation of sales using the shortcut method.

Solution 3.7

Let the sales be represented by x , then

$$\begin{aligned}
 S^2 &= \frac{\sum x_i^2 - n(\bar{x})^2}{n-1} \\
 \bar{x} &= \frac{9 + 5 + 9 + 7 + 10 + 14 + 12 + 10 + 6 + 17}{10} \\
 &= \frac{99}{10} = 9.9 \\
 \sum x^2 &= 9^2 + 5^2 + 9^2 + 7^2 + 10^2 + 14^2 + 12^2 + 10^2 + 6^2 + 17^2 \\
 &= 1101 \\
 \therefore S^2 &= \frac{1101 - 10 \times (9.9)^2}{10 - 1} \\
 &= \frac{1101 - 10 \times 98.01}{9} = \frac{1101 - 980.1}{9} \\
 &= \frac{120.9}{9} = 13.433 \\
 S &= \sqrt{\frac{\sum x_i^2 - n(\bar{x})^2}{n-1}} = \sqrt{13.433} = 3.6651
 \end{aligned}$$

Example for Grouped Data

X	F	x^2	fx_i^2	fx
5	6	25	150	30
8	7	64	448	56
11	5	121	605	55
14	4	196	784	56
17	3	289	867	51
	25	695	2854	248

$$\begin{aligned}
 S^2 &= \frac{\sum fx_i^2 - n(\bar{x})^2}{\sum f - 1} = \frac{\sum fx_i^2 - (\sum fx)^2/n}{n - 1} \quad \text{---3.2.3.7} \\
 &= \frac{2854 - (248)^2/25}{25 - 1} \\
 &= \frac{2854 - 61504/25}{24} \\
 &= \frac{2854 - 2460.16}{24} \\
 &= \frac{393.84}{24} = 16.41 \\
 S &= \sqrt{\frac{\sum fx_i^2 - (\sum fx)^2/n}{n - 1}} \\
 &= \sqrt{16.41} = 4.0509
 \end{aligned}$$

3.2.4 Semi – Interquartile Range (SIR)

Semi – Interquartile Range is obtained from quantile range and it is denoted and computed by:

$$\text{SIR} = \frac{Q_3 - Q_1}{2} \quad \text{--- 3.2.4.1}$$

where Q_1 = First Quantile, Q_3 = Third Quantile and the numerator ($Q_3 - Q_1$) is known as the quantile range.

Illustration 3.8

Determine the semi – interquartile range for the income distribution of SAO company employees as given in the following table:

Income class in ₹'000	10 – 20	20 – 30	30 – 40	40 – 50
Frequency (f)	7	10	8	5

Solution 3.8

Income Class	Frequency	CF
10 – 20	7	7
20 – 30	10	17
30 – 40	8	25
40 – 50	5	30
	30	

The first quartile Q_1 is in the class containing the $30/4 = 7.5^{\text{th}}$ item. This is the class 20 – 30

$$\therefore L_1 = 20, \quad F_1 = 7; \quad f_1 = 10, \quad C = 20 - 30 = 10, \quad \frac{N}{4} = 7.5$$

$$\begin{aligned} Q_1 &= L_1 + C\left(\frac{N}{4} - F_1\right) \\ &= 20 + \frac{f_1}{10}(7.5 - 7) \\ &= 20 + 1.0(7.5 - 7) \\ &= 20 + 0.50 = 20.50 \end{aligned}$$

The third quartile (Q_3) is in the class containing the $\frac{3(30)}{4}$ i.e 22.5^{th} item. This is the class 30 – 40.

$$\therefore L_3 = 30, \quad f_3 = 8, \quad F_3 = 17, \quad C = 30 - 40 = 10, \quad \frac{3N}{4} = 22.5$$

$$\begin{aligned} Q_3 &= L_3 + C\left(\frac{3N}{4} - F_3\right) \\ &= 30 + \frac{f_3}{8}(22.5 - 17) \\ &= 30 + 1.25(5.5) \\ &= 30 + 6.875 = 36.875 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Semi - Interquartile range} &= \frac{1}{2}(Q_3 - Q_1) \\
 &= \frac{1}{2}(36.875 - 20.50) \\
 &= \frac{1}{2}(16.375) \\
 &= 8.1875
 \end{aligned}$$

3.3 Coefficient of Variation and of Skewness

3.3.1 Coefficient of Variation (CV)

This is a useful statistical tool. It shows degree of variation between two sets of data. It is a dimensionless quantity and defined as:

$$\begin{aligned}
 \text{CV} &= \frac{\text{Standard Deviation}}{\text{Mean}} \times 100\% \\
 &= \frac{\sigma}{\bar{x}} \times 100\%
 \end{aligned}$$

By a way of interpretation, the smaller the C.V of a data set, the higher the precision of that set. Also in comparison, the data set with least C.V has a better reliability than the other.

Illustration 3.9:

The following data consists of the ages of twelve banks in year (last birthday) during their last anniversary in Oyo state. Calculate the coefficient of variation: 15, 14, 8, 12, 11, 10, 16, 14, 12, 11, 13, 8

Solution 3.9

$$\text{Coefficient of Variation} = \frac{S \times 100\%}{\bar{x}} = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100\%$$

$$\begin{aligned}
 \bar{x} &= \frac{15 + 14 + 8 + 12 + 11 + 10 + 16 + 14 + 12 + 11 + 13 + 8}{12} \\
 &= \frac{144}{12} = 12
 \end{aligned}$$

$$S_x^2 = \frac{\sum f x_i^2 - n \bar{x}^2}{\sum f - 1}, \text{ where}$$

$$\sum f x_i^2 = 15^2 + 14^2 + 8^2 + 12^2 + 11^2 + 10^2 + 16^2 + 14^2 + 12^2 + 11^2 + 13^2 + 8^2 = 1800$$

$$\begin{aligned}
 \therefore S_x^2 &= \frac{1800 - 12(12)^2}{12 - 1} \\
 &= \frac{1800 - 12 \times 144}{11} \\
 &= \frac{1800 - 1728}{11} \\
 &= \frac{72}{11} = 6.5455
 \end{aligned}$$

$$\begin{aligned}
 S &= \sqrt{6.5455} \\
 &= 2.5584 \\
 \therefore CV &= \frac{S}{\bar{x}} = \frac{2.5584}{12} \\
 &= 0.2132
 \end{aligned}$$

3.3.2 Coefficient of Skewness

If a frequency curve is not symmetrical, its skewness is the degree of asymmetry of the distribution.

The Pearsonian coefficient of skewness is given as:

$$\text{Coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

OR it can also be obtained as

$$\text{Coefficient of skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

The coefficient of skewness is zero for a symmetry or normal distribution. For a positively skewed distribution, the mean is larger than the mode, while for a negatively skewed distribution the mean is smaller than the mode. The median takes a value between the mean and the mode.

Illustration 3.10:

Given the following arithmetic means, the medians and the standard deviations of two distributions:

- i. Arithmetic mean = 20, Median = 22 and Standard Deviation = 8
- ii. Arithmetic mean = 20, Median = 23 and Standard deviation = 10

Determine which of the distributions is more skewed.

Solution 3.10

By the Pearson measure of skewness, we have

$$\text{Skewness} = \frac{3(\text{Arithmetic mean} - \text{Median})}{\text{Standard Deviation}}$$

$$\text{i. Skewness} = \frac{3(20 - 22)}{8} = \frac{-6}{8} = -0.75$$

$$\text{ii. Skewness} = \frac{3(20 - 23)}{10} = \frac{-9}{10} = -0.90$$

since $|-0.90| > |-0.75|$, the second (ii) distribution is more skewed.

3.4 Chapter Summary

Measure of variation has been described as measure of spread. It is commonly used to obtain the spread about the average and to make comparison of spreads for two sets of data. Among these measures of dispersion discussed, we have the range, mean deviation, variance/standard deviation and semi – interquartile range. Here, both ungrouped and grouped data are considered.

Also, the concepts of coefficient of variation and of skewness are discussed.

MULTIPLE-CHOICE AND SHORT ANSWERS QUESTIONS

1. For a set of data, the difference between the highest and the lowest number is known as _____
 - A. Mean
 - B. Variance
 - C. Range
 - D. Interquartile range
 - E. Mean deviation

2. The main difference between the mean deviation and variance is the _____
 - A. Differences between the data set and mean are zero.
 - B. Differences between the data set and the mean are squared before being summarized in variance.
 - C. Square roots of differences between the data set and the mean are obtained
 - D. Differences in the order of arrangement.
 - E. Differences between the data set and the mean are in geometric order.

3. Semi – Interquartile range is determined by

- A. $\frac{Q_3 - Q_2}{2}$
- B. $\frac{Q_3 - Q_1}{2}$
- C. $\frac{Q_3 + Q_2}{2}$
- D. $\frac{Q_3 + Q_1}{2}$
- E. $\frac{Q_4 - Q_1}{2}$

4. For a normal distribution, the coefficient of skewness is

- A. 1
- B. -1,
- C. 0
- D. 2,
- E. -2

Use the following information to answer questions 5 to 10. given the following set of data: 3, 7, 2, 8, 5, 6, 4

- 5. Determine the range.
- 6. Determine the mean deviation.
- 7. Determine the variance.
- 8. Determine Q_1 .
- 9. Determine Q_3 .
- 10. Determine Semi-interquantile range.

SOLUTIONS TO MULTIPLE-CHOICE AND SHORT ANSWER QUESTIONS

1. C,
2. B,
3. B,
4. C,
5. 5,
6. 2
7. ${}^{40}/_7 = 5.714$
8. $Q_1 = 2$,
9. $Q_3 = 6$
10. $\frac{Q_3 - Q_1}{2} = \frac{6 - 2}{2} = 2$

CHAPTER FOUR

MEASURE OF RELATIONSHIP AND REGRESSION

CHAPTER CONTENTS

- (a) Types of Correlation;
- (b) Measure of Correlation;
- (c) Simple Regression Line; and
- (d) Coefficient of Determination.

OBJECTIVES

At the end of the chapter, readers should be able to

- (a) understand the concepts of univariate and bivariate;
- (b) understand the concept of scatter diagram;
- (c) know the various types of correlation coefficients;
- (d) compute Pearson product moment correlation coefficient and make necessary interpretations;
- (e) compute Spearman's rank correlation coefficient and make necessary interpretations;
- (f) understand the concept of regression;
- (g) fit a simple linear regression model to data;
- (h) fit a simple linear regression line using
 - (i) graphical method
 - (ii) formula (the least squares method)
- (i) make necessary interpretation of the regression constant and regression coefficient;
- (j) forecast (estimation) by the use of fitted regression line; and
- (k) know the meaning and computation of coefficient of determination.

4.1 Introduction

The type of data we have been using so far is the univariate type, i.e one variable. But in this chapter we shall be dealing with bivariate data, i.e two variables.

The statistical analysis that requires the use of bivariate data is generally termed the measure of relationship and regression analysis which is the focus of this chapter. In this analysis, the interest is usually on relationship or pattern of the relationship. Here, one

can look at how students' performance in one subject (Mathematics) affects another subject (Accountancy). Another good example is the consideration of how an increase in the income affects spending habits or savings.

In the measure of relationship, the usual practice is to quantify or qualify and represent the bivariate by letters. Taking for instance, the marks scored by a set of students in

Mathematics and Accountancy can be represented by x and y variables respectively and can be written as point (x,y) , where x , and y stand for independent and dependent variables respectively.

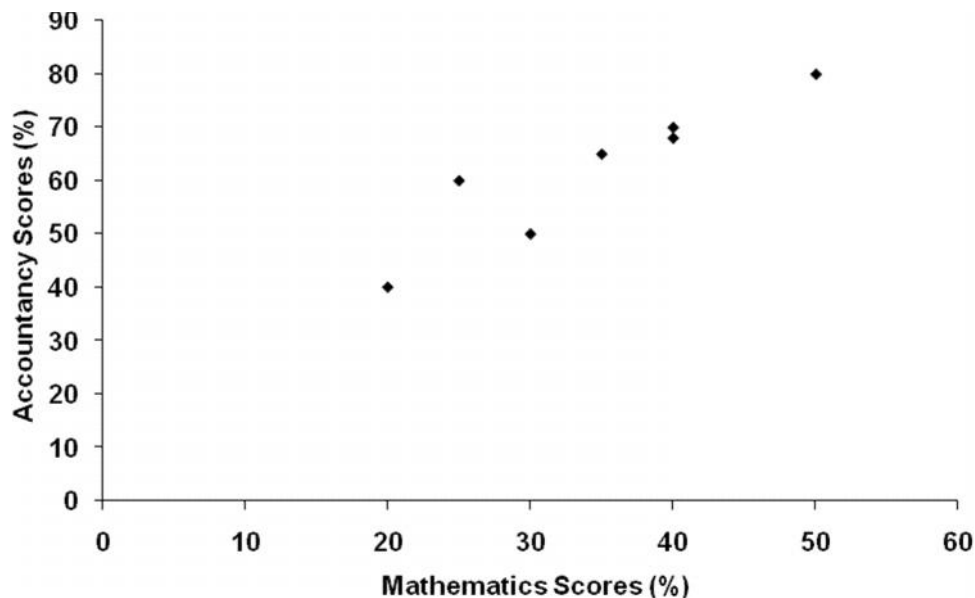
A graphical presentation of bivariate data on a two-axis coordinate graph is known as the SCATTER DIAGRAM. Here, the bivariate data are plotted on a rectangular coordinate system in order to see the existing relationship between the two variables under study. The following bivariate data will be used as an illustration:

Illustration 4.1

Draw the scatter diagram for the following data:

Mathematics Scores (X_i) in %	30	40	35	40	20	25	50
Accountancy scores (Y_i) in %	50	70	65	68	40	60	80

SCATTER DIAGRAM



4.2 Types of Correlation

Correlation measures the degree of association (relationship) between two variables. Therefore, two variables are said to be correlated or related when one variable changes in value, it also changes the value of the other variable. The degree of correlation (r) between two variables x and y is expressed by a real number which lies between -1.0 and $+1.0$ (i.e. $-1 \leq r \leq 1$) and it is called correlation coefficient or coefficient of correlation.

Simple correlation is basically classified according to the value of its coefficient. Hence, there are:

- i. Positive correlation;
- ii. Negative correlation; and
- iii. Zero correlation.

The following scatter diagrams (fig. 4.2.1) depict the above types of correlation:

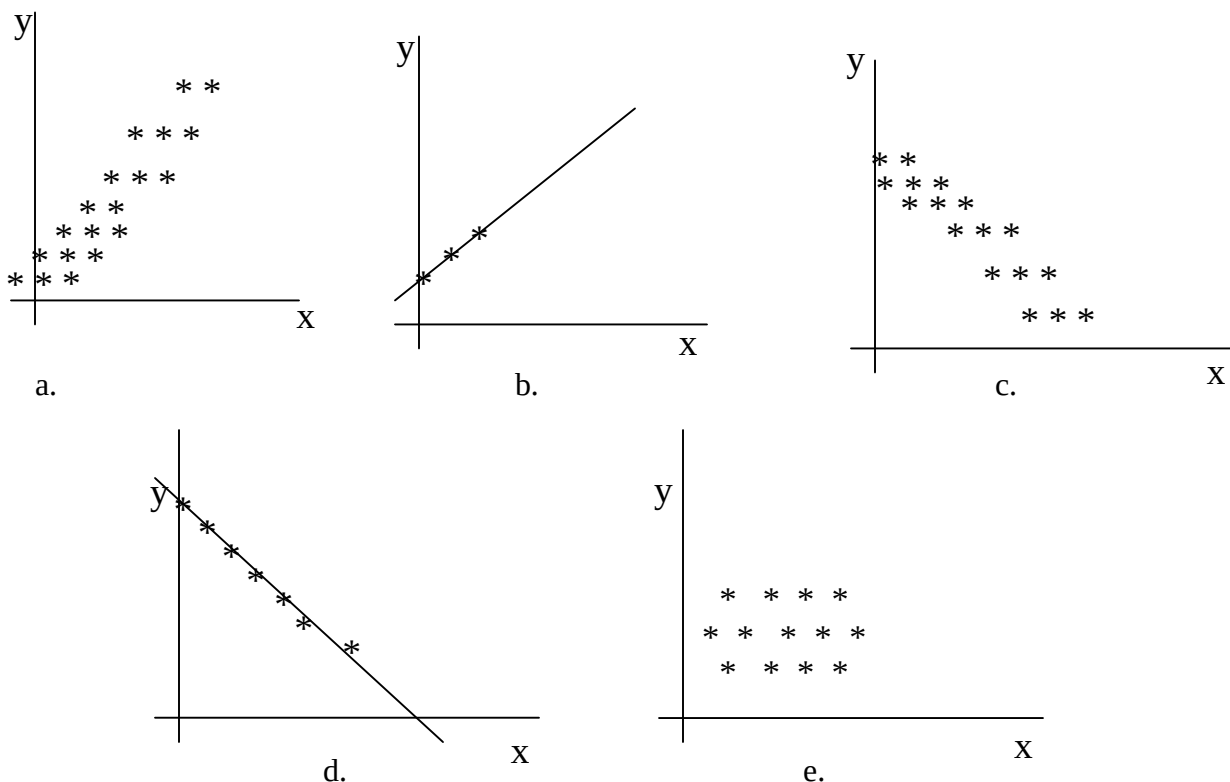


Figure 4.2.1

Figure 4.2.1(a) depicts positive correlation, where the two variables involved are directly proportional to each other. It means that when one variable increases (decreases), the other variable also increases (decreases). Therefore, there is positive correlation when two variables are increasing or decreasing together. Examples of paired variables that give positive correlation include:

- i. Price and supply of a commodity.

- ii. Income and expenditure of a family

Figure 4.2.1(b) depicts a perfect positive correlation where the changes in the two related variables are exactly proportional to each other and in the same direction.

In figure 4.2.1(c), there is Negative correlation. Here is a situation where one variable increases (decreases), the other variable decreases (increases). Therefore, the two variables move in opposite directions. Examples of paired variable that give negative correlation include:

- i. Number of labourers and time required to complete field work.
- ii. Demand and price for a commodity.

Figure 4.2.1(d) depicts a perfect negative correlation where the changes in the two related variables are exactly proportional to each other but at reverse (opposite) directions.

Figure 4.2.1(e) depicts the zero correlation, where there is no correlation existing between the two variables. Here, no fixed pattern of movement can be established.

4.3 Measures of Correlation

4.3.1 The measure of correlation can be determined by using the following methods:

- a. Product moment correlation coefficient method (Karl Pearson's Co-efficient of correlation):

The degree of correlation between variables x and y is measured by the product moment correlation co-efficient r, defined as

$$\begin{aligned}
 r &= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \\
 &= \frac{\sum xy - (\sum x)(\sum y) / n}{\sqrt{(\sum x^2 - (\sum x)^2 / n)(\sum y^2 - (\sum y)^2 / n)}} \\
 &= \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2] [n \sum y^2 - (\sum y)^2]}} \quad \text{---4.3.1}
 \end{aligned}$$

where

n = the number of items (in pairs)

The numerator of equation 4.3.1 is called co-variance and the denominator

is the square root of the product of the variances of variables x and y

$$r = \frac{\text{Covariance}(x, y)}{\sqrt{\text{Var}(x)\text{Var}(y)}}$$

Illustration 4.2

The marks scored by seven students in Mathematics (x) and Accounts (y) are given below. If the maximum scores obtainable in mathematics and accounts are respectively 50 and 100, determine the Pearson correlation coefficient for these scores.

X	30	40	35	40	20	25	50
Y	50	70	65	68	40	60	80

Solution:

x	y	xy	x^2	y^2
30	50	1500	900	2500
40	70	2800	1600	4900
35	65	2275	1225	4225
40	68	2720	1600	4624
20	40	800	400	1600
25	60	1500	625	3600
50	80	4000	2500	6400
240	433	15595	8850	27849

Now, Pearson Correlation Coefficient (r)

$$\begin{aligned}
 r &= \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}} \\
 &= \frac{(7)(15595) - (240)(433)}{\sqrt{[7(8850) - (240)^2][7(27849) - (433)^2]}} \\
 &= \frac{109165 - 103920}{\sqrt{[4350][7454]}} \\
 &= \frac{5245}{\sqrt{32424900}} \\
 &= \frac{5245}{5694.28661} = 0.9211
 \end{aligned}$$

Illustration 4.3

The costs on advertisement (x) and revenues (y) generated by a company for 10 months are given below:

Advertisement (x) (₦'000)	45	70	32	24	75	16	28	43	60	15
Revenue (y) (Million)	42	51	38	39	44	20	22	46	47	36

Determine the product moment correlation coefficient for the table.

Solution 4.3:

Let x and y represent the cost of advertisement and revenue respectively.

x	y	xy	x ²	y ²
45	42	1890	2025	1764
70	51	3570	4900	2601
32	38	1216	1024	1444
24	39	936	576	1521
75	44	3300	5625	1936
16	20	320	256	400
28	22	616	784	484
43	46	1978	1849	2116
60	47	2820	3600	2209
15	35	525	225	1225
408	384	17171	20864	15700

Now, Pearson Correlation Coefficient (r_k)

$$\begin{aligned}
 r &= \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2] [n\sum y^2 - (\sum y)^2]}} \\
 &= \frac{(10)(17171) - (408)(384)}{\sqrt{[10(20864) - (408)^2] [10(15700) - (384)^2]}} \\
 &= \frac{171710 - 156672}{\sqrt{[208640 - 166464] [157000 - 147456]}}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{15038}{\sqrt{[42176] [9544]}} \\
&= \frac{15038}{\sqrt{402527744}} \\
&= \frac{15038}{20063.0941} \\
&= 0.7495
\end{aligned}$$

(b.) Rank correlation (Spearman's Rank Correlation)

There are occasions when we wish to put some object in an ordinal scale without giving actual marks to them. For instance, during a beauty contest the judges place the contestants in some order like first, second, third etc. without actually giving specific marks to them. This process is called ranking and the ordinal scale is used.

Ranking occurs where either for lack of time, money or suitable measurements may be impossible to quantify the items. In dealing with a correlation problem where the values are in ranks, rank correlation methods are used.

The best known of such methods is the Spearman's Rank correlation co-efficient which is defined as:

$$R = \frac{1 - 6\sum d_i^2}{n(n^2 - 1)}$$

where d_i = difference in each pair of ranks

n = number of objects being ranked.

R is defined in such a way that when the ranks are in perfect agreement,

R equals +1 and when they are in perfect disagreement

R equals -1; otherwise $-1 < R < 1$.

Illustration 4.4: If two judges P and Q ranked 10 contestants in a beauty show as follows:

Contestant	H	I	J	K	L	M	N	O	P	Q
Rank by P	4	2	5	6	1	8	3	7	10	9
Rank by Q	3	4	2	8	1	9	5	6	10	7

Obtain the Spearman's Rank correlation coefficient for the given data.

Solution

Contestant	H	I	J	K	L	M	N	O	P	Q
Rank by P(x)	4	2	5	6	1	8	3	7	10	9
Rank by Q(y)	3	4	2	8	1	9	5	6	10	7
$d = x - y$	1	-2	3	-2	0	-1	-2	1	0	2
d^2	1	4	9	4	0	1	4	1	0	4

$$\Sigma d^2 = 28$$

$$R = 1 - \frac{6\Sigma d_i^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6(28)}{10(99)} = 1 - \frac{168}{990}$$

$$= 1 - 0.169696969 = 0.83030303$$

Illustration 4.5: The following are the marks obtained by five students in Mathematics (x) and Accounts (y). Determine the rank correlation coefficient of the data.

x	11	12	13	15	19
y	16	12	14	20	18

Solution

Since the marks are not given in ranks, we need to first rank the marks

x	y	Ranking x	Ranking y	$d = x - y$	d^2
11	16	5	3	2	4
12	12	4	5	-1	1
13	14	3	4	-1	1
15	20	2	1	1	1
19	18	1	2	-1	1
					8

Note that ranking can be done in ascending order or descending order. What is important is consistency in the order chosen to use.

$$\Sigma d^2 = 8$$

$$\begin{aligned}
 R &= 1 - \frac{6\sum d_i^2}{n(n^2 - 1)} \\
 &= 1 - \frac{6 \times 8}{5 \times 24} = 1 - \frac{(8)}{5(4)} = 1 - \frac{8}{20} \\
 &= 1 - 0.4 = 0.6
 \end{aligned}$$

Remark 1:

Note that when ties of ranks occur (i.e when two or more of the ranks are the same), each value of the ties is given the mean of that ranks. For example, if two places are ranked second then each place takes the value of $1/2 (2+3) = 5/2$, since for the ties one place would have been second while the other one would have been third. Similarly if three places are ranked sixth then each one is ranked $1/3(6+7+8) = 21/3$. Having done this, the procedure for getting R is the same as outlined before.

Illustration 4.6: The following are marks obtained in Mathematics (x) and Accounts (y) by 10 students. Calculate the rank correlation coefficient of the data.

Students	H	I	J	K	L	M	N	O	P	Q
x	14	12	18	14	17	16	20	11	13	19
y	13	11	15	18	15	19	15	12	14	20

Solution

x	y	Ranking x	Ranking y	d = x - y	d ²
14	13	6.5	8	-1.5	2.25
12	11	9	10	-1	1
18	15	3	5	-2	4
14	18	6.5	3	3.5	12.25
17	15	4	5	-1	1
16	19	5	2	3	9
20	15	1	5	-4	16
11	12	10	9	1	1
13	14	8	7	1	1
19	20	2	1	1	1
					48.5

Note that in variable x , there are two persons with the same mark 14 and the positions to be taken by the two of them are 6 and 7th positions. Hence, the average position is $= (6 + 7)/2 = 6.5$

For the variable y , there are three persons with the same mark of 15 and the positions to be taken by the three of them are 4th, 5th and 6th positions. Hence, the average position is $15/3$ i.e 5.

$$\begin{aligned}
 \Sigma d^2 &= 48.5 \\
 R &= 1 - \frac{6 \Sigma d_i^2}{n(n^2 - 1)} \\
 &= 1 - \frac{6(48.5)}{10(10^2 - 1)} = 1 - \frac{6(48.5)}{10(99)} \\
 &= 1 - \frac{291}{990} \\
 &= 1 - 0.2939394 = 0.7060606
 \end{aligned}$$

Illustration 4.7: Using the table in example 4.3.2, compute its rank correlation coefficient.

X	y	Rank of x	Rank of y	$d_i = R_x - R_y$	d_i^2
45	42	7	6	1	1
70	51	9	10	-1	1
32	38	5	4	1	1
24	39	3	5	-2	4
75	44	10	7	3	9
16	20	2	1	1	1
28	22	4	2	2	4
43	46	6	8	-2	4
60	47	8	9	-1	1
15	35	1	3	-2	4
					30

$$\begin{aligned}
 R &= 1 - \frac{6 \Sigma d_i^2}{n(n^2 - 1)} \\
 &= 1 - \frac{6(30)}{10(10^2 - 1)}
 \end{aligned}$$

$$\begin{aligned}
&= 1 - \frac{180}{990} \\
&= \frac{990-180}{990} \\
&= \frac{810}{990} = 0.8182
\end{aligned}$$

Remark: By comparing the two coefficients in examples 4.3.2 and 4.3.6, it could be seen that the Spearman's Rank correlation coefficient is on the high side compared with the product moment correlation coefficient. It therefore, implies that the Spearman's Rank correlation coefficient is not as accurate as the product moment correlation, but it is much easier to calculate

4.3.2 Interpretation of Correlation

1. Interpretation of product moment correlation is as follows:
 - (a.) if r is close to ± 1 , there is a perfect relationship between the two variables x and y . i.e. there is good correlation.
 - (b.) when r is close to zero(0), there is no relationship, that is, there is poor or non – existence of correlation but it is much easier to calculate.

For example:

r	=	+0.92 }	Good correlation
r	=	-0.96 }	Good correlation
r	=	-0.12 }	Poor or non-existent correlation
r	=	0 }	Non-existent correlation or no correlation at all
r	=	+0.26 }	Poor or non- existent correlation

The same interpretation(s) apply to rank correlation R .

4.4 Simple Regression Line

In the previous section, we discussed how to measure the degree of association. In this section, the nature of the relationship that exists between two variables x and y will be looked at, the interest may be about the students' performance who offered mathematics and accounts. We want to know whether their performance in a subject like accounts is affected by how well they do in mathematics. In this case, our variable y represents the accounts (the dependent or response variable), while mathematics represents the

independent or explanatory variable. The relationship between these two variables is characterized by mathematical model called a Regression Equation.

For a simple linear regression and for regression of y on x , we have:

$$y = a + bx$$

where y is the dependent variable and x is the independent variable

a is the intercept of the line on the y -axis (the point where the line cuts the y -axis).

b is known as the regression coefficient. This is, the slope or gradient of the regression line.

Note: It is important to distinguish between the independent and dependent variables which is not necessary for correlation.

4.4.1 Methods of Obtaining or Fitting Regression Line

We have two methods of fitting the regression line. These are

- (i) graphical and
- (ii) Algebraic.

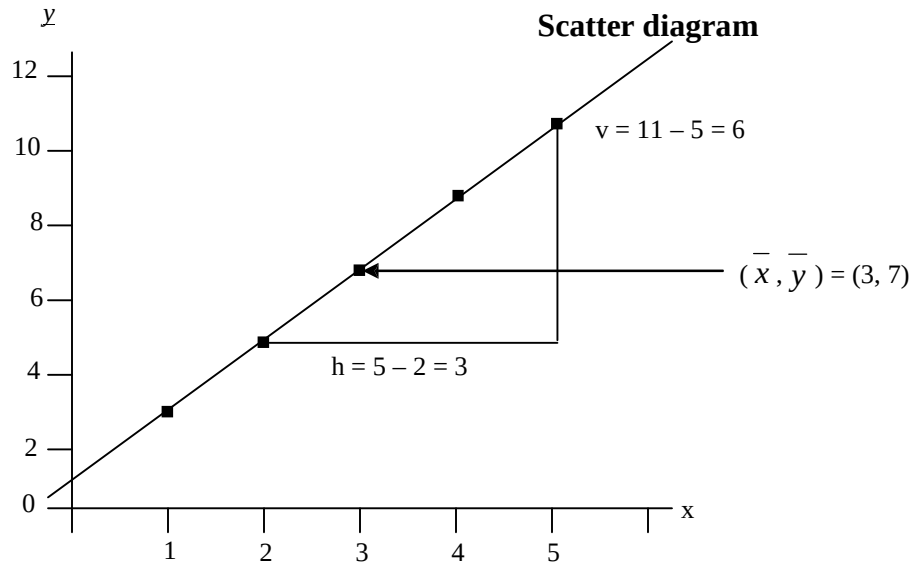
(a.) Graphical method: The following steps are to be taken:

- i. Draw the scatter diagram for the data.
- ii. Look at two points that a straight line will pass through on the diagram.
One of the points ought to be $(\bar{x}, \bar{y})$.
- iii. Estimate constants a and b from the graph.
 a = intercept on the y – axis of the drawn straight line.
 b = slope or gradient of the line drawn i.e slope = $\frac{\text{vertical length}}{\text{horizontal length}}$.
- iv. Regression line $y = a + bx$ is stated.

Illustration 4.8: Suppose we are required to find the relationship between two variables y and x on which we have the following data:

$\underline{x}$	1	2	3	4	5
$\underline{y}$	3	5	7	9	11

Solution



$$\begin{aligned}
 \text{Slope (b)} &= \frac{\text{Vertical length}}{\text{Horizontal length}} \\
 &= \frac{v}{h} \\
 a &= 1, \quad b = \frac{v}{h} = \frac{4}{3} = 1.33 \\
 \therefore y &= a + bx \quad \Rightarrow \quad y = 1 + 1.33x
 \end{aligned}$$

(b.) Algebraic method: In the algebraic method, we use the “normal equation” which is derived by the least squares method. The said normal equations are:

$$na + b\sum x = \sum y \quad \text{-----4.4.1.1(a)}$$

$$a\sum x + b\sum x^2 = \sum xy \quad \text{-----4.4.1.1(b)}$$

which are used to fit the regression line of y on x as $y = a + bx$

It should be noted that when equations 4.4.1.1 are solved simultaneously, we have the following estimates of a and b :

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$a = \bar{y} - b\bar{x}$$

Illustration 4.9: Using the last question (under the graphical approach), calculate or fit the regression line of y on x .

Solution:

x	y	xy	x^2
1	3	3	1
2	5	10	4
3	7	21	9
4	9	36	16
5	11	55	25
15	35	125	55

$$5a + b(15) = 35 \quad \text{---1}$$

$$a(15) + b(55) = 125 \quad \text{---2}$$

$$5(a) = 35 - b(15)$$

$$a = \frac{35 - b(15)}{5}$$

$$= 7 - b(3) \quad \text{OR}$$

$$15(7 - b(3)) + b(55) = 125$$

$$105 - b(45) + b(55) = 125$$

$$105 - 45b + 55b = 125$$

$$-45b + 55b = 125 - 105$$

$$10b = 20$$

$$b = 20/10 = 2$$

$$5a + b(15) = 35$$

$$5a + 2(15) = 35$$

$$5a + 30 = 35$$

$$5a = 35 - 30$$

$$5a = 5$$

$$a = 5/5 = 1$$

$\therefore y = 1 + 2x$ is the regression line of y on x

Note that when one compares the results of examples 4.4.1 and 4.4.2, the algebraic method gives a better result.

Illustration 4.10:

The table below shows the income and expenditure (in ₦'000) of a man for 10 months.

Income (x)	8	18	52	38	26	60	40	50	82	75
Expenditure (y)	2	4	5	7	9	11	13	15	20	23

Fit simple linear regression line $y = a + bx$ to the data.

Solution

x	y	xy	x^2	y^2
8	2	16	64	4
18	4	72	324	16
52	5	260	2704	25
38	7	266	1444	49
26	9	234	676	81
60	11	660	3600	121
40	13	520	1600	169
50	15	750	2500	225
82	20	1640	6724	400
75	23	1725	5625	529
449	109	6143	25261	1619

From the model

$$y = a + bx$$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$a = y - bx$$

$$\begin{aligned}
 b &= \frac{(10)(6143) - (449)(109)}{(10)(25261) - (449)^2} \\
 &= \frac{12489}{51009} \\
 &= 0.2448
 \end{aligned}$$

Also,

$$a = \bar{y} - b\bar{x}$$

$$\bar{y} = \frac{\sum y}{n} = \frac{109}{10} = 10.9$$

$$\bar{x} = \frac{\sum x}{n} = \frac{449}{10} = 44.9$$

Now,

$$\begin{aligned}a &= 10.9 - (0.2448)(44.9) \\&= 10.9 - 10.9915 \\&= -0.0915\end{aligned}$$

Hence,

The model is

$$\begin{aligned}y &= a' + bx \\&= -0.0915 + 0.2448x\end{aligned}$$

4.4.2 Computation or Fitting Regression Line of Variable (x) on Variable (y).

In section 4.4.1, fitting the regression of variable y on variable x was discussed. But in this section, we present the regression of variable x on variable y which gives the regression line.

$$\begin{aligned}x &= a' + b'y \\ \text{where } b' &= \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2} \\ a' &= \bar{x} - b' \bar{y}\end{aligned}$$

We are to note that variables x and y are now dependent and independent variables respectively.

Illustration 4.11:

Using the data in example 4.4.1.3, determine the regression line of variable x on variable y .

Income (x)	8	18	52	38	26	60	40	50	82	75
Expenditure (y)	2	4	5	7	9	11	13	15	20	23

Fit simple linear regression line to the table.

Solution

x	y	xy	x^2	y^2
8	2	16	64	4
18	4	72	324	16
52	5	260	2704	25
38	7	266	1444	49
26	9	234	676	81
60	11	660	3600	121
40	13	520	1600	169
50	15	750	2500	225
82	20	1640	6724	400
75	23	1725	5625	529
449	109	6143	25261	1619

$$\begin{aligned}
 b' &= \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2} \\
 &= \frac{10 \times 6143 - 449 \times 109}{10 \times 1619 - (109)^2} \\
 &= \frac{61430 - 48941}{16190 - 11881} \\
 &= \frac{12489}{4309} \\
 b' &= 2.8984 \\
 a' &= \frac{\sum x}{n} - b' \frac{\sum y}{n} \\
 &= \frac{449}{10} - \frac{2.8984(109)}{10} \\
 &= 44.9 - 2.8984(10.9) \\
 &= 44.9 - 31.5926 \\
 &= 13.3074
 \end{aligned}$$

∴ The model is $x = 13.3074 + 2.8984y$

- If this result is compared with the one obtained in example 4.4.1.3, you can see that the two results have nothing in common; it is not a question of change of subjects. Thus, regression of y on x is DEFINITELY different from regression of x on y .

4.4.3 Application and Interpretation of Linear Regression

1. Since regression coefficient (b) is the slope of the regression line, its magnitude gives an indication of the steepness of the line and it can be interpreted as thus.
 - i. If $b = 0$, it means the line is parallel to x axis.
 - ii. If b is high and positive, it gives very steep and upward slopping regression.
 - iii. If b is negative, it gives downward sloping of the regression line.
2. The regression line can be used for prediction. When the regression equation or regression line is used to obtain y value corresponding to a given x value, we say that x is used to predict y . We can equally use y to predict x .

Illustration 4.12: If $y = 16.94 + 0.96x$, what is y when $x = 50$?

Solution

$$\begin{aligned}y &= 16.94 + 0.96(50) \\&= 16.94 + 48 \\&= 64.94\end{aligned}$$

4.5 Coefficient of Determination.

The concept of “coefficient of determination” demonstrates the relationship between the coefficient of correlation and regression coefficients. Coefficient of determination, usually denoted by r^2 , is defined by:

$$r^2 = b \times b'$$

where r , b and b' are earlier defined.

Illustration 4.13

Using the results in examples 4.4.1.3 and 4.4.2.1, determine

- a. coefficient of determination.
- b. correlation coefficient of the data.

Solution:

a. Since $r^2 = b \times b'$

$$b = 0.2448$$
$$b' = 2.8984$$
$$\therefore r^2 = 0.2448 \times 2.8984$$
$$= 0.7095$$

b. Correlation coefficient of the data is

$$r^2 = 0.7095$$
$$r = \sqrt{0.7095}$$
$$= 0.8423$$

4.6 Chapter Summary

Treatment of bivariate data by the use of correlation and regression analyses is presented. Correlation measures the degree of association while regression gives the pattern of the relationship between the two variables. Two types of correlation coefficients, namely: the Pearson's product moment correlation and Spearman's rank correlation are considered.

Regression line fitting by graphical and calculation methods are considered. The use of regression predict, correlation coefficient and coefficient of determination is also presented.

MULTIPLE-CHOICE AND SHORT ANSWER QUESTIONS

1. Correlation measures
 - A. Pattern of relationship
 - B. Degree of association
 - C. Degree of statistical analysis
 - D. Pattern of statistical analysis
 - E. Nature of statistical analysis
2. When the rank correlation is -1 , it means
 - A. Perfect agreement
 - B. Directly proportional relationship
 - C. Perfect disagreement
 - D. Indirect relationship
 - E. Unstable relationship

3. Which of the following is the regression coefficient in $y = a + bx$, where x and y are variables?
- A. $a+b$,
 - B. $a - b$,
 - C. ab ,
 - D. a
 - E. b

Use of the following table to answer questions 4 and 5

X	1	2	3	4
Y	2	3.5	1	3.5

4. Assuming variables x and y are marks, the product moment correlation coefficient is _____
5. Assuming variables x and y are ranks, the rank correlation coefficient is _____

Use the following table to answer questions 6, 7 and 8

x	2	3	4
y	6	4	2

6. Using the regression line $y = a + bx$, the value of b is _____
7. Using the regression line $y = a + bx$, the value of a is _____
8. If correlation coefficient is 0.95, the value of the regression coefficient of regression line x on y is _____

Use the following information to answer questions 9 and 10. Given the regression line $y = 5 + 1.5x$,

9. the value of y when $x = 6$ is _____.
10. the value of x when $y = 11$ is _____.

Answers to Chapter Four

1. B

2. C

3. E

$$\begin{aligned}
 4. \quad r &= \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}} \\
 &= \frac{4(26) - 10(10)}{\sqrt{[4(30) - (10)^2][4(29.5) - (10)^2]}} = \frac{4}{\sqrt{(20)(18)}} = 0.2108
 \end{aligned}$$

$$\begin{aligned}
 5. \quad R &= 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6(7.5)}{4(4^2 - 1)} = 1 - \frac{45}{60} \\
 &= 1 - 0.75 = 0.25
 \end{aligned}$$

$$6. \quad b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{3(32) - 9(12)}{3(29) - 9^2} = \frac{96 - 108}{87 - 81} = -2$$

$$7. \quad a = \bar{y} - b\bar{x} = 4 - (-2)3 = 10$$

$$8. \quad \text{Coefficient of determination } r^2 = (0.95)^2 = 0.9025$$

$$\text{i.e } 0.9025 = b \cdot b' = -2b'$$

$$\Rightarrow b' = \frac{0.9025}{-2} = -0.4513$$

where b' is the regression coefficient of x on y.

$$9. \quad y = 5 + 1.5(6) = 14$$

$$10. \quad y = 11 = 5 + 1.5\underline{x} \Rightarrow 1.5\underline{x} = 6 \Rightarrow \underline{x} = 4$$

CHAPTER FIVE

TIME SERIES ANALYSIS

CHAPTER CONTENT

- (a) Introduction;
- (b) Basic Components of a Time Series;
- (c) Time Series Analysis;
- (d) Estimation of Trend; and
- (e) Estimation of Seasoned Variation.

OBJECTIVES

At the end of the chapter, readers should be able to

- a) know the meaning of a time series
- b) know the basic components of a time series;
- c) estimate the trend by the use of moving averages and least squares methods;
- d) carry-out exponential smoothening; and
- e) estimate seasonal index.

5.1 Introduction

A time series is a set of data that are successively collected at regular intervals of time.

The regular interval time can be daily, weekly, monthly, quarterly or yearly. Examples of some time series data include:

- a) Monthly production of a company;
- b) Daily sales at a medical store;
- c) Amount of annual rainfall over a period of time; and
- d) Money deposited in a bank on various working days.

Furthermore, it is essential to know that when a time series is analysed, it has the following benefits:

- i. Understanding the past behaviour of a variable and be able to:
 - determine direction of periodic fluctuations; and
 - predict future tendencies of the variable.

- ii. Determining the impact of the various forces influencing different variables which then facilitate their comparison, such as:
 - the differences that may have to do with price of commodities;
 - the physical quantity of goods produced, marketed or consumed in order to make a comparison between periods of time, schools, places and etc.
- iii. Knowing the behaviour of the variables in order to iron out intra-year variations as control events.

5.2 BASIC COMPONENTS OF A TIME SERIES

In time series, the existence of fluctuations from time to time is caused by composite forces which are at work constantly;

These factors have four components viz:

- a) Secular Trend or Secular Variation (T);
- b) Seasonal Fluctuations or Variation (S);
- c) Cyclical Variation (C); and
- d) Irregular or Random Variations (I).

5.2.1 Secular Trend or Secular Variation (T)

The pattern of time series trend may be linear or non-linear. It is linear when the series values are concentrated along a straight line on the time –plot. Time – plot is known as the graph of time series values against different time points. The sketches of time plot showing typical examples of a secular trend are given below.

Fig 5.1(a)

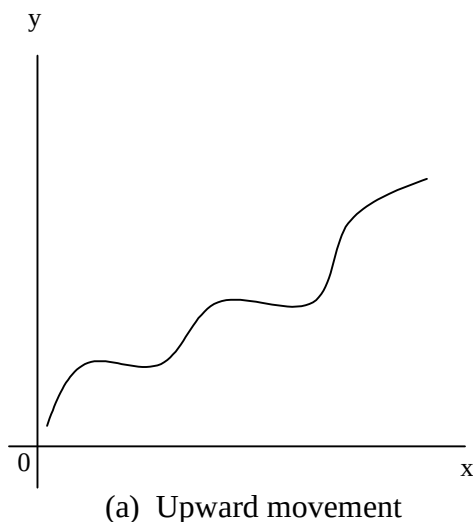


Fig. 5.1(b)

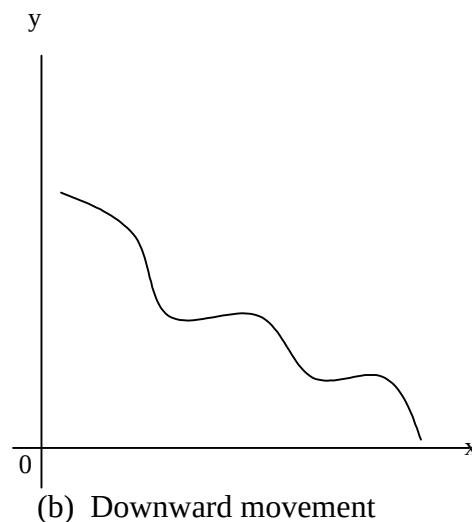


Illustration 5.1

The following table shows the number of cartons of Malt drink sold by a retailer in Lagos over twelve consecutive months in a year.

Table 1

Month	1	2	3	4	5	6	7	8	9	10	11	12
Sales	14	24	42	19	29	38	43	58	48	43	64	74

Draw the time – plot of the table above

Solution TIME PLOT OF MALT DRINK SOLD IN TWELVE MONTHS

Fig 5.2

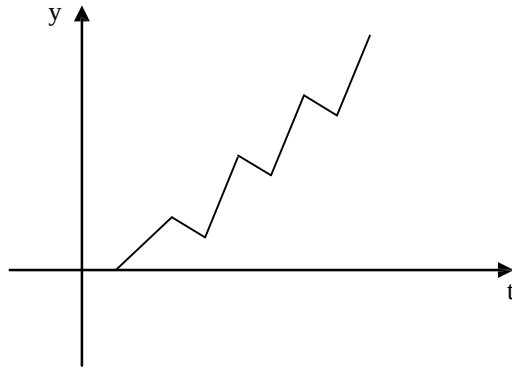


5.2.2 Seasonal Fluctuations or Variations

This is a variation that repeatedly occurs during a corresponding month or period of successive years. It is an annual reoccurring event which a time series appears to follow in a particular period or time of the year.

Data on climate such as rainy season, sales of goods during Christmas are good examples of time series with seasonal variations. The figure below depicts a time-plot showing the presence of seasonal variation.

Fig. 5.3

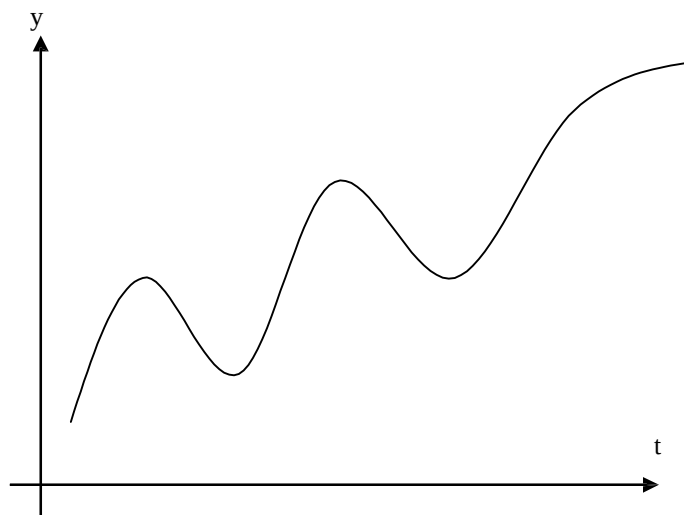


5.2.3 Cyclical Variation (C)

This is a long-term oscillation or wavelike fluctuation about the trend line of a time series. It is similar to seasonal variation with a difference of reoccurring in more than one year period. Cyclical variations are called business cycles in the sense that periods of prosperity followed by recession or depression and then recovery are caused by aggregate economic conditions rather than seasonal effects. The length of the cycle varies between four and seven years.

The cyclical variation is less predictable. The figure depicted below shows the pattern of cyclical variation.

Fig. 5.4



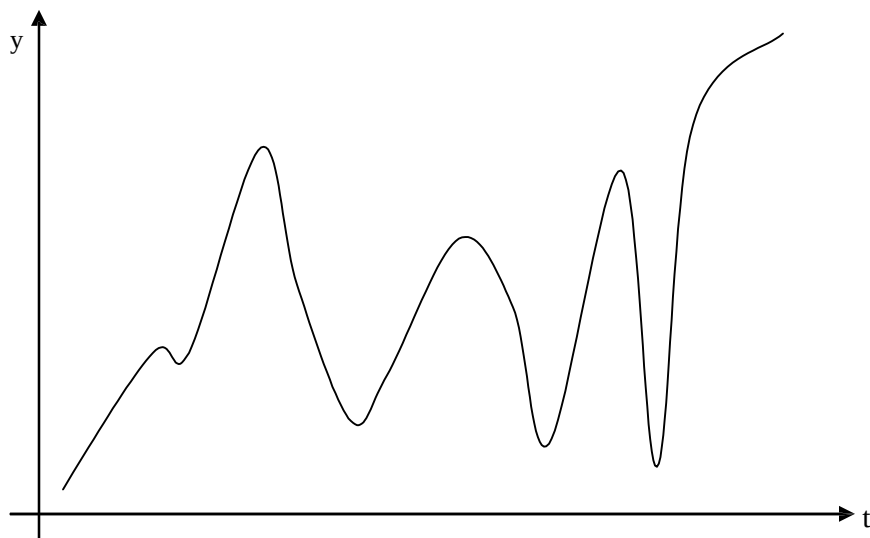
5.2.4 Irregular or Random Variation (I)

This is a variation caused by sporadic events (unpredictable events) such as floods, strikes, disasters, wars.

Random variation represents the residual variation in time series which can not be accounted for by the three other components (i.e Trend, Seasonal, Cyclical variations).

See the figure below for a typical random variation.

Fig 5.5



5.3 TIME SERIES ANALYSIS

This is an act of analysing and interpreting time series data. It can be said to be an investigative method into the time series components; which at times is also referred to as decomposition of a time series.

The first step in the time series analysis is the drawing of the “time-plot”. This will clearly show the pattern of the series movement.

Let Y represent the series by convention and T, S, C, I as components earlier indicated in paragraph 5.2. The two models of time series are additive and multiplicative models. For the additive, it is written as

$$Y = T + S + C + I \quad 5.3.1$$

while in multiplicative model, it is written as;

$$Y = TSCI \quad 5.3.2$$

Most of the time the data available can be used to estimate only the trend and seasonal variation.

5.4 ESTIMATION OF TREND

There are various methods of estimating trend. However, this study pack will consider only two methods, namely:

- a) Moving average method, and
- b) Least Squares method.

5.4.1 Moving Average Method

This is a popular method of trend estimation. The trend is obtained in this method by smoothing out the fluctuation. The procedure of computing the trend by the Moving Average (M.A) method depends on whether the period of moving average desired, atimes called the order, is even or odd:

For a given time series $Y_1, Y_2, Y_3, \dots, Y_n$, moving averages of order n are given by.

$$\begin{array}{ll} \frac{Y_1 + Y_2 + Y_3 + \dots + Y_n}{n}, & \frac{Y_2 + Y_3 + Y_4 + \dots + Y_{n+1}}{n} \\ \frac{Y_3 + Y_4 + Y_5 + \dots + Y_{n+2}}{n}, & \text{e.t.c} \end{array} \quad (5.4.1.1)$$

where the moving averages are expected to be written against the middle items considered for each average and the numerators of equation 5.4.1.1 are called the moving totals.

The following are procedural steps of computing trend by moving average (M.A):

- a. Determine the order to be used whether odd or even.
- b. When it is odd, directly apply equation 5.4.1.1 to get the desired moving average (M.A); and
- c. If it is even, first obtain the moving totals of order n from the given series and then obtain a 2 combined n moving totals of earlier obtained moving totals. The moving totals lastly obtained will be divided by $2n$ to give the desired M.A. The purpose of using two moving totals is to overcome the problem of placing the M. A against middle items considered. By convention, M. A. must be written against middle item of the items

considered. Hence, moving totals make this possible.

The following examples illustrate the trend estimation by Moving Average (M.A) method:

Illustration 5.2

Using the data in Table 1 of Illustration 5.1

Month(x)	1	2	3	4	5	6	7	8	9	10	11	12
Sales(y)	14	24	42	19	29	38	43	58	48	43	64	74

- Determine
- (a) the moving average of order 3.
 - (b) the moving average of order 4

Solution 5.2

a.

Month(x)	Sales(y)	3 – Month moving total	3 – month moving average
1	14		
2	24	80	
3	42	80	26.67
4	19	85	28.33
5	29	90	30.00
6	38	86	28.67
7	43	110	36.67
8	58	139	46.33
9	48	149	49.67
10	43	149	49.67
11	64	155	51.67
12	74	181	60.33

b.

C ₁	C ₂	C ₃	C ₄	C ₅
Month(x)	Sales(y)	4 – Month moving total	2 of 4 – month moving total	4 – month average = C ₄ ÷ 8
1	14			
2	24			
3	42	99	213	26.63
4	19	114	242	30.25
5	29	128	257	31.13
6	38	129	297	37.13
7	43	168	355	44.38
8	58	187	379	47.38
9	48	192	405	50.63
10	43	213	442	55.25
11	64	229		
12	74			

5.4.2 Merits and Demerits of Moving Average (M.A) Method

(A) Merits:

- (i) The method is simple if compared with least squares method;
- (ii) The effect of cyclical fluctuations is completely removed if the period of Moving Average (M.A) is equal to the average period of cycles; and
- (iii) It is good for a time series that reveals linear trends.

(B) Demerits

- (i) The extreme values are always lost by Moving Average (M.A) method;
- (ii) The method is not suitable for forecasting; and
- (iii) It is not good for non-linear trend.

5.4.3 Least Squares Method (L.S.M)

Recall that by least squares method, fitting a linear regression of variable (y) on variable (x) gives $y = a + bx$, where a and b are constants and are the intercept and regression coefficient respectively. This method can equally be extended to a time series data by taking the time period t as variable x and the time series value as variable y . Hence, the trend line by L.S.M is given as :

$$y = a + bx \quad 5.4.2.1$$

$$\text{where } b = \frac{\sum xy - \sum x \sum y / n}{\sum x^2 - (\sum x)^2 / n} \quad 5.4.2.2$$

$$\text{and } \bar{a} = \bar{y} - b \bar{x} \quad \text{sss} 5.4.2.3$$

The following examples give illustrations of Trend estimation by L.S.M.

Illustration 5.3

Using the data in table in Illustration 5.1, estimate the trend and fit the trend by least squares approach.

Month(x)	1	2	3	4	5	6	7	8	9	10	11	12
Sales(y)	14	24	42	19	29	38	43	58	48	43	64	74

Solution 5.3

Month(x)	Sales(y)	x^2	xy	Trend $\tilde{Y} = 62.0604 - 1.7657x$
1	14	1	14	60.2947
2	24	4	48	58.5290
3	42	9	126	56.7633
4	19	16	76	54.9976
5	29	25	145	53.2319
6	38	36	228	51.4662
7	43	49	301	49.7005
8	58	64	301	47.9348
9	48	81	432	46.1691
10	43	100	430	44.4034
11	64	121	704	42.6377
12	74	144	888	40.8720
78	607	650	3693	

From

$$Y = a + bx$$

where

$$\begin{aligned}
 b &= \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \\
 &= \frac{(12)(3639) - (78)(607)}{(12)(650) - (78)^2} \\
 &= \frac{44316 - 47346}{7800 - 6084} \\
 &= \frac{-3030}{1716} \\
 \therefore &= -1.7657
 \end{aligned}$$

Also

$$\begin{aligned}
 a &= \bar{y} - b\bar{x} \\
 \bar{y} &= \frac{\sum y}{n} \\
 &= \frac{607}{12} = 50.5833 \\
 \bar{x} &= \frac{\sum x}{n} \\
 &= \frac{78}{12} = 6.5
 \end{aligned}$$

Now,

$$\begin{aligned}
 a &= \bar{y} - b\bar{x} \\
 a &= 50.5833 - (-1.7657)(6.5) \\
 &= 50.5833 + 11.47705 \\
 &= 62.06035 \\
 &= 62.0604
 \end{aligned}$$

Recall

$$\begin{aligned}
 y &= a + bx \\
 &= 62.0604 + (-1.7657)x \\
 &= 62.0604 - 1.7657x
 \end{aligned}$$

REMARK: As the data in time series involve large number of period and data, the use of L.S.M will require us to reduce the computational tediousness by coding the period aspect of the data.

The principle of the coding is achieved by the change of variable x to t using the relationship: $t_i = x_i - x_m$, $i = 1, 2, \dots, n$

Where x_m is the median of x_i . By this idea, the new variable t_i will give $\sum t_i = 0$. If this condition ($\sum t_i = 0$) is imposed on any time series data, then the normal equations of

equations 5.4.2.2 and 5.4.2.3 will reduce to:

$$b = \frac{\sum y_i t_i}{\sum t_i^2} \quad \text{.....5.4.2.4}$$

$$a = \frac{\sum y_i}{n} = \bar{y} \quad \text{.....5.4.2.5}$$

for the Trend $T = a + bt$

Illustration 5.4

Use the Least Square Method of coding to fit the regression line to the following table:

Year	Q ₁	Q ₂	Q ₃	Q ₄
2000	20	40	25	45
2001	30	50	37	60
2002	42	60	45	65

Solution 5.4

Let's consider the following arrangement of the data

Year	Quarter	x	$t = x - 5.5$	y	ty	t^2
2000	1	0	-5.5	20	-110	30.25
	2	1	-4.5	40	-180	20.25
	3	2	-3.5	25	-87.5	12.25
	4	3	-2.5	45	-112.5	6.25
2001	1	4	-1.5	30	-40	2.25
	2	5	-0.5	50	-25	0.25
	3	6	0.5	37	18.5	0.25
	4	7	1.5	60	90	2.25
2002	1	8	2.5	42	105	6.25
	2	9	3.5	60	210	12.25
	3	10	4.5	45	202.5	20.25
	4	11	5.5	65	357.5	30.25
			0	519	428.5	143

* Note that the mean of variable $x = 5.5$

Least squares equation is $y = a + bx$, where

$$b = \frac{\sum ty}{\sum t^2} = \frac{428.5}{143} = 2.9965$$

$$\text{and } a = y - bt$$

$$\bar{y} = \frac{519}{12} = 43.25$$

$$\begin{aligned}
 \text{Hence, } a &= 43.25 - 2.9965(5.5) \\
 &= 43.25 - 16.4808 \\
 &= 26.7692
 \end{aligned}$$

$$\begin{aligned}
 \therefore y &= 26.7092 + 2.9965(x - 5.5) \\
 &= 26.7692 + 2.9965x - 16.4808 \\
 &= 10.2884 + 2.9965x
 \end{aligned}$$

a) Merits of LSM

- i. No extreme values are lost in this method as in the case of the M. A.
- ii. The method is free from subjective error.
- iii. The method can be used for forecasting.

b) Demerits

- i. It requires more time for computation.

5.4.4 Exponential Trend (Exponential Smoothing)

When the trend shows an exponential function, where the x and y variables are in arithmetic and geometric progressions respectively, smoothening the trend is the appropriate approach. This is achieved as follows:

For an exponential function

$$Y = ab^x \quad \dots 5.4.3.1$$

where a and b are constants, taking the logarithm of both sides of equation 5.4.3.1 yields

$$\text{Log } Y = \text{Log } a + x \text{Log } b \quad \dots 5.4.3.2$$

If $\text{Log } Y = z$, $\text{Log } a = A$ and $\text{Log } b = B$, equation 5.4.3.2 becomes

$$z = A + Bx \quad \dots 5.4.3.3$$

Which is now in linear form of variables z and x . The normal equations generated for the estimation of A and B are given as:

$$\Sigma z = nA + B \Sigma x \quad \dots 5.4.3.4$$

$$\Sigma zx = A \Sigma x + B \Sigma x^2 \quad \dots 5.4.3.5$$

Solving equations 5.4.3.4, and 5.4.3.5 we finally get $a = \text{Anti-Log } A$; $b = \text{antilog } B$.

The estimated exponential trend is obtained by putting the estimated values of a and b in equation 5.4.3.1.

Illustration 5.5

Given the population censuses of a country for certain number of periods as:

Census Year (x)	1950	1960	1970	1980	1990
Population in Million (y)	25.0	26.1	27.9	31.9	36.1

Fit an exponential trend $Y = ab^x$ to the above data using the Least Squares Method (L.S.M).

Solution 5.5

Census (x)	Population (y)	$u = (x - 1970)/10$	$v = \log y$	u^2	uv
1950	25.0	-2	1.3979	4	-2.7958
1960	26.1	-1	1.4166	1	-1.4166
1970	27.9	0	1.4456	0	0
1980	31.9	1	1.5038	1	1.5038
1990	36.1	2	1.5575	4	3.1150
		$\Sigma u = 0$	7.3214	10	0.4064

$$Y = ab^x$$

$$\log y = \log ab^x$$

$$\log y = \log a + x \log b$$

$$v = A + Bx$$

let $\log y = v$, $\log a = A$, $\log b = B$. then

$$\therefore u = (x - 1970)/10$$

So that $\Sigma u = 0$

$$\therefore A = \frac{\Sigma v}{n} = \frac{7.3214}{5} = 1.4643 \rightarrow a = \text{antilog } A = 29.1259$$

$$B = \frac{\Sigma uv}{\Sigma u^2} = \frac{0.4064}{10} = 0.04064 \rightarrow b = \text{Antilog } B = 1.0981$$

From $Y = ab^x$

$$Y = 29.1259(1.0981)^{\left(\frac{x-1970}{10}\right)}$$

To obtain the trend values Y for different x, we use the linear trend

$$v = A + Bu$$

$$v = 1.4643 + 0.04064u$$

5.5 Estimation of Seasonal Variation

Estimation of Seasonal variation involves the use of original time series data and the trend obtained. Here, the approach depends on the model assumed. The model can be additive or multiplicative as stated in equations 5.3.1 and 5.3.2.

For the additive model, the following steps are to be adhered to:

- a) Arrange the original given time series(Y) in column;
- b) The Trend T to be placed as next column;
- c) Obtain the seasonal variation (S) in another column by subtracting T from Y; i.e
 $S = Y - T - (C + I)$, from equation 5.3.1. Here, it is assumed that
 $C + I = 0$, hence $S = Y - T$
(Note that T(trend) can be obtained by either M. A. or L. S.M);
- d) Obtain the average seasonal variation for each period/season (weeks, months, quarters, half-yearly, etc). This is known as seasonal index (S.I).
- e) Check whether the (S.I) obtained in step 4 is a balanced one. Add the indices obtained and if the total is equal to zero, it implies balance; otherwise adjust them to balance.
- f) For the adjustment of S. I., divide the difference (to zero) by the number of period/season and then add to or subtract from S. I. obtained in step 4 as dictated by the sign of the difference in order to make the sum total zero.

For the multiplicative model, the following steps are to be followed:

- a) The same as step 1 in the additive model;
- b) The same as step 2 in the additive model;
- c) Obtain the seasonal variation (S) in another column by dividing Y by T;
i.e

$$S = \frac{Y}{T}, \text{ from equation 5.3.2, It is also assumed that } CI = 1,$$
$$\text{hence } S = Y/T.$$

- d) The same as step 4 in the additive model;
- e) Check whether the S. I. obtained in step 4 is balanced or not by having the total of the indices as number of period/season. For example, a quarterly period will have total of indices as 4, for half-yearly 2, etc. atimes when indices are expressed in percentages, it will be 400 for quarter, 200 for half-yearly, etc. If it is not balanced, there is need to adjust the indices to

the balanced form; and

- f) The same as step 6 in the additive model with modification of difference to 4 or 2 as the period indicates.

Illustration 5.6:

A company secretary preparing for his retirement within the next six years decided to invest quarterly in the purchase of shares either private placement or through public offers. The table below shows his quarterly investments (₦'0000) in 2000, 2001, 2002 and 2003

<i>Year</i>	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2000	15	35	40	20
2001	25	45	55	30
2002	37	55	60	40
2003	47	62	72	58

Estimate the trend by least squares method. Hence, compute seasonal variation.

Solution 5.6

Define x as follows

Quarter 1 2000, $x = 0$

Quarter 2 2000, $x = 1$

Quarter 3 2000, $x = 2$

Quarter 4 2000, $x = 3$

Quarter 1 2001, $x = 4$ e.t.c

Computation of least square trend line (Directed method)

X	y	x^2	xy	Trend $y = a - bx$ $= 23.6692 + 2.6441x$
0	15	0	0	23.6692
1	35	1	35	26.0133
2	40	4	80	28.6574
3	20	9	60	31.3015
4	25	16	100	33.9456
5	45	25	225	36.5897
6	55	36	330	39.2338
7	30	49	210	41.8779
8	37	64	296	44.5220
9	55	81	495	47.1661
10	60	100	600	49.8102
11	40	121	440	52.4543
12	47	144	564	55.0984
13	62	169	806	57.7425
14	72	196	1008	60.3866
15	58	225	870	63.0307
120	696	1240	6119	

From

$$y = a + bx$$

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$n = 16, \sum xy = 6119, \sum x = 120, \sum y = 696$$

$$\begin{aligned}
 b &= \frac{(16)(6119) - (120)(696)}{(16)(1240) - (120)^2} \\
 &= \frac{97904 - 83520}{19840 - 14400} \\
 &= \frac{14384}{5440} \\
 &= 2.6441
 \end{aligned}$$

Also,

$$a = \bar{y} - b\bar{x}$$

$$\bar{y} = \frac{\sum y}{n} = \frac{696}{16} = 43.5$$

$$\bar{x} = \frac{\sum x}{n} = \frac{120}{16} = 7.5$$

$$\begin{aligned}
 a &= 43.5 - (2.6441)(7.5) \\
 &= 43.5 - 19.8308 \\
 &= 23.6692
 \end{aligned}$$

Hence,

$$y = a + bx$$

$$\text{Trend}(T) = y = 23.6692 + 2.6441x$$

x	y	Trend $T = a + bx$	Variation by multiplicative model (Y/T)
0	15	23.6692	0.6337
1	35	26.0133	1.3455
2	40	28.6574	1.3958
3	20	31.3015	0.6389
4	25	33.9456	0.7475
5	45	36.5897	1.2299
6	55	39.2338	1.4019
7	30	41.8779	0.7164
8	37	44.5220	0.8310
9	55	47.1661	1.1661
10	60	49.8102	1.2046
11	40	52.4543	0.7626
12	47	55.0984	0.8530
13	62	57.7425	1.0737
14	72	60.3866	1.1923
15	58	63.0307	0.9202

Seasonal Index Table

Year	Quarter 1(Q₁)	Quarter 1(Q₂)	Quarter 1(Q₃)	Quarter 1(Q₄)
2000	0.6337	1.3455	1.3958	0.6389
2001	0.7475	1.2299	1.4019	0.7164
2002	0.8310	1.1661	1.2046	0.7626
2003	0.8530	1.0737	1.1923	0.9202
Total	3.0652	4.8152	5.1946	3.0381
(Index) Average	0.7663	1.2038	1.2987	0.7595
Adjustment	-0.007075	-0.007075	-0.007075	-0.007075
Adjusted Index	0.759225	1.196725	1.291625	0.752425

$$\begin{aligned}\text{Total of Averages} &= 0.7663 + 1.2038 + 1.2987 + 0.7595 \\ &= 4.0283\end{aligned}$$

Since the total of averages is supposed to be 4, we need to adjust.

$$\text{Adjustment by } \frac{4 - 4.0283}{4} = \frac{-0.0283}{4} = -0.007075$$

5.6 Chapter Summary

The time series is described as a set of data collected at regular intervals of time. Analysis of time series reveals its principal components as Trend, Seasonal, Cyclic and Random variations. The methods of obtaining trend, as discussed in this book, are moving averages and least squares methods. The concept of seasonal index is considered from the perspective of additive and multiplicative models. Exponential smoothening is also discussed with numerical example.

MULTIPLE-CHOICE AND SHORT-ANSWER QUESTIONS

1. Which of the following is not a time series data?
 - A. Monthly production of a company.
 - B. Daily sales at a medicine store.
 - C. Expenditure at home.
 - D. Daily deposits in a bank.
 - E. Annual rainfall.
2. For monthly time series data, the order of the moving average is_____
 - A. 2;
 - B. 4;
 - C. 6;
 - D. 10;
 - E. 12
3. Using the conventional symbol of time series components, which of the following is the additive model?
 - A. $P = T + C + I + S$;
 - B. $Y = T + S + C + I$
 - C. $Y = TSCI$
 - D. $P = TSCI$
 - E. $Y = ab^x$

4. Which of the following is NOT true of moving average method?
- The method is simple if compared with Least Squares Method.
 - The effect of cyclical fluctuations is completely removed by the method.
 - The extreme values are always lost.
 - The method is suitable for forecasting.
 - It is not good for non-linear trend.

Use the following table to answer questions 5 – 8.

Time (t)	Value of Series (Y)	Moving Average of order 3
1	24	
2	30	a
3	25	b
4	35	c
5	33	d
6	38	

- Find **a**
- Find **b**
- Find **c**
- Find **d**

Use the following information to answer questions 9 and 10

Time (t)	Series (Y)	Trend by LSM $Y = 30 + 3t$	Seasonal Variation assuming additive model
1	30		p
2	32		
3	37		q

- Find p .
- Find q .

Answers

1. C,

2. E,

3. B,

4. D

$$5. \quad a = \frac{24 + 30 + 25}{3} = 26.33$$

$$6. \quad b = \frac{30 + 25 + 25}{3} = 30$$

$$7. \quad c = \frac{25 + 35 + 33}{3} = 31$$

$$8. \quad d = \frac{35 + 33 + 38}{3} = 35.33$$

$$9. \quad p = 30 - \{30 + 3(1)\} = -3$$

$$10. \quad q = 37 - (30 + 3(3)) = -2$$

CHAPTER SIX

INDEX NUMBERS

CHAPTER CONTENTS

- (a) Introduction;
- (b) Construction Methods of Price Index Number;
- (c) Unweighted Index Number; and
- (d) Weighted Index Number.
- (e) Construction of Quantity Index Number

OBJECTIVES

At the end of the chapter, readers and students are expected to

- a) understand the concept of Index numbers;
- b) differentiate between price indices and price relatives;
- c) know the differences between unweighted index numbers and weighted index numbers; and
- d) compute and handle problems on index numbers by the use of weighted methods such as Laspeyre, Paasche, Fisher and Marshall Edgeworth methods.

6.1 Introduction

It is a usual practice in business, economy and other areas of life to find the average changes in price, quantity or value of related group of items or variables over a certain period of time, or for geographical locations. Index number is the statistical concept or device that is usually used to measure the changes,

Spiegel defined Index number as “A statistical measure designed to show changes in a variable or a group of variables with respect to time, geographical location or other characteristics”.

By the principle of index number, the statistical device measures

- a) the differences in the marginal of a group of related variables;
- b) the differences that may have to do with price of commodities;

- c) the physical quantity of goods produced, marketed or consumed in order to make a comparison between periods of time, schools, places, etc.

Based on the above principle of the index numbers, it can be classified in terms of the variables that it tends to measure. Hence, it is broadly categorized into:

- (i) Price Index Number consisting of retail price.
- (ii) Quantity or Volume Index numbers.
- (iii) Value Index Numbers.

6.1.1 Uses / Applications of Index Numbers

It is important and of great benefits to state the following uses of Index numbers:

- a) To deflate a value series in order to convert it into physical terms.
- b) To keep abreast of current business condition i.e it acts as business or economic barometer.
- c) To give the trend movement in business or economy.
- d) To forecast by using series of the indices.
- e) To assess the worth of purchasing power of money.
- f) To compare the standard of living in various areas/countries or geographical locations.
- g) To compare students' intelligence in various schools or countries.

6.1.2 Problems Associated with Constructions of Index Numbers

The following are among the problems usually encountered in the construction of index numbers:

- a. Definition of the purpose for which index number is being compiled or constructed.
- b. Selection of commodities/items to include in the index. Here, one has to decide what type of item, what quantity and quality are to be selected.
- c. Selection of sources of data. The data source must be reliable, hence, utmost care must be taken in selecting the source.
- d. Method of collecting data: Once the source of data has been determined, the next line of action is to decide on an efficient and effective method of data collection. This will facilitate accurate and reliable results. Here, the method depends on the source whether it is of primary or secondary type of data.

- e. Selection or choice of base year: The base year is the reference point or period that other data are being compared with. It is essentially important that the period of choice as the base year, is economically stable and free from abnormalities of economy such as inflation, depression, famines, boom etc. The period should not be too far from the current period.
- f. Methods of Combining data: The use of appropriate method of combining data depends on the purpose of the index and the data available. The choice of appropriate method dictates the formula to be used.
- g. Choice of Weight: The weights in index number refer to relative importance attached to various items. It is therefore necessary to take into account the varied relative importance of items in the construction of index numbers in order to have a fair or an accurate index number.

6.2 Construction Methods of Price Index Numbers

The Price Index number measures the changes in the general level of prices for a given number or group of commodities. It could be wholesale price index or retail price index, or index for prices of manufactured products.

The construction methods of price index numbers can be broadly classified into two:

- (i) the use of unweighted price index numbers and
- (ii) use of weighted price index numbers.

6.3 Unweighted Price Index Number Method

In the unweighted index number, it is observed that equal importance is attached to all items in the index.

The following unweighted indices shall be considered in this section:

- (a) Simple Price Relative Index Number.
- (b) Simple Aggregate Price Index.
- (c) Simple Average of Relative Method.

6.3.1 Simple Price Relative Index Numbers (SPRI)

This Index number is the simplest form of all index numbers. It can be simply defined as the ratio of the price of a single commodity in a current or given year to the price of the same commodity in the base year.

This can be expressed mathematically as follows:

$$SPRI = \frac{P_{ti}}{P_{oi}} \times 100$$

where P_{ti} = current or given year price for item i.

P_{oi} = Base year price for item i.

6.3.2 Simple Aggregate Price Index (SAPI)

This index measures the changes in price level over time, using only the arithmetic mean and ignoring differences in the relative importance of the commodities. It is expressed as the total prices of commodities in a current or given year as a percentage of the total prices of the base year of the same commodities. Symbolically, it is given as

$$SAPI = \frac{\sum P_{ti}}{\sum P_{oi}} \times 100$$

where $\sum P_{ti}$ = the sum of prices for item i at period t.

$\sum P_{oi}$ = the sum of prices for item i at the base year.

6.3.3 Simple Average of Relative Price Index (SARPI)

This approach is to remove the shortcomings of the method of SAPI. The index is computed by the following formula:

$$SARPI = \frac{\sum \left(\frac{P_{ti}}{P_{oi}} \times 100 \right)}{n} \quad \text{or} \quad \frac{\sum \left(\frac{P_{ti}}{P_{oi}} \right)}{n} \times 100$$

where $\sum$ stands for summation

n = the total number of items/commodities

P_{ti} and P_{oi} are as defined in SPRI.

Illustration 6.1

Determine the SPRI, SAPI and SARPI for the following table using 2004 as base year.

Items / commodities	Price per unit (Naira/Cedi)		
	Year 2004	<i>Year 2005</i>	Year 2006
A	20	30	60
B	30	42	55
C	10	15	20
D	6	8	15

Solution 6.1

i. $SPRI = \frac{P_{ti}}{P_{oi}} \times 100$

$$SPRI_{2005} = \frac{P_1}{P_0} \times 100 \quad SPRI_{2006} = \frac{P_2}{P_0} \times 100$$

$P_0(2004)$	$P_1(2005)$	$P_2(2006)$	$SPRI_{2005}$ $P_1/P_0 \times 100$	$SPRI_{2006}$ $P_2/P_0 \times 100$
20	30	60	150	300
30	42	55	140	183
10	15	20	150	200
6	8	15	130	250

ii. $SAPI$

Items / commodities	Price per unit (Naira/Cedi)		
	Year 2004	<i>Year 2005</i>	Year 2006
A	20	30	60
B	30	42	55
C	10	15	20
D	6	8	15
	66	95	150

$$SAPI = \frac{\sum P_{ti}}{\sum P_{oi}} \times 100$$

using 2004 as base year i.e $\Sigma P_{oi} = 66$

$$\begin{aligned} SAPI_{2005} &= \frac{\Sigma P_{ti}}{\Sigma P_{oi}} \times 100 \\ &= \frac{95}{66} \times 100 \\ &= 143.9394 \\ &\approx 144 \end{aligned}$$

$$\begin{aligned} SAPI_{2006} &= \frac{\Sigma P_{ti}}{\Sigma P_{oi}} \times 100 \\ &= \frac{150}{66} \times 100 \\ &= 227.2727 \\ &\approx 227 \end{aligned}$$

$$\text{iii. SARPI} = \sum \left[\left(\frac{P_{ti}}{P_{oi}} \right) \times 100 \right] / n$$

Items	P_0 Year 2004	P_1 Year 2005	P_2 Year 2006	$\frac{P_1}{P_0}$ P_0	$\frac{P_2}{P_0}$ P_0
A	20	30	60	1.5	3.00
B	30	42	55	1.4	1.83
C	10	15	20	1.5	2.00
D	6	8	15	1.3	2.50
	66	95	150	5.70	9.33

$$\begin{aligned} SARPI_{2005} &= \frac{\Sigma (P_1/P_0 \times 100)}{n} \\ &= \frac{5.7 \times 100}{4} \\ &= 142.5 \approx 143 \end{aligned}$$

$$\begin{aligned} SARPI_{2006} &= \frac{\Sigma (P_2/P_0 \times 100)}{n} \\ &= \frac{9.33 \times 100}{4} \\ &= 233.25 \approx 233 \end{aligned}$$

6.4 Weighted Index Numbers

In this index, weights are attached to each item or commodity on the assumption that such weights denote the relative importance of the items. The weighted index number can be broadly categorized into

- (a) Weighted Aggregative Indices
- (b) Weighted Average of Relative Indices.

6.4.1 Weighted Aggregative Indices

As weights give relative importance to the group of items, various methods of the weighted aggregative index involve different weighing techniques. Some of the methods usually encountered, include:

- a. Laspeyre
- b. Paasche
- c. Fisher
- d. Marshall Edgeworth

(a) Laspeyre's Index

In 1864, Laspeyre invented this method of weighted aggregative index by making use of the base year quantities as weights. The method is commonly and widely used because it is easy to compute being based on fixed weights of the base year. It is computed using the formula

$$La-P_{oi} = \frac{\sum p_i q_o}{\sum p_o q_o} \times 100$$

where Σ stands for summation

p_i = price of the current or given year.

q_o = quantity of the base year.

This method has an upward bias.

(b) Paasche Index

The German statistician, Paasche, introduced this method in the year 1874. Its focus is on the usage of current or given year quantities as weights. The formula for its construction is given by

$$Pa-P_{oi} = \frac{\sum p_i q_i}{\sum p_o q_i} \times 100$$

where p_i = current or given year price

p_o = base year price.

q_i = current or given year quantity.

The method is tedious to compute and of downward bias.

(c) Fisher Ideal Index

This method is based on the geometric mean of Laspeyre's and Paasche indices. The method is theoretically better than other methods because it overcomes the shortcomings of these other methods. It is computed by

$$\text{Fis-P}_{oi} = \sqrt{\frac{\sum p_i q_o}{\sum p_o q_o} \times \frac{\sum p_i q_i}{\sum p_o q_i} \times 100}$$

OR

$$= \sqrt{\{(\text{La-P}_{oi}) \times (\text{Pa-P}_{oi})\}}$$

where p_i, q_i, p_o, q_o are as defined before.

(d) Marshall – Edgeworth Index

The method uses the average of base and current/given year quantities as weights. It constructs the index by the formula:

$$\text{MaxEd-P}_{oi} = \frac{\sum p_i \{(q_o + q_i)/2\}}{\sum p_o \{(q_o + q_i)/2\}} \times 100$$

OR

$$= \frac{\sum (p_i q_o + p_i q_i)}{\sum (p_o q_o + p_o q_i)} \times 100$$

where p_i, q_i, p_o, q_o are as defined before.

Illustration 6.2

Use the following table to calculate price index numbers for the year 2006 by taking 2001 as the base year and using the following methods:

- a) Laspeyre Index.
- b) Paasche Index
- c) Fisher Index
- d) Marshall – Edgeworth Index

Commodity Item	2001		2006	
	Price	Quantity	Price	Quantity
A	14	45	20	35
B	13	15	19	20
C	12	10	14	10
D	10	5	12	8

Solution 6.2

Commodity	2001		2006	
Item	p _o Price	q _o Quantity	p _i Price	q _i Quantity
A	14	45	20	35
B	13	15	19	20
C	12	10	14	10
D	10	5	12	8

$$La-P_{oi} = \frac{\sum p_i q_o}{\sum p_o q_o} \times 100$$

Items	p _o	q _o	p _i	q _i	p _o q _o	p _i q _o	p _i q _i	p _o q _i	p _i q _o + p _i q _i	p _o q _o + p _o q _i
A	14	45	20	35	630	900	700	490	1600	1120
B	13	15	19	20	195	285	380	300	665	495
C	12	10	14	10	120	140	140	100	280	220
D	10	5	12	8	50	60	96	40	156	90
					995	1385	1316	930	2701	1925

$$a) \quad La-P_{oi} = \frac{\sum p_i q_o}{\sum p_o q_o} \times 100$$

$$\begin{aligned}
 \text{Laspeyre Index}_{(2006)} &= \frac{\sum p_i q_o}{\sum p_o q_o} \times 100 \\
 &= \frac{1385}{995} \times 100 \\
 &= 1.3920 \times 100 \\
 &= 139.1960 \approx 139
 \end{aligned}$$

$$b) \quad \text{Paasche Index}_{(2006)} = \frac{\sum p_i q_i}{\sum p_o q_i} \times 100$$

$$\begin{aligned}
 &= \frac{1316}{930} \times 100 \\
 &= 1.4151 \times 100 \\
 &= 141.5054 \approx 142
 \end{aligned}$$

$$\begin{aligned}
 \text{c) Fisher Index} &= \sqrt{\frac{\sum p_i q_0}{\sum p_0 q_0} \times \frac{\sum p_i q_i}{\sum p_0 q_i} \times 100} \\
 \therefore \text{Fisher Index}_{(2006)} &= \sqrt{\frac{\sum p_i q_0}{\sum p_0 q_0} \times \frac{\sum p_i q_i}{\sum p_0 q_i} \times 100} \\
 &= \sqrt{\frac{1385}{995} \times \frac{1280}{930} \times 100} \\
 &= \sqrt{1.9158 \times 100} \\
 &= 1.3841 \times 100 \\
 &= 138.4128 \approx 138
 \end{aligned}$$

$$\begin{aligned}
 \text{d) Marshall-Edgeworth Index} &= \frac{\sum p_i \{(q_0 + q_i)/2\}}{\sum p_0 \{(q_0 + q_i)/2\}} \times 100 \\
 &= \frac{\sum (p_i q_0 + p_i q_i)}{\sum (p_0 q_0 + p_0 q_i)} \times 100
 \end{aligned}$$

OR

$$\begin{aligned}
 \text{Marshall-Edgeworth Index}_{(2006)} &= \frac{\sum (p_i q_0 + p_i q_i)}{\sum (p_0 q_0 + p_0 q_i)} \times 100 \\
 &= \frac{2701}{1925} \times 100 \\
 &= 1.4031 \times 100 \\
 &= 140.3117 \approx 140.
 \end{aligned}$$

6.4.2 Weighted Average of Relative Index

This index overcomes the drawback of simple average of relative index where equal importance is given to the items or commodities in the group. Here, weighted average can be based on arithmetic mean (A.M) or geometric mean (G.M). Based on weighted A.M, the price index is given by

$$P_{oi} = \frac{\sum (p_i/p_0) w_i}{\sum w_i} \times 100$$

where $w_i = p_0 q_0$

Illustration 6.3

Obtain the Weighted Average Relative Price Index for the table used in illustration 6.2

Items	2001		2006		$W_i = p_0 q_0$	p_i/p_0	$(p_i/p_0)w_i$	$p_i q_0$
	Price p_0	Qty q_0	Price p_i	Qty q_i				
A	14	45	20	35	630	$20/14 = 1.4286$	900.0180	900
B	13	15	19	20	195	$19/13 = 1.4615$	284.9925	285
C	12	10	14	10	120	$14/12 = 1.1667$	140.0040	140
D	10	5	12	8	50	$12/10 = 1.2000$	60.0000	60
					995		1385.0145	1385

$$\begin{aligned}
 P_{oi} &= \frac{\sum (p_i/p_0)w_i}{\sum w_i} \times 100 = \frac{\sum p_i q_0}{\sum p_0 q_0} \times 100 \\
 &= \frac{1385.0145}{995} \times 100 = \frac{1385}{995} = 1.39
 \end{aligned}$$

6.5 Construction of Quantity Index Number

The procedures for the construction of price index numbers discussed above are also applicable for the construction of quantity index numbers. All that need to be done is to replace price(p) by quantity(q) and q by p in the formula. For instance, in the formulae for price index numbers (p_{oi}), the corresponding quantity index number is Q_{oi} . That is,

$$\text{If } P_{oi} = \frac{\sum p_i}{\sum p_0} \times 100, \quad Q_{oi} = \frac{\sum q_i}{\sum q_0} \times 100$$

$$\text{If } P_{oi} = \frac{\sum p_i q_0}{\sum p_0 q_0} \times 100, \quad Q_{oi} = \frac{\sum q_i p_0}{\sum q_0 p_0} \times 100$$

6.6 Chapter Summary

The concept of Index Number is an important statistical concept which is used to measure changes in a variable or group of variables with respect to time and other characteristics. The chapter covered the unweighted (price relative) and weighted indices. Among the weighted indices considered are Laspeyre, Paasche, Fisher Ideal and Marshall-Edgeworth indices.

MULTIPLE-CHOICE QUESTIONS AND SHORT – ANSWER QUESTIONS

1. Which of the following is **NOT** one of the uses of an index number?
 - A. To deflate a value series in order to convert it into real physical terms
 - B. To compare student's intelligence in various schools or countries.
 - C. To select the commodities sources.
 - D. To forecast by using series of the indices.
 - E. To assess the worth of purchasing power of money.

2. Which of the following is **NOT** a problem in the construction of an index number?
 - A. Selection of sources of data.
 - B. Definition of the purpose for which index is needed.
 - C. Method of collecting data for index.
 - D. Method of combining the data.
 - E. Unweighted average price index.

3. Which of the following is **NOT** a weighted price index number?
 - A. Laspeyre Index.
 - B. Simple Aggregate Price Index
 - C. Fisher Ideal Index
 - D. Marshall-Edgeworth Index
 - E. Paasche Index

4. The Weighted Aggregative Index with a backward bias is _____
 - A. Laspeyre Index
 - B. Simple Aggregate Price Index
 - C. Fisher Ideal Index
 - D. Marshall-Edgeworth Index
 - E. Paasche Index

5. Which of the following is Unweighted Price Index?

- A. $\Sigma q_i/q_o \times 100$
- B. $\frac{(\Sigma p_i/p_o)}{2} \times 100$
- C. $\frac{\Sigma p_i q_o}{\Sigma p_o q_o} \times 100$
- D. $\frac{\Sigma p_i q_i}{\Sigma p_o q_o} \times 100$
- E. $\frac{\Sigma p_i \{(q_o + q_i)/2\}}{\Sigma p_o \{(q_o + q_i)/2\}} \times 100$

Use the following table to answer Questions 6 and 7

Items	2000 price	2005 price
A	50	80
B	70	100
C	90	100
D	100	120

- 6. The Simple Aggregate Price Index of the above table using 2000 as base year is ____
- 7. The Simple Average of Relative Price Index with year 2000 as base year is ____

Use the following table to answer Questions 8 to 10

Items	2001		2006	
	Price	Quantity	Price	Quantity
A	80	50	100	60
B	90	60	100	70
C	100	70	120	90

- 8. Using year 2001 as base year, the Laspeyre price index for year 2006 is ____
- 9. Using year 2001 as base year, the Paasche price Index for year 2006 is ____
- 10. Fisher's Ideal price index of the above table is ____

ANSWERS

1. C

2. E

3. B

4. E

5. D

$$6. \quad \text{SAPI} = \frac{\sum p_i}{\sum p_0} \times 100 = \frac{400}{310} \times 100 = 129.03$$

$$7. \quad \text{SARPI} = \frac{\{\sum (p_i/p_0)\}}{n} \times 100 = \frac{1.6 + 1.429 + 1.1111 + 1.2}{4} \times 100$$

$$= \frac{5.3401}{4} \times 100 = 133.50$$

Items	p_0	q_0	p_i	q_i	$p_i q_i$	$p_i q_0$	$p_i q_0$	$p_0 q_0$
A	80	50	100	60	6000	4800	5000	4000
B	90	60	100	70	7000	6300	6000	5400
C	100	70	120	90	10800	9000	8400	7000
					23800	20100	19400	16400

$$8. \quad \text{Lasp.} = \frac{\sum p_i q_0}{\sum p_0 q_0} \times 100 = \frac{194000}{16400} \times 100 = 119.29$$

$$9. \quad \text{Paasche} = \frac{23800}{20100} \times 100 = 118.41$$

$$10. \quad \text{Fisher} = \sqrt{\{(\text{Lasp.}) \times (\text{Paasch})\}}$$

$$= \sqrt{(118.29 \times 118.41)} = 118.35$$

CHAPTER SEVEN

PROBABILITY

CHAPTER CONTENTS

- a) Introduction;
- b) Concept of Probability Theory;
- c) Addition Law of Probability;
- d) Conditional Probability and Independence;
- e) Multiplication Law of Probability;;
- f) Mathematical Expectation; and
- g) Some Special Probability Distributions.

OBJECTIVES

At the end of the chapter, readers should be able to:

- (a) understand the concept of probability and be able to handle simple probability problems;
- (b) compute probabilities by using the additive and multiplicative laws;
- (c) use mathematical expectations in discrete probability distribution; and
- (d) compute probabilities using Binomial, Poisson and Normal Distributions.

7.1 Introduction

As various activities of business and life in general depend on chance and risks, the study of probability theory is an essential tool to make correct and right decisions.

7.2 Concept of Probability Theory

As earlier said, probability theory is majorly concerned with chance and calculated risks in the face of uncertainty. This is achieved by building a mathematical concept, based on the study of some samples of the theoretical or imaginary population, which leads to formulation of some simple rules for the purpose of inference about the corresponding population.

7.2.1 Importance and Uses of Probability

Without any doubt, the importance of probability is felt in large percentage of human endeavour, most especially in business, commerce and other related areas where problems of risks are usually involved.

However, some of the uses of probability are itemized below:

- a. Probability theory is used as quantitative analysis of some problems arising from business and other areas.
- b. Probability theory is used as the basis of statistical inference.
- c. Probability theory plays vital roles in insurance and statistical quality control.

7.2.2 Definitions of Some Useful Terms

- a. **An experiment** (a random experiment) means performing an act which involves unpredictable outcome. For example, tossing of coins, throwing a die, etc.
- b. **An outcome** is one of the possible results that can happen in a trial of an experiment.

Illustration 7.1:

- i. The outcomes of a fair coin = {H, T} , where H and T stand for Head and Tail respectively.
 - ii. The outcomes of an unbiased die = {1, 2, 3, 4, 5, 6}.
- c. **Sample space** is the list or set of all possible outcomes of an experiment, while each outcome is called SAMPLE POINT.
For example, the sample space in tossing a die is {1, 2, 3, 4, 5, 6} , while 3 is a sample point.
 - d. **An event** is a collection of sample points which have certain quality or characteristics in common.

In set theory, it can be defined as a subset A of the points in the sample space S.

Examples of sample spaces for random experiments

The result/ outcome of a random experiment is called a sample space. The type of experiment determines the nature of the sample space. Some of the time, more than one step may be necessary to obtain the sample space.

a) When a coin is tossed once

$S = \{ H, T \}$ i.e 2 sample points

b) When a die is tossed once

$S = \{ 1, 2, 3, 4, 5, 6 \}$ i.e 6 sample points

c) When a coin is tossed twice (tossing two coins together)

	H	T	
H	HH	HT	$S = \{ HH, HT, TH, TT \}$ i.e 4 (2x2) sample points
T	TH	TT	

d) When 3 coins are tossed (tossing a coin Thrice)

	H	T		HH	HT	TH	TT
H	HH	HT	H	HHH	HHT	HTH	HTT
T	TH	TT	T	TTH	THT	TTH	TTT

$S = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$
i.e 8 (2x2x2) sample points

e) When a coin and a die are tossed

	1	2	3	4	5	6	
H	H1	H2	H3	H4	H5	H6	i.e 12 (2 x 6) sample points
T	T1	T2	T3	T4	T5	T6	

f) When two coins and a die are tossed

	1	2	3	4	5	6	
HH	HH1	HH2	HH3	HH4	HH5	HH6	
HT	HT1	HT2	HT3	HT4	HT5	HT6	
TH	TH1	TH2	TH3	TH4	TH5	TH6	
TT	TT1	TT2	TT3	TT4	TT5	TT6	i.e. 24 (2 x 2 x 6) sample points

e) When two dice are tossed

	1	2	3	4	5	6	
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)	
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)	
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)	
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)	
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)	i.e 36 (6 x 6) sample points

7.2.3 Definition of Probability

Probability is a statistical concept that measures the likelihood of an event happening or not. As a measure of chance, which an event E is likely to occur, it is convenient to assign a number between 0 and 1. Therefore, the probability of an Event E written as $P(E)$ is a positive number which can be formally expressed as $0 \leq P(E) \leq 1$. Note that

- If $P(E) = 0$, then it is sure or certain that the event E will not occur or happen;
- If $P(E) = 1$, then it is sure or certain that the event E will occur or happen.

There are two schools of thought about the procedures of getting the estimates for the probability of an event. These are the

- Classical or a Priori approach:** In this approach, it is assumed that all the n trials of an experiment are equally likely and the outcomes are mutually exclusive (can not happen together). If an event E can occur in r different ways out of n possible ways, then the probability of E is written as

$$P(E) = \frac{r}{n} \text{ i.e. } \frac{n(E)}{n(S)}$$

Illustration 7.2: If a fair coin is tossed once, what is the probability of obtaining Head?

Solution:

By a fair coin, it means the coin is not loaded or biased in any way.

Let E be the event of a Head, then $n(E) = 1$ and $n(S) = 2$.

$$\begin{aligned} P(E) &= \frac{\text{Number of times head is obtained}}{\text{Number of possible outcomes}} \\ &= \frac{1}{2} = 0.5 \end{aligned}$$

Illustration 7.3: If a fair die is cast once, what is the probability of obtaining an even number?

Solution:

Let the sample space(S) = {1, 2, 3, 4, 5, 6 }

$n(S) = 6$

Let E be set of even numbers = {2, 4, 6}

$n(E) = 3$

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = 0.5$$

- b) Frequency or Posteriori approach:** It is assumed in this approach that after n repetitions of an experiment, where n is very large, an event E is observed to occur in r of these, then the probability of E is

$$P(E) = \frac{r}{n}$$

Illustration 7.4: SAO Bank Plc gave out loans to 50 customers and later found that some were defaulters while only 10 repaid as scheduled. Determine the probability of repayment of loan in that bank.

Solution:

$$\begin{aligned}
 P(\text{repayment}) &= \frac{\text{Number that repaid}}{\text{Total number of people that took loan.}} \\
 &= \frac{10}{50} = 0.2
 \end{aligned}$$

Remark:

However, the two above approaches are found to have difficulties because of the two underlined terms: (equally likely and very large) which are believed to be vague and even being relative terms to an individual. Due to these difficulties, a new approach tagged “Axiomatic Approach” was developed. This axiomatic approach is based on the use of set theory.

Axiomatic definition of Probability

Suppose there is a sample space S and A is an event in the sample space. Then to every event A , there exists a corresponding real value occurrence of the event A which satisfies the following three axioms:

- i. $0 \leq P(A) \leq 1$
- ii. $P(S) = 1$
- iii. If $A_1, A_2, \dots, A_k$ are mutually exclusive events,

$$P(A_1 \text{ or } A_2 \text{ or } \dots \text{ or } A_k) = P(A_1 \cup A_2 \cup \dots \cup A_k)$$

$$\begin{aligned}
 \text{i.e } P(A) &= \text{sum of the probabilities of the simple events,} \\
 &\quad \text{comprising the event } A \\
 &= \frac{\text{Number of sample points in } A}{\text{Total number of sample points in } S} \\
 &= \frac{n(A)}{n(S)}
 \end{aligned}$$

where n is defined as the number of sample points or elements.

Note: Two or more events are said to be mutually exclusive if they can not occur at the same time.

Illustration 7.5: If a fair die is cast, determine the probability of

- a. each sample point.
- b. the sum total of all the sample points.

Solution:

- a. Since the die is fair, each sample point is equally chanced.

$$\text{Sample space (S)} = \{1, 2, 3, 4, 5, 6\}$$

$$\therefore P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

(This confirms the axiom (i) above)

$$\begin{aligned} \text{b. } P(S) &= P(1) + P(2) + P(3) + P(4) + P(5) + P(6) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1 \end{aligned}$$

(This also confirms axiom (ii) above)

Illustration 7.6: A ball is drawn at random from a box containing 8 green balls, 5 yellow balls and 7 red balls. Determine the probability that it is

- green,
- yellow,
- red,
- not green
- green or yellow.

Solution

Let G, Y and R represent green, yellow, and red balls respectively.

Then, the sample points in each event is

$$n(G) = 8, \quad n(Y) = 5 \quad \text{and} \quad n(R) = 7$$

Also, the sample space (S) consists of $8 + 5 + 7 = 20$ sample points

$$\text{i.e } n(S) = 20$$

$\therefore$ the probability of:

$$\begin{aligned} \text{a. } P(G) &= \frac{n(G)}{n(S)} = \frac{8}{20} = 0.4 \\ \text{b. } P(Y) &= \frac{n(Y)}{n(S)} = \frac{5}{20} = 0.25 \\ \text{c. } P(R) &= \frac{n(R)}{n(S)} = \frac{7}{20} = 0.35 \\ \text{d. } P(\text{not green}) &= P(G') = 1 - P(G) \\ &= 1 - 0.4 = 0.6 \\ \text{e. } P(\text{green or yellow}) &= P(G \cup Y) = P(G) + P(Y) \\ &= 0.4 + 0.25 = 0.65 \end{aligned}$$

Alternate method

$$P(G \cup Y) = P(S) - P(R) = 1 - 0.35 = 0.65$$

7.2.4 Use of Tree Diagram

There is a simple way of determining the probability of issues involving experiments. This can be achieved by the use of Tree diagram. A tree diagram can be described as a set or group of connected lines with each line looking like a branch of a tree. It represents the trials or stages of an experiment in a diagram. Here, the weight attached to the outcome of trial or stage is written on the branch, while the corresponding outcome is at end of the branch.

Illustration 7.7: Draw a tree diagram to determine the probability of obtaining Head (H) in a coin tossed once.

Solution

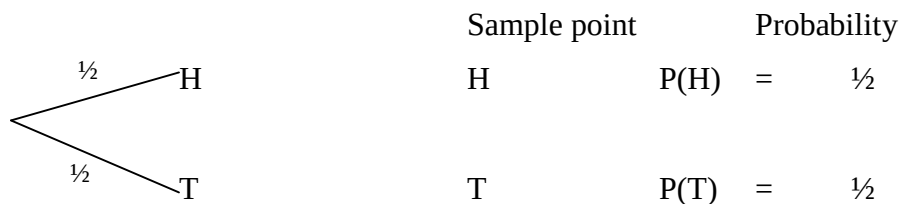
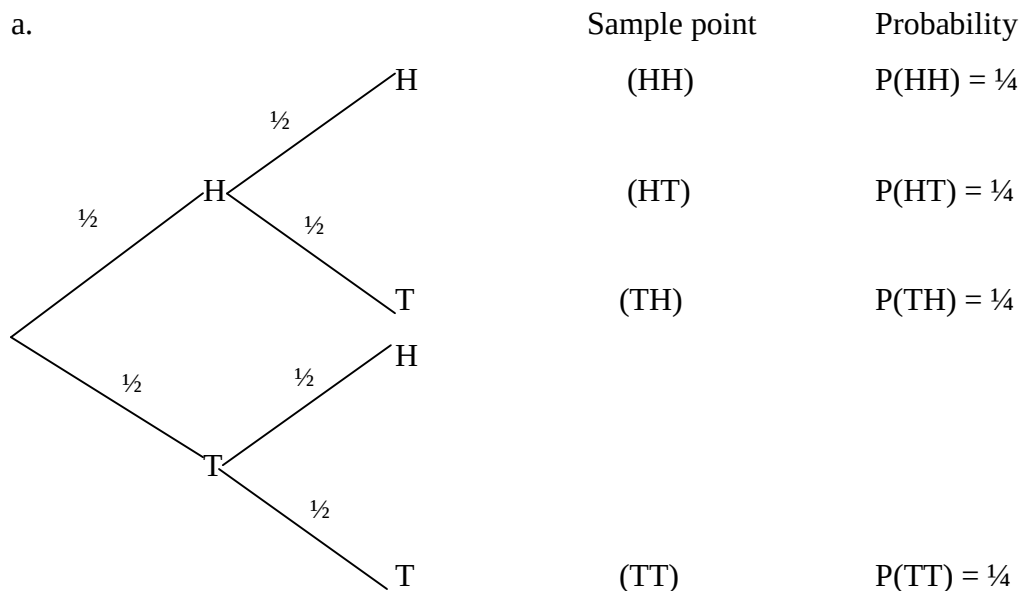


Illustration 7.8: Draw a tree diagram and obtain

- the probabilities in the case of tossing a fair coin twice;
- the probability of having a Tail (T) in the outcome.

Solution



b. $\text{Pr}(\text{of having T}) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

7.3 Addition Law of Probability

If E_1, E_2 are two events in a sample space S , then the probability of E_1 or E_2 occurring is given by

$$\begin{aligned} P(E_1 \text{ or } E_2) &= P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) \text{ by set theory} \\ &= \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} - \frac{n(E_1 \cap E_2)}{n(S)} \end{aligned}$$

This is an addition rule of probability for any two events. We should note further that in set theory “ $\cap$ ” is interpreted as “and” while “ $\cup$ ” is “or”.

By extension, the addition rule for three events E_1, E_2 and E_3 can be written as:

$$\begin{aligned} P(E_1 \cup E_2 \cup E_3) &= P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) \\ &\quad + P(E_1 \cap E_2 \cap E_3) \end{aligned}$$

Lastly, when events E_1, E_2 are mutually exclusive, $P(E_1 \cap E_2) = 0$ and the addition rule becomes:

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

In particular, E and E^1 are mutually exclusive $\therefore P(E_1 \cup E_1^1) = P(E_1) + P(E_1^1)$. $(E_1 \cup E_1^1) = S$ i.e $P(E_1) + P(E_1^1) = P(S) = 1$ i.e $P(E_1) = 1 - P(E_1^1)$

Similarly for three mutually exclusive events E_1, E_2 and E_3 , the addition rule also becomes:

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$$

Illustration 7.9:

From the following data 1, 2, 3, 4, 5, 6, 7, 8, 9, 10; obtain the probability of the following

- (a) An even number
- (b) A prime number
- (c) A number not greater than 5.
- (d) An even number or a prime number.

Solution:

Let E_1, E_2 and E_3 represent even number, prime number and numbers not greater than 5 respectively

Sample Space = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

(a) Let $E_1 = \{2, 4, 6, 8, 10\}$ = Even numbers

Then $P(E_1) = 5/10 = 1/2$ or 0.5.

(b) Let $E_2 = \{2, 3, 5, 7\}$ = Prime numbers

Then, $P(E_2) = 4/10 = 2/5$ or 0.4

(c) Let $E_3 = \{1, 2, 3, 4, 5\}$ number not greater than 5

Then $P(E_3) = 5/10 = 1/2$

(d) $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

where $E_1 \cap E_2 = \{2\} \Rightarrow P(E_1 \cap E_2) = \frac{1}{10}$

$\therefore P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \{5/10 + 2/5 - 1/10\}$

$$= \frac{5}{10} + \frac{2}{5} - \frac{1}{10}$$

$$= \frac{5 + 4 - 1}{10}$$

$$= \frac{9}{10} - \frac{1}{10}$$

$$= \frac{9 - 1}{10} = \frac{8}{10} = 0.8$$

OR

$$n(E_1 \cup E_2) = \{2, 3, 4, 5, 6, 7, 8, 10\} = 8$$

$$\therefore p(E_1 \cup E_2) = 8/10 = 0.8$$

Illustration 7.10: There are 24 female students in a class of 60 students. Determine the probability of selecting a student who is either a male or female.

Solution:

Let E_1 and E_2 represent female and male respectively.

Then $P(E_1 \cup E_2) = P(E_1) + P(E_2)$, because E_1 and E_2 are mutually

$= \frac{24}{60} + \frac{36}{60}$ exclusive i.e $p(E_1 \cap E_2) = 0$

$$= 1$$

7.4 Conditional Probability and Independence

Let E_1 and E_2 be two events such that there is an intersection between E_1 and E_2 , and that $P(E_1) > 0$. then, we denote the conditional probability by $P(E_2/E_1)$; and it can be read “the probability of E_2 given that E_1 has occurred or happened . The symbol (/) represents given. Here, event E_1 is known or assumed to have occurred or happened and it affects the sample space. By definition, the conditional probability is expressed as

$$P(E_2/E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)}, \text{ or}$$

$$P(E_1 \cap E_2) = P(E_1)P(E_2/E_1)$$

A practical example of conditional probability is seen in situations where two events are involved and must be satisfied; e.g probability of a student offering physics given that he must be a male.

The concepts of dependent and independent events go a long way to explain how the occurrence of an event affects another one. Here, if the events E_1 and E_2 occurrence or non-occurrence does not affect each other, the events are said to be independent; otherwise they are dependent events. Examples of independent events are:

- (a) obtaining Head in coin tossing,
- (b) passing of QA and Economics in ATS examination..

Also, examples of dependent are:

- (c) Weather condition and sales of minerals;
- (d) drug taken and rate of recovery from illness.

By applying the idea of independent event to our conditional probability $P(E_2/E_1)$, it becomes $P(E_2/E_1) = P(E_2)$ because the probability of E_2 occurring or happening is not affected by the occurrence or non-occurrence of E_1 . Hence

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$$

By extension for independent events E_1 , E_2 and E_3 ,

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3)$$

Illustration 7.11:

If the probability that Dayo will be alive in 20 years is 0.8 and the probability that Toyin will be alive in 20 years is 0.3, what is then the probability that they will both be alive in 20 years?

Solution

$$P(\text{that Dayo will be alive in 20 years}) = 0.8$$

$$P(\text{that Toyin will be alive in 20 years}) = 0.3$$

$$P(\text{that they will both be alive in 20 years}) = 0.8 \times 0.3 = 0.24$$

Note: Both are independent events.

Illustration 7.12:

Suppose that a box contains 5 white balls and 4 black balls. If two balls are to be drawn randomly one after the other without replacement, what is the probability that both balls drawn are black.

Solution:

Let E_1 be the event “first ball drawn is black” and E_2 be the event “second ball drawn is black”, where the balls are not replaced after being drawn. Here E_1 and E_2 are dependent events and conditional probability approach is an appropriate method.

The probability that the first ball drawn is black is

$$P(E_1) = \frac{4}{5 + 4} = \frac{4}{9}$$

The probability that the second ball drawn is black, given that the first ball drawn was black, is $P(E_2/E_1)$

$$P(E_2/E_1) = \frac{3}{5 + 3} = \frac{3}{8}$$

Thus, the probability that both balls drawn are black is

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2/E_1) = \frac{4}{9} \cdot \frac{3}{8} = \frac{12}{72} = \frac{1}{6}$$

7.5 Multiplication Law of Probability

Let us recall that when combining events in the addition rule, we dealt with the case of “OR” where we use the set symbol $\cup$ (union) to represent it. In a similar vein, the multiplication rule of probability is dealing with “AND” when combining events; and this is represented by the set symbol $\cap$ (intersection).

From conditional probability of events E_1, E_2 ;

$$P(E_2/E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)},$$

$$\text{or } P(E_1 \cap E_2) = P(E_2/E_1) \times P(E_1)$$

$$\text{it is also true that } P(E_1 \cap E_2) = P(E_1/E_2) \times P(E_2)$$

This expression represents the multiplication rule of the probability where two events E_1 and E_2 are involved.

Note : If E_1 and E_2 are independent events $P(E_2/E_1) = P(E_2)$ $P(E_1/E_2) = P(E_1)$
 $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2/E_1) = P(E_2)$ $P(E_1/E_2) = P(E_1) \cdot P(E_2)$

Illustration 7.13:

The probability that an employee comes late to work in any given day is 0.14. The probability that employee is a female in the company is 0.275. Obtain the probability that an employee selected from the company is a female late comer?

Solution:

Let E_1 = event of employee comes late

E_2 = event of employee is a female

Then, $P(E_1 \cap E_2)$ is required

$$\begin{aligned} P(E_1 \cap E_2) &= p(E_2) \cdot p(E_1/E_2) \\ &= 0.275 \times 0.14 \\ &= 0.0385 \end{aligned}$$

Note: E_1, E_2 are independent events.

Illustration 7.14: A box contains 6 black and 8 yellow balls. If two successive draws of one ball are made, determine the probability that the first drawn ball is black and the second drawn ball is yellow if (a) the first drawn ball is replaced before the second draw; (b) the first drawn ball is not replaced.

Solution:

Let B and Y represent black and yellow balls respectively.

- (a) This is a case of selection with replacement, and the events involved are independent.

$$\begin{aligned} \Pr(B \cap Y) &= P(B) \times P(Y) \\ \text{where } P(B) &= \frac{6}{6+8} = \frac{6}{14} = \frac{3}{7}, \text{ and} \\ P(Y) &= \frac{8}{6+8} = \frac{8}{14} = \frac{4}{7}, \text{ and} \\ \therefore P(B \cap Y) &= \frac{3}{7} \times \frac{4}{7} = \frac{12}{49} \end{aligned}$$

- (b) This is selection without replacement;

$$\begin{aligned} \Pr(BY) &= P(B) \times P(Y/B) \\ \text{where } P(B) &= \frac{6}{6+8} = \frac{6}{14} = \frac{3}{7}, \text{ and} \\ P(Y/B) &= \frac{8}{5+8} = \frac{8}{13} \\ \therefore P(BY) &= \frac{3}{7} \times \frac{8}{13} = \frac{24}{91} \end{aligned}$$

7.6 Mathematical Expectation

Mathematical expectation is an important concept in the study of statistics and probability in the sense that it can be used to determine some statistics like the mean and the variance,

Let X be a discrete random variable having the possible values $x_1, x_2, x_3, \dots, x_n$ with the probabilities $P_1, P_2, P_3, \dots, P_n$ respectively where $\sum p_i = 1$. Then the mathematical expectation, at times called “expected value” or “expectation”, of X is denoted by $E(X)$ and is defined as:

$$\begin{aligned} E(X) &= x_1 P_1 + x_2 P_2 + \dots + x_n P_n \\ &= \sum_{i=1}^n x_i P_i \\ \text{or } E(X) &= \sum_{i=1}^n x_i f(x_i) \text{ or simply as } \sum x f(x) \quad \text{--- 7.8.1} \end{aligned}$$

where $P_i = f(x_i)$

A special class of equation 7.8.1 is where the probabilities are all equal to give

$$E(X) = \frac{x_1 + x_2 + \dots + x_n}{n}, \text{ which is the arithmetic mean.}$$

Similarly for a continuous random variable X having density function $f(x)$, the expected value is defined by:

$$\int_{-\infty}^{\infty} x f(x) dx$$

$$E(X) = \int xf(x) dx$$

We have to note the following expectation as defined below:

$$(a) \quad E(X) = \sum xf(x) = \mu, \text{ the mean.}$$

$$(b) \quad E(X^2) = \sum x^2 f(x)$$

$$(c) \quad \text{Variance}(X) = \sigma^2 = \sum (x - \mu)^2 f(x) = E(X^2) - [E(X)]^2$$

where (i) is the expectation of variable X and (ii) is the expectation of the square of X.

The following are some of the properties of expectation

a. If C is any constant, then

$$E(CX) = CE(X)$$

b. If X and Y are any random variables, then

$$E(X + Y) = E(X) + E(Y)$$

c. If X and Y are independent random variables,

$$E(XY) = E(X) E(Y).$$

Illustration 7.15:

Find: a. $E(x)$, b. $E(x^2)$ and c. $E[(x - \mu)^2]$ for the probability distribution shown in the following table:

X	6	10	14	18	22
P(X)	1/6	5/36	1/4	1/3	1/9

Solution:

$$\begin{aligned} a. \quad E(X) &= \sum x \cdot P(x) \\ &= 6 \times \frac{1}{6} + 10 \times \frac{5}{36} + 14 \times \frac{1}{4} + 18 \times \frac{1}{3} + 22 \times \frac{1}{9} \\ &= 14.3333 \end{aligned}$$

This represents the mean of the distribution.

$$\begin{aligned} b. \quad E(x^2) &= \sum x^2 \cdot P(x) \\ &= 6^2 \times \frac{1}{6} + 10^2 \times \frac{5}{36} + 14^2 \times \frac{1}{4} + 18^2 \times \frac{1}{3} + 22^2 \times \frac{1}{9} \\ &= 230.6667 \end{aligned}$$

$$\begin{aligned} c. \quad V(X) &= E(X^2) - [E(X)]^2 \\ &= 230.6667 - (14.3333)^2 \\ &= 25.22 \quad \text{this represents the variance of the distribution.} \end{aligned}$$

Illustration 7.16: A company organized a lottery where there are 200 prizes of ₦50, 20 prizes of ₦250 and 5 prizes of ₦1000. If the company is ready to issue and sell 10,000 tickets, determine a fair price to pay for a ticket.

Solution: Let X and $f(x)$ represent the amount of money to be won and density function respectively. Then the following table depicts the distribution:

$X(\text{₦})$	50	250	1000	0
Frequency	200	20	5	$10,000 - 225$ $= 9775$
$f(x)$	$200 / 10,000$ $= 0.02$	$20 / 10,000 =$ 0.002	0.0005	0.9775

Note: $f(x) = \frac{\text{Number of prizes in a category}}{\text{Total Number of available tickets}}$

Then, the fair price

$$\begin{aligned}
 &= E(X) = 50(0.02) + 250(0.002) + 1000(0.0005) + 0(0.9775) \\
 &= \text{₦}2
 \end{aligned}$$

Illustration 7.17: Determine the expected value of the sum of points in tossing a pair of fair dice.

See the sample 7 on page 125 which gives the table below with X representing the sum

X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$	$6/36$	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$

$$\begin{aligned}
 E(X) &= \sum x.P(x) \\
 &= 2x\frac{1}{36} + 3x\frac{2}{36} + 4x\frac{3}{36} + 5x\frac{4}{36} + 6x\frac{5}{36} + 7x\frac{6}{36} + 8x\frac{5}{36} + 9x\frac{4}{36} + 10x\frac{3}{36} + 11x\frac{2}{36} + 12x\frac{1}{36} \\
 &= \frac{1}{18} + \frac{1}{6} + \frac{1}{3} + \frac{5}{9} + \frac{5}{36} + \frac{7}{6} + \frac{10}{9} + 1 + \frac{5}{6} + \frac{11}{18} + \frac{1}{36} \\
 &= 7
 \end{aligned}$$

Aliter: Let X and Y represent the points showing on the two dice. Then

$$E(X) = E(Y) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + \dots + 6\left(\frac{1}{6}\right) = \frac{7}{2}$$

$$\text{By the expectation property (b), } E(X + Y) = E(X) + E(Y) = \frac{7}{2} + \frac{7}{2} = 7$$

7.7 Some Special Probability Distributions

The following distributions are special in the sense that they have practical applications and being used frequently in computations of probability.

7.7.1 Binomial Distribution

Binomial probability distribution or Bernoulli distribution is the most fundamental and important discrete probability distribution in statistics and is defined by the probability function:

$$P(X) = \frac{n!}{x!(n-x)!} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n \quad \text{-- 7.9.1}$$

where $\frac{n!}{x!(n-x)!} = {}^nC_x = {}^nC_x$, (expansion principle in permutation and Combination which is read as n combination x)

$n! = n(n-1)(n-2) \dots n(n-1)$ (read as n factorial);

$x! = x(x-1)(x-2) \dots (1)$ (read as x factorial);

$0! = 1$,

e.g $8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

p is the constant probability of success,

q is the constant probability of failure (i.e $1 - p = q$) and

x is the number of successes in a series of n

independent trials ($x < n$)

with the distribution (Binomial) parameters n and p , the following are some of its properties:

- (a) Mean (μ) = np
- (b) Variance (σ) = npq
- (c) Standard deviation = $\sqrt{npq}$
- (d) Coefficient of skewness = $\frac{q-p}{\sqrt{npq}}$

Illustration 718:

If 30% of the bolts produced by a machine are defective, determine the probability that out of 4 bolts chosen at random,

- (a.) 1; (b.) 0 and (c.) at most 2 bolts will be defective

Solution:

The probability of a defective bolt is $p = 0.3$, and the probability of a non-defective bolt is $q = 1 - p = 1 - 0.3 = 0.7$

$$\begin{aligned}
 \text{(a.) } \Pr\{1 \text{ defective bolt out of } 4\} &= {}^4C_1(0.3)^1(0.7)^3 \\
 &= 4 \times 0.3 \times 0.343 \\
 &= 0.4116
 \end{aligned}$$

$$\begin{aligned}
 \text{(b.) } \Pr\{0 \text{ defective bolt out of } 4\} &= {}^4C_0(0.3)^0(0.7)^4 \\
 &= 1 \times 1 \times 0.2401 \\
 &= 0.2401
 \end{aligned}$$

$$\begin{aligned}
 \text{(c.) } \Pr\{2 \text{ defective bolts out of } 4\} &= {}^4C_2(0.3)^2(0.7)^2 \\
 &= 6 \times 0.09 \times 0.49 \\
 &= 0.2646
 \end{aligned}$$

Thus,

$$\begin{aligned}
 \Pr\{\text{at most 2 defective bolts}\} &= \Pr\{0 \text{ defective bolt}\} + \\
 &\quad \Pr\{1 \text{ defective bolt}\} + \\
 &\quad \Pr\{2 \text{ defective bolts}\} \\
 &= 0.2401 + 0.4116 + 0.2646 \\
 &= 0.9163
 \end{aligned}$$

Illustration 7.19: A bank conducted a survey of its customers and found out that 35% use savings account. If ten customers who deployed cash are observed,

- (a) determine the probability that
 - a. none will deposit in savings account.
 - b. only one will deposit in savings account
 - c. only two will deposit in savings account.
 - d. not more than two will deposit in savings account.
- (b) determine the mean and variance of the customers involved in savings account.

Solution: Let X represent the number of customers who deposit in savings account. Then we have:

$$\begin{aligned}
 \text{a. } P(X = x) &= \frac{n!}{x!(n-x)!} p^x q^{n-x} = \frac{10!}{x!(10-x)!} (0.35)^x (0.65)^{10-x} \\
 \text{(i) } P(X = 0) &= \frac{10!}{0!(10-0)!} (0.35)^0 (0.65)^{10-0} = 0.013463 \\
 \text{(ii) } P(X = 1) &= \frac{10!}{1!(10-1)!} (0.35)^1 (0.65)^{10-1} = 0.07249169 \\
 \text{(iii) } P(X = 2) &= \frac{10!}{2!(10-2)!} (0.35)^2 (0.65)^{10-2} = 0.175653 \\
 \text{(iv) } P(X < 2) &= P(X=0) + P(X=1) + P(X=2) \\
 &= 0.013463 + 0.0724916 + 0.175653 \\
 &= 0.2616 \\
 \text{b. Mean} &= np = 10(0.35) = 3.5 \quad \text{and} \\
 \text{Variance} &= npq = 10(0.35)(0.65) = 2.275
 \end{aligned}$$

7.7.2 Poisson Distribution

Siemon Poisson (1781 – 1840) derived the Poisson distribution as a limiting case of binomial distribution in 1837 in sequel to the following conditions:

- i. p , the constant probability of success for each trial is very closed and tends to zero (i.e $p \rightarrow 0$).
- ii. n , the number of trials is very large and even tends to infinity (i.e. $n \rightarrow \infty$)
- iii. $np = m$ (say), is the mean, finite and constant.

Therefore, the probability function of the Poisson distribution is defined by:

$$P(X) = \frac{e^{-m} m^x}{x!} \quad x = 0, 1, 2, \dots$$

With the Poisson parameter m , we have the following properties:

- (a) Mean (μ) = m ii. Variance (σ^2) = m
- (b) Standard deviation (σ) = $\sqrt{m}$
- (c) Coefficient of skewness = $1/\sqrt{m}$

Illustration 7.20

If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs:

- a. 0; b. 1; c. 2; d. 3;
bulbs will be defective

Solution:

$$P(X=x) = \frac{e^{-m} m^x}{x!} \quad x = 0, 1, \dots$$

$$P = 3\% = 0.03$$

$$n = 100$$

$$m = np = 0.03 \times 100 = 3$$

$$\text{a. } P(X=0) = \frac{e^{-3} 3^0}{0!} = e^{-3} \times 1 = e^{-3} = 0.0498$$

$$\text{b. } P(X=1) = \frac{e^{-3} 3^1}{1!} = e^{-3} \times 3 = 0.0498 \times 3 = 0.1494$$

$$\text{c. } P(X=2) = \frac{e^{-3} 3^2}{2!} = \frac{e^{-3} \times 9}{2} = \frac{0.0498 \times 9}{2} = 0.2240$$

$$\text{d. } P(X=3) = \frac{e^{-3} 3^3}{3!} = \frac{e^{-3} \times 27}{6} = \frac{0.0498 \times 9}{2} = 0.2240$$

7.7.3 Normal Distribution

Normal distribution is a continuous probability distribution and it is commonly used. It is also known as Gaussian distribution.

If x is a continuous random variable following the normal distribution with mean μ and standard deviation σ , then probability density function of Normal distribution is given by

$$P(X) = f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

where constants $\pi = 3.1416$ and $e = 2.7183$

If we compute $\int_a^b f(x)dx$ or $\int_{-\infty}^{\infty} f(x)dx$, it gives us the area under the normal curve. For

instance, $\int_{-\infty}^{\infty} f(x)dx = 1$. We are to note further that the area under the normal curve can

be obtained within certain range. Some of the properties of Normal distribution are given below:

- a. Mean = Mode = Median; and this makes it not to be skewed.
- b. Its graph is symmetric about the mean and this makes it to be bell shaped.
- c. Normal distribution is a continuous probability distribution with parameters μ and σ .
- d. The total area under the normal curve is equal to 1.
- e. The sub – areas under the normal curve correspond to some probabilities. This is achieved by the use of standard Normal variate (standard score) and the Normal Z table. The standard score Z is obtained by the expression

$$Z = \frac{x - \mu}{\sigma}$$

and to evaluate the sub-areas on the normal curve, the following steps are to be adhered to:

- i. The area required is sketched on the graph.
- ii. Determine the Z – value for each end point of the required area.
- iii. Sketch the corresponding area of the required one on the Z – graph.

A typical graph of Z – score is given below

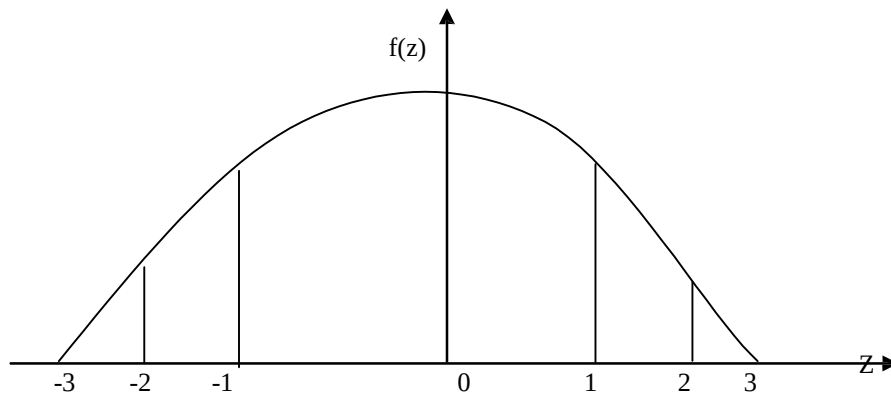


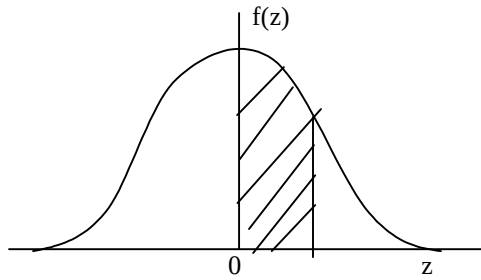
Fig. 7.7.3.1

- iv. Use the Normal Z table in the appendix to evaluate the required area. For instance,

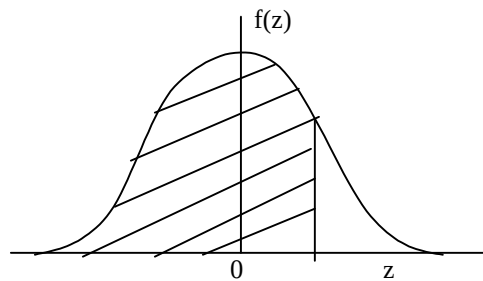
$$P(-1 \leq Z \leq 1) = 0.6827, \quad P(-2 \leq Z \leq 2) = 0.9545.$$

Note that the area under the curve $0 \leq z \leq 1 = -1 \leq z \leq 0$. These areas are probabilities.

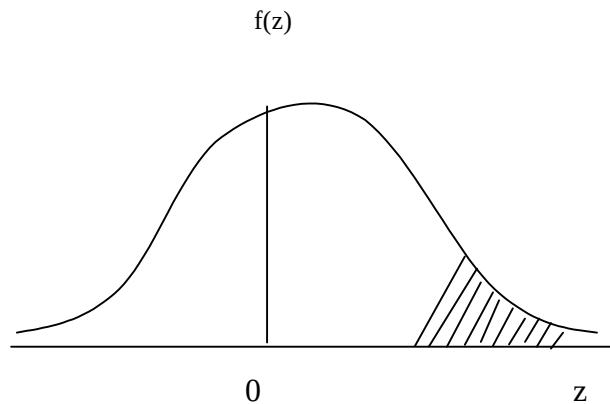
We are to note further that the Normal Z table can be read from the following forms and types.



In this type of graph, the table is read directly for the shaded area.



In this type of graph, the table is read by adding 0.5 to the table value for the shaded area.



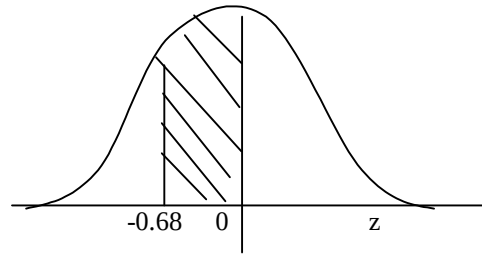
In this type of graph, the table is read by subtracting the table value from 1 for the shaded area.

Illustration 7.21: Determine the area under the standard normal curve:

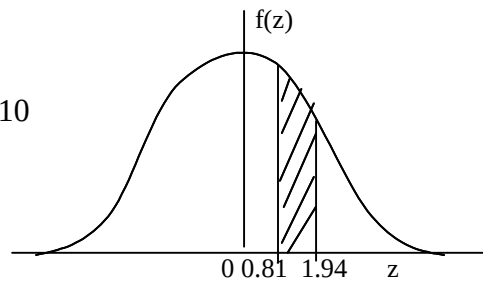
- between $Z = -0.68$ and $Z = 0$.
- between $Z = 0.81$ and $Z = 1.94$
- to the right of $Z = -1.2$

Solution

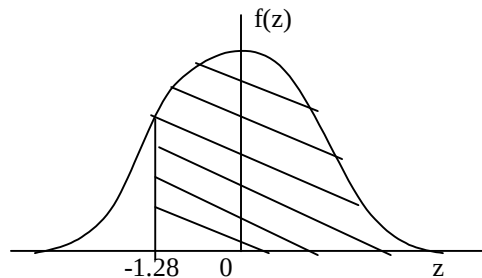
$$\begin{aligned}\text{a. } & P(-0.68 \leq Z \leq 0) \\ &= 0.2517 - 0.0 \\ &= 0.2517\end{aligned}$$



$$\begin{aligned}\text{b. } & P(0.81 \leq Z \leq 1.94) \\ &= 0.4738 - 0.2910 \\ &= 0.1828\end{aligned}$$



$$\begin{aligned}\text{c. } & P(Z \geq -1.28) \\ &= 0.3997 + 0.5 \\ &= 0.8997\end{aligned}$$

**Illustration 7.22:**

The mean inside diameter of a sample of 200 washers produced by a machine is 0.502 inches(in), and the standard deviation is 0.005inch. The purpose for which these washers is intended is to allow a maximum tolerance in the diameter of 0.480 to 0.516; otherwise, the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming that the diameters are normally distributed.

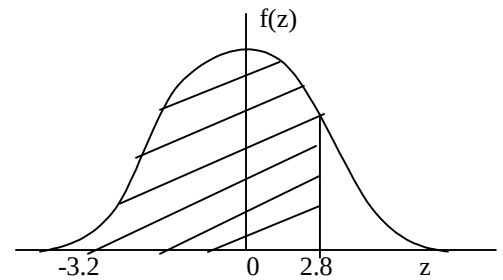
Solution:

$$\text{Standard score } Z = \frac{x - \mu}{\sigma}$$

$$\begin{aligned} 0.476 \text{ in standard units (Z)} &= \frac{0.486 - 0.502}{0.005} \\ &= \frac{-0.016}{0.005} = -3.2 \end{aligned}$$

$$\begin{aligned} 0.516 \text{ in standard units} &= \frac{0.516 - 0.502}{0.005} \\ &= \frac{0.014}{0.005} = 2.8 \end{aligned}$$

$$\begin{aligned} P(-3.2 < x < 2.8) \\ &= P(-3.2 < z < 2.8) \\ &= 1 - \phi(3.2) - \phi(2.8) \\ &= 1 - 0.4993 - 0.4974 \\ &= 0.0033 \\ &= 0.33\% \end{aligned}$$

**7.7.4. Approximations to Binomial Distribution****(a) Approximation to the Binomial distribution**

Recall that the probability function of a Binomial distribution is

$$P(X) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

where n and p are parameters specifying the distribution.

Computations with binomial when n is large usually become tedious and at times, they create problems. In order to overcome these problems, the idea of approximations to binomial was established.

i) Poisson approximation to the Binomial distribution

Poisson approximation to Binomial distribution is made possible on the following conditions:

- o the sample size n must be large ($n > 30$).
- o p , the probability of success, must be small and close to zero so that np is less than 5 ($np < 5$).
- o the mean of Poisson (m) is obtained by $m = np$.

(Note: some authors use λ for the mean).

ii) Normal approximation to the Binomial distribution

Normal approximation to Binomial distribution is made possible on the following conditions:

- the sample size n is large.
- the probabilities of success (p) and failure (q) are not small so that np and nq are greater than 5.
- that mean and variance of Normal distribution are obtained by $\mu = np$ and ($\sigma^2 = npq$).
- that by Central Limit Theorem, the standardized random variable

$$Z = \frac{q - p}{\sqrt{npq}}$$

is obtained for the Normal approximation.

The following examples illustrate the said approximations:

Illustration 7.23: Five percent of the goods produced in a manufacturing company turn out to be defective. Determine the probability that in a sample of 20 goods selected at random exactly two will be defective by using:

- the Binomial distribution.
- the Poisson approximation to the Binomial distribution.

Solution

- Let X represent the number of defective goods with the given probability of defective $p = 0.05$.

$$\therefore P(X=2) = \frac{20!}{2!(18)!} (0.05)^2(0.95)^{18} = 0.1888$$

- Mean of Poisson $m(\text{or } \lambda) = np = 20 \times 0.05 = 1$

$$P(X=x) = \frac{e^{-m} m^x}{x!} \quad m = 0, 1, \dots$$

$$\therefore P(X=2) = \frac{e^{-1} 1^2}{2!} = 0.1839$$

Note that the answers of (a) and (b) are very close.

Illustration 7.24: If a fair coin is tossed 10 times, determine the probability of getting between 4 and 6 heads inclusive by using the

- binomial distribution;
- normal approximation to the binomial distribution

Solution

- Let X represent the random variable of the number of heads.

$$P(X = 4) = \frac{10!}{4!6!} (0.5)^4(0.5)^6 = \frac{105}{512} = 0.2051$$

$$P(X = 5) = \frac{10!}{5!5!} (0.5)^5(0.5)^5 = \frac{63}{256} = 0.2461$$

$$P(X = 6) = \frac{10!}{6!4!} (0.5)^6(0.5)^4 = \frac{105}{512} = 0.2051$$

$$\therefore P(4 \leq x \leq 6) = 0.2051 + 0.2461 + 0.2051 = 0.6563$$

- By normal approximation, where $\mu = np = 10(0.5)^5 = 0.3125$

and $\sigma = \sqrt{npq} = \sqrt{10(0.5)(0.5)} = 1.58$, the standard scores are.

(Note that the data is now treated as continuous variable type, hence 4 to 6 heads shall be treated as 3.5 to 6.5 heads.)

$$3.5 \text{ in standard score} = (3.5 - 5)/1.58 = -0.95$$

$$6.5 \text{ in standard score} = (6.5 - 5)/1.58 = 0.95$$

$$\begin{aligned} \text{Probability of } (-0.95 \leq Z \leq 0.95) &= 0.8289 - 0.1711 \\ &= 0.6578 \end{aligned}$$

$$\therefore P(4 \leq x \leq 6) \text{ by normal approximation} = 0.6578$$

Also, the answers of (a) and (b) are very close.

(b) Normal approximation to the Poisson distribution

Without any doubt, there is a relationship between the Binomial and Normal distributions as treated in section 7.9.4.1. Hence, it is expected that there will also be a relationship between the Poisson and Normal distributions.

If X is the Poisson random variable, then the standardized random variable is

$$Z = \frac{x - m}{\sqrt{m}}$$

$$\sqrt{m}$$

where $\underline{m}$ is the Poisson mean and the Poisson distribution approaches the normal distribution as $\underline{m} \rightarrow \infty$ i.e as $\underline{m}$ is getting larger.

7.8 Chapter Summary

The principle of elementary probability and its applications are treated. The additive and multiplicative laws, and even the conditional probabilities are considered.

Treatment of some special distributions such as Binomial, Poisson and Normal are considered with their approximations.

MULTIPLE CHOICE AND SHORT – ANSWER QUESTIONS

Use the following information to answer the next three questions. Given a set of numbers

$$S = \{2, 4, 7, 10, 13, 16, 22, 27, 81, 102\}$$

If a number is picked at random, determine the probability that the number is

1. even
 - A. 0.4
 - B. 0.2
 - C. 0.6
 - D. 0.8
 - E. 0.9

2. odd
 - A. 0.4
 - B. 0.8
 - C. 0.2
 - D. 0.9
 - E. 0.6

3. prime

- A. 0.2
- B. 0.5
- C. 0.3
- D. 0.6
- E. 0.4

Use the following information to answer the next 2 questions:

Supposing there is a lottery game in which 2000 tickets are issued and that there are 5 major and 50 minor prizes to be won. If I buy a ticket, the probability that it will win

4. a major prize is

- A. 0.25
- B. 0.025
- C. 0.0025
- D. 0.5
- E. 0.05

5. a minor prize is

- A. 0.25
- B. 0.025
- C. 0.0025
- D. 0.5
- E. 0.05

6. Using the item of question 4, the probability of winning a prize in the lottery game is

7. In a three diagram of method of calculating the probability of an event, the weight attached to the outcome of trial or stage is written on the branch while the corresponding outcome is at the of the branch

Use the following statement to answer questions 8 to 10: If a ball is selected randomly from a box containing 6 white balls, 4 blue balls and 5 red balls, obtain the probability that:

8. White ball is selected.....
9. Not white ball is selected.....
10. Blue or red ball is selected.....

Answers

1. C
2. A
3. C
4. C
5. B
6. 0.0275 i.e. $\frac{5}{2000} + \frac{50}{2000} = 0.0275$
7. end
8. $\text{Pr}(\text{white}) = \frac{6}{15} = 0.4,$
9. $\text{Pr}(\text{Not white}) = 1 - \text{Pr}(\text{white}) = 1 - 0.4 = 0.6$
10. $\text{Pr}(\text{Blue or Red}) = \frac{4}{15} + \frac{5}{15} = \frac{9}{15} = 0.6$

CHAPTER EIGHT

STATISTICAL INFERENCE

CHAPTER CONTENTS

- (a) Introduction;
- (b) Basic Concepts of Estimation;
- (c) Point Estimation of the Population Mean;
- (d) Point Estimation of the Population;
- (e) Useful Concepts in Hypothesis Testing;
- (f) Testing Hypothesis About a Population Parameter; and
- (g) Testing Hypotesis Abourt a Population Proportion

OBJECTIVES

At the end of the chapter, readers should be able to:

- (a) know that statistical inference consists of estimation and significance test;
- (b) demonstrate good understanding of both point and interval estimates;
- (c) estimate the population mean and population proportion;
- (d) understand test of hypothesis on population mean and population proportion for both large and small samples;

8.1 Introduction

An important study of statistical theory, which is commonly used in decision making, is the concept of statistical inference. Here, estimates and inference (both inductive and deductive) are made in order to make decisions. These decision makings are carried out on the basis of samples taken from the population. For instance, a research worker performs an experiment and obtains some data. On the basis of the data, some estimates and certain decisions or conclusions are made. These decisions are then generalized to similar situations of that experiment. In order words, the researcher may generalize from a particular experiment to all the classes of all similar experiments.

Succinctly, statistical inference can be broadly classified into two; viz:

- a) Estimation theory and;
- b) Test of Hypothesis.

8.2 Basic Concepts of Estimation

8.2.1 Population Parameter

This is a quantity whose numerical value is known for a population. It is this specified quantity that allows one to make probability statement about the distribution of the given population. Therefore, a population is completely known when we know the probability distribution $f(x)$ with the value(s) of its parameter(s) for a given random variable X . For instance, in binomial distribution, the value of P (probability of success) must be known before any meaningful statement can be made about the population distribution it is representing.

8.2.2 Sample Statistics

In practice, the population parameter is not usually known. Therefore, efforts are to be made to have an approximate value called estimate. All we need to do is to take random samples from the target population and then use these samples to obtain values which serve to estimate the population parameter. This set of estimates serves as a representative of population parameter. Therefore, any quantity obtained from a sample for the purpose of estimating a population parameter is called a sample statistic, or briefly statistic.

The following table shows the usual symbols used for population parameter and sample statistics.

Description of Measure	Population parameter symbol	Sample statistics symbol
Mean	μ	$\bar{x}$
Variance	σ^2	S^2
Number of items	N	n
Proportion	P	p

8.2.3 Estimate and Estimator

An approximate value obtained from a sample is called an estimate; while an estimator is the expression used for computing or determining the estimate. For example, in computation of mean for variable X, we have

$$\bar{x} = \frac{\sum x}{n}$$

$\bar{x}$ is the estimate while $\sum x/n$ is the estimator.

Further, an estimate can be point or interval. It is point estimate when the population parameter is given by a single number; while an estimate of population parameter is given between the range of two numbers.

8.2.4 Sampling Distribution

This is an important concept which makes inference from sample to population possible. By sampling distribution, we mean the probability distribution of a sample statistic. This can be achieved by considering all possible samples of size n from N (population size); and for each sample, calculate or determine the statistic estimate. Next obtain the frequency distribution of the statistic estimates in order to have what is called “sampling distribution”.

For sampling distribution, we have various types such as: mean, variance, proportion, to mention a few.

8.3 Point Estimation of the Population Mean

The point estimation of the population mean can be obtained by adhering to the following steps:

- a) First obtain the sampling distribution of means for all the possible samples of size n .
- b) Next, compute the sample mean $\bar{x}$ by finding the average of set of means obtained in (i) i.e finding mean of means to give the desired sample mean $\bar{x}$.

The following example will illustrate the procedures stated above.

Illustration 8.1:

If all possible samples of size two are drawn with replacement from a population consisting of numbers 7, 8, 12, 13, determine the:

- mean of the population (μ);
- standard deviation of the population;
- the mean of the sampling distribution of means ($\bar{x}$);
- the standard deviation of the sampling distribution of means.

Solution

$$\text{a. Population mean } \mu = \frac{7 + 8 + 12 + 13}{4} = \frac{40}{4} = 10$$

- b. Standard deviation of population (σ),

$$\begin{aligned}\sigma^2 &= \frac{(7-10)^2 + (8-10)^2 + (12-10)^2 + (13-10)^2}{4} \\ &= \frac{9 + 4 + 4 + 9}{4} = \frac{26}{4} = \frac{13}{2} = 6.5 \\ \therefore \sigma &= \sqrt{\sigma^2} = \sqrt{6.5} = 2.55\end{aligned}$$

- c. Selection of 2 samples with replacement:

(7, 7)	(8, 7)	(12, 7)	(13, 7)
(7, 8)	(8, 8)	(12, 8)	(13, 8)
(7, 12)	(8, 12)	(12, 12)	(13, 12)
(7, 13)	(8, 13)	(12, 13)	(13, 13)

The corresponding sample means:

7	7.5	9.5	10.0
7.5	8.0	10.0	10.5
9.5	10.0	12	12.5
10.0	10.5	12.5	13.0

$\therefore$ The sampling distribution of means can be stated as:

Number ($\bar{x}$)	7	7.5	8.0	9.5	10.0	10.5	12.0	12.5	13.0
Frequency(f)	1	2	1	2	4	2	1	2	1

The mean of sampling distribution of means ($\bar{x}$) is equal to $\bar{x} = \frac{\sum f\bar{x}}{\sum f}$

$$= \frac{7 + 2(7.5) + 8.0 + 2(9.5) + 4(10) + 2(10.5) + 12 + 2(12.5) + 13}{16}$$

$$= \frac{160}{16} = 10$$

d. $S^2 = \frac{(7-10)^2 + (7.5-10)^2 + \dots + (12.5-10)^2 + (13-10)^2}{16}$

$$= \frac{52}{16} = 3.25$$

$\therefore S = \sqrt{S^2} = \sqrt{3.25} = 1.8$

Remark: From the above computations, we could note that:

- (a) the results of (a) and (c) are equal. This illustrates the fact that $E(\bar{x}) = \mu = 10$; i.e $\bar{x}$ is an unbiased estimator of μ .
- (b) the result of (b) and (d) demonstrates a relationship between the variances of the population mean and sample mean.

Here, it is established that $\underline{S}^2 = \sigma^2/n$ for infinite population or sampling with replacement.

i.e $\underline{S}^2 = 3.25$ and $\sigma^2 = 6.5$ where $n = 2$ (as the sample size)

8.4 Point Estimation of Population Proportion.

Here, it is assumed that the population is infinite and binomially distributed. With these assumptions, the same procedures used in point estimation of population mean shall be equally applied. That is,

- a) to first obtain the sampling distribution of proportions and next;
- b) to compute the mean of this set of proportions.

For instance, a consideration of all possible samples of size n drawn from the population and for each sample, the proportion statistic (p) is determined leading to the sampling distribution of proportions with mean (μ_p)

$$\mu_p = p \quad \text{and} \quad \sigma_p = \sqrt{\frac{pq}{n}} = \sqrt{\frac{p(1-p)}{n}}$$

Illustration 8.2

If each person of a group of 400 people tosses a fair die 90 times, how many people should be expected to report that between 20% and 30% of their tosses resulted in having die face 5?

Solution:

The probability p of die face 5 = $1/6 = 0.1667$ while $q = 1 - p$

$$q = 1 - 1/6 = 5/6 = 0.8333$$

$$\mu = np = 90(1/6) = 15 \quad \text{and} \quad \sigma = \sqrt{npq} = \sqrt{90(1/6)(5/6)} = 3.54$$

Normal approximation to Binomial shall be used for this problem

The probability that face 5 in 90 tosses will be between 20% and 30% is desired, 20% of 90 = 18 and 30% of 90 = 27. Since these numbers (18 and 27) are discrete, we need to adjust them to continuous variable for 17.5 and 27.5.

∴ The standard scores of these adjusted numbers are:

$$17.5 \text{ in standard units (z)} = \frac{17.5 - 15}{3.54} = z(0.7062)$$

$$27.5 \text{ in standard units (z)} = \frac{27.5 - 15}{3.54} = z(3.53)$$

$$\begin{aligned} \Pr(0.7062 \leq 3.53) &= 0.9998 - 0.7612 \\ &= 0.2386 \end{aligned}$$

$$\therefore 23.86\% \text{ of } 400 = 95.44 \approx 95$$

About 95 people would report that their experiment resulted in between 20% and 30% of die face 5.

8.5 Useful Concepts in Hypothesis Testing

The following terms are useful in the principle of Hypothesis testing.

8.5.1 Hypothesis: The concept of hypothesis in statistics aids decision making. Hypothesis can be defined as the assumption or guess about the population parameters involved.

There are two types of hypothesis; namely, the

- a) Null hypothesis
- b) Alternative hypothesis

An hypothesis which states that there is no difference between the procedures, results from samples and other phenomenon is called Null Hypothesis and it is denoted by H_0 , while any hypothesis which differs from the null hypothesis or given hypothesis is known as Alternative Hypothesis and it is denoted by H_1 . For instance, if the population mean $\mu = 5.0$ and the sample mean $\bar{x} = 6.0$, we set our hypothesis as thus:

$$\begin{array}{ll} H_0 : \mu = \bar{x} & \text{i.e } H_0 : 5.0 = 6.0 \\ H_1 : \mu \neq \bar{x} & \text{i.e } H_1 : 5.0 \neq 6.0 \text{ or} \\ H_1 : \mu < \bar{x} & \text{i.e } H_1 : 5.0 < 6.0 \end{array}$$

8.5.2 Errors:

There are two types of errors in hypothesis testing, namely,

- (a) Type I error
- (b) Type II error.

When a hypothesis which is supposed to be accepted is rejected, we call that error a type I error. On the other hand, if we accept a hypothesis which is supposed to be rejected, a type II error is committed.

The above statements can be summarized in the following table:

Decisions	H_0 true	H_0 False
Accept H_0	Correct decision	Type I error
Reject H_0	Type II error	Correct decision

8.5.3 Level of Significance

In hypothesis testing, the maximum probability of the willingness to risk a type I error is called the level of significance or simply significance level and it is denoted by α . It therefore implies that the investigator has $(1 - \alpha)$ confidence that he/she is making right decision

In practice, this must be stated first in any hypothesis testing. The common levels of usage in practice are 1%(or 0.01) and 5% (or 0.05).

8.5.4 Test Statistic:

A test statistic is the computed value, which, when compared with the tabulated value, enables one to decide whether to accept or reject hypothesis and hence determine whether the figures of the observed samples differ significantly from that of the population. It is also called *test of hypothesis*, test of significance or rules of decision.

8.5.5 Critical Region:

This is a region or critical value of one side of the distribution of the case study, with area equal to the level of significance. It is used to decide whether to accept a hypothesis or not.

8.5.6 One-tailed and Two-tailed Tests

The type of tailed test depends on the stated alternative hypothesis (H_1). It is one-tailed test or one-sided test when the alternative hypothesis is one directional. For example, we have the following hypothesis for one-tailed test:

$$H_0: \mu = \bar{x}$$

$$H_1: \mu > \bar{x}$$

$$H_1: \mu \geq \bar{x}$$

$$H_1: \mu < \bar{x}$$

$$H_1: \mu \leq \bar{x}$$

For a two-tailed test or two-sided test, the hypothesis is of two directions e.g

$$H_0: \mu = \bar{x}$$

$$H_1: \mu \neq \bar{x}$$

The resultant effect of the type of tailed test is on the significance level to be used in obtaining the critical value from the table. If the significance level is α , then critical values for α and $\alpha/2$ will be obtained respectively for one-tailed and two-tailed tests.

For instance, if $\alpha = 0.05$, we check critical values at 0.05 and $0.05/2 = 0.025$ respectively for the one-tailed and two-tailed tests.

8.6 Testing Hypothesis About a Population Parameters

The procedure for carrying out significance test about a population parameter is given as follows:

- a) Set up the hypothesis about the parameter of interest and state its significance level.
- b) Set the hypothesis by using appropriate test statistic.
- c) Draw conclusion by making decision on the result of your computed value in step (ii) above. Here, table value at a significance level is compared with computed value.

8.7 Testing Hypothesis About the Population Mean

This section will present testing hypothesis under two conditions namely; (i) when we have large samples and (ii) when we have small samples.

For large samples ($n \geq 30$) it is usually assumed that the sampling distribution of the desired statistic is normally distributed or approximately normal. It is for this reason that tests concerning large samples assumed normal and used the Z test. Therefore the test statistic for large sample is

$$Z_{\text{cal}} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

where $\bar{x}$ = sample mean;

μ_0 = hypothesis value of μ .

σ = the given value of the standard deviation; but when this not given, one can use sample standard deviation (S) for large samples.

n = sample size

When we have small samples ($n < 30$), the test – statistic is the t – test and it is defined as thus:

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

where $\bar{x}$, μ_0 , $\underline{n}$ are as defined above and

$$S^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

The final decision on the tests is made by comparing the computed values (Z_{cal} or t_{cal})

with the table values. The normal table is used for the Z – test (for large samples) while t – distribution table at n – 1 degrees of freedom is used for small samples.

When $Z_{\text{cal}} > Z_{\text{table}}$, and $t_{\text{cal}} > t_{\text{table}}$, we reject the null hypothesis (H_0); otherwise, accept H_0 .

Illustration 8.3

In a University, a sample of 225 male students was taken in order to find the average height. From the samples, the computed average height was 184.0cms while the mean of actual population height was 178.5cms with a standard deviation of 120cms. You are required to show if the sample mean height is significantly different from the population mean at 5% significant level.

Solution:

Hypothesis

$$H_0: \mu = x \Rightarrow H_0: 178.5 = 184.0$$

$$H_1: \mu \neq x \Rightarrow H_1: 178.5 \neq 184.0$$

$$\begin{aligned} Z_{\text{cal}} &= \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{184.0 - 178.5}{\frac{120}{\sqrt{225}}} = \frac{5.5}{8} \\ &= 0.6875 \end{aligned}$$

Table value of Z at 5% level of significance (for two tailed test) = 1.96

Decision: since $Z_{\text{cal}} < Z_{\text{table}}$ ($0.6875 < 1.96$), we accept H_0 and concluded that no significance difference between the sample mean and population mean.

Illustration 8.4

SAO is a manufacturing company with 5000 workers. The company is interested in knowing the average number of items sold per week per person by the company's workers. While the quality control manager thinks that the average number of items sold per worker per week is 11, the company secretary thinks that the true value should be more. The quality control manager subsequently selected 11 workers at random and got the following results as number of items sold in a week, 13, 4, 17, 9, 3, 20, 16, 12, 8, 18, 12.

You are required to set up a suitable hypothesis and test it at 5% level of significance.

Solution:

$$H_0 : \mu = 11$$

$$H_1 : \mu > 11$$

$$\alpha = 0.05$$

Since the sample size (11) is small, we use the t – test

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{S / \sqrt{n}} \quad \mu_0 = 11, n = 11$$

$$\bar{x} = \frac{\sum x}{n} \quad \text{and} \quad S^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\begin{aligned} \text{mean} = \bar{x} &= \frac{\sum x}{n} = \frac{13 + 4 + 17 + 9 + 3 + 20 + 16 + 12 + 8 + 18 + 12}{10} \\ &= \frac{132}{11} = 12 \end{aligned}$$

The computation of S^2 is set out in the table below

x	$x - \bar{x}$	$(x - \bar{x})^2$
13	1	1
4	-8	64
17	5	25
9	-3	9
3	-9	81
20	8	64
16	4	16
12	0	0
8	-4	16
18	6	36
12	0	0
		<u>312</u>

$$\begin{aligned} S^2 &= \frac{\sum (x - \bar{x})^2}{n - 1} \\ &= \frac{312}{11 - 1} = \frac{312}{10} = 31.2 \end{aligned}$$

$$\begin{aligned} \therefore S^2 &= 31.2, \rightarrow S = \sqrt{31.2} \\ &= 5.5857 \\ &\approx 5.6 \end{aligned}$$

using all these pieces of information, our test statistic becomes

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

$$\begin{aligned}
&= \frac{12-11}{5.6/\sqrt{11}} \\
&= 0.5923 \\
&\approx 0.59
\end{aligned}$$

Table value of t at 5% significance level (for one – tailed test) = 1.81

Decision: the problem is of one – tail test (see the alternative hypothesis), and $t_{\text{cal}} < t_{\text{table}}$ i.e $0.59 < 1.81$, we accept H_0 and conclude that no significance difference between the sample mean and the population mean.

8.8 Testing Hypothesis About the Population Proportion

In the significance test for a proportion, there is the need to know the population proportion (P_0) and then compute the sample proportion (P) from the sample drawn. The sample proportion is obtained by taking a number of observations (x) (possessing an attribute) out of the total number n so that $p = \frac{x}{n}$

As we have done for the mean, we set up the hypothesis as thus:

$$H_0 : P_0 = P$$

$$H_1 : P_0 \neq P \text{ (or } P_0 > P \text{ or } P_0 < P)$$

and the significance level given is α .

Then, the test statistic is

$$Z_{\text{cal}} = \frac{P - P_0}{\sqrt{\frac{P(1-P)}{n}}} \quad \text{or} \quad \frac{\frac{x}{n} - P_0}{\sqrt{\frac{\frac{x}{n}(1 - \frac{x}{n})}{n}}}$$

and we decide by comparing the Z_{cal} with the table value Z_{table} at the given α

Illustration 8.5

A demographer claims that pupils in all the primary schools in a State constitute 30% of the total population of the state. A random sample of 400 pupils from all primary schools in the Local Government Areas of the State shows that 25% of them are pupils of primary school.

Test at 5% level of significance the validity or otherwise of the demographer's claim.

Solution:

Let P be the population proportion of pupils in the primary schools in the State and $\hat{P}$ the estimated proportion.

$$H_0 : P_0 = P \quad \text{i.e } H_0 : 0.30 = 0.25$$

$$H_1 : P_0 \neq P \quad \text{i.e } H_1 : 0.30 \neq 0.25$$

$$\begin{aligned} \therefore Z_{\text{cal}} &= \frac{P - P_0}{\sqrt{\frac{P(1-P)}{n}}} \\ &= \frac{0.25 - 0.30}{\sqrt{\frac{0.30(1-0.30)}{400}}} = \frac{-0.05}{\sqrt{\frac{0.30(0.70)}{400}}} = \frac{-0.05}{0.0229} \\ &= -2.1834 \end{aligned}$$

It is a two-sided test

$\therefore$ for $\alpha = 0.05$, then $\alpha/2 = 0.025$ and table value of $Z = 1.96$

Decision: since $|Z_{\text{cal}}| > Z_{\text{table}}$ i.e $2.1834 > 1.96$,

H_0 is rejected and conclude that the data can not support the demographer's claim.

8.9 Chapter Summary

Statistical inference has been discussed under two broad headings, namely: the estimation theory and test of hypothesis.

In the estimation theory, point and interval estimates are defined. The principle of sampling distribution is also presented as a basis for obtaining point estimates of mean and proportion. Important concept such as hypothesis, errors, significance level, test statistics, one-tailed and two-tail tests are discussed in the hypothesis testing. Tests concerning the mean and proportion are finally presented.

MULTIPLE CHOICE QUESTIONS (MCQ)

- In estimation theory, a statistic of a distribution is a quantity computed from_____
 - A population,
 - The normal distribution
 - A sample,
 - The poisson distribution
 - Average of the data from both sample and population.

2. Significance level is referred to as the risk of _____
- A. Sampling error,
 - B. Non-sampling error,
 - C. Bias error
 - D. Type i error,
 - E. Type ii error.
3. Which of the following hypothesis is not for one – tailed test?
- A. $H_1 : \mu > 0$,
 - B. $H_1 : \mu \geq 0$,
 - C. $H_1 : \mu < 0$
 - D. $H_1 : \mu \leq 0$,
 - E. $H_1 : \mu \neq 0$.

SHORT ANSWER QUESTIONS (SAQ)

4. The test – statistic for a large sample in the hypothesis testing of mean is _____
5. The degree of freedom for t – test with n samples is _____
6. The test – statistic for proportion is _____

Use the following information to answer questions 7 to 10

A sample of 4 items are taken randomly from a population sample, their weights (in Kgs) are recorded as follows: 6, 8, 12, and 14.

7. Determine the mean weight of the sample.
8. Determine the standard deviation of sample.
9. Compute the standard error of the mean.
10. If the population mean $\mu = 9$, compute the test – statistic for the data

Answers

1. C

2. D

3. E

$$4. \quad Z_{\text{cal}} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}},$$

5. $\underline{n} - 1$

$$6. \quad Z_{\text{cal}} = \frac{P - P_0}{\sqrt{\frac{P(1-P)}{n}}},$$

7. 10,

8.

$$= \sqrt{\frac{(6-10)^2 + (8-10)^2 + (12-10)^2 + (14-10)^2}{4}}$$

$$= \sqrt{\frac{16 + 4 + 4 + 16}{4}}$$

$$= 3.16$$

$$9. \quad 3.16 / \sqrt{4} = 1.58$$

$$10. \quad t_{\text{cal}} = \frac{\bar{x} - \mu_0}{S / \sqrt{n}} = \frac{10 - 9}{1.58} = 0.63$$

